

Regularity of the Time-to-Collision Map and Convergence of Learned Hamilton–Jacobi Solutions

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Abstract. Controllers for autonomous vehicles that depend solely on inter-vehicle spacing can behave overly aggressively by ignoring relative velocity. Time-to-collision (TTC) addresses this, but classical kinematic TTC assumes a frozen system state, making it non-predictive and unreliable as feedback. We introduce a dynamically-informed TTC, defined as the hitting time of the collision set under an optimal-velocity prediction flow with merely measurable (L^∞) leader acceleration. This makes the collision estimate consistent with the dynamics, at the cost of a circularity dynamical model. We establish well-posedness and collision avoidance of the resulting model through barrier-domain and near-collision boundary analysis. The analytical core is a regularity theory for TTC on the domain of initial data. We prove That TTC map is in fact C^1 , on compact subsets, with an explicit gradient. All constants are governed by a single uniform transversality margin that degenerates as tangential collisions are approached. Since the dynamic TTC admits no closed form, online simulation introduces delay that undermines safety guarantees. Recasting the TTC map as value function, we prove a two-sided convergence result for a physics-informed formulation; the L^2 error of the learned value function is bounded by the square root of the total loss in HJB framework, and at zero residual the approximation coincides pointwise with the viscosity solution. A two-stage architecture enables delay-free platoon operation, and simulations on a six-vehicle platoon confirm stability across diverse scenarios.

Key words. Time-to-collision; car-following dynamics; hitting-time regularity; Hamilton–Jacobi–Bellman equations; viscosity solutions; physics-informed neural networks; Carathéodory solutions; collision avoidance

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1. Introduction and Related Works. Microscopic traffic models describe vehicle-to-vehicle interaction through ordinary differential equations in which each vehicle responds to its predecessor. Classical formulations [46, 42], and in particular the Optimal Velocity Model (OVM) [4], reproduce observed phenomena such as stop-and-go waves and spontaneous congestion [16, 27], have been analysed extensively for stability [59, 61, 37, 38], and underpin driver-assistance and autonomous-vehicle control [53, 51]. Numerous refinements improve behavioural realism [22, 24, 36, 39, 66, 65]. A limitation shared by this family is that the control depends only on inter-vehicle spacing, and therefore carries no information about the rate of

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closure: the response grows exaggerated as spacing decreases even when the leader is pulling away and collision risk is low [65]. Time-to-collision (TTC) supplies exactly the missing information, and empirical studies confirm that drivers respond more strongly to TTC than to spacing alone [62, 31, 44, 55, 56].

A circularity in kinematic TTC. The classical TTC of Hayward [21] is the gap divided by the closing speed, and the refinements that followed — Potential TTC [54], Modified TTC [45, 63], two-dimensional extensions [60, 40] — relax the constant-speed assumption only through limited acceleration projections. All remain *frozen-state*: TTC is computed as though no control acts from the present moment onward [21, 8, 17, 9, 23, 2]. Embedded in a feedback law this is self contradictory — the controller is designed to prevent precisely the collision that the estimate presumes will be met passively — and several OVM-based models inherit the inconsistency by retaining an instantaneous TTC term [25, 35, 30, 1, 26, 32].

We therefore define TTC through the system’s own predicted evolution. Writing Ξ for the inter-vehicle gap and Λ for the relative velocity, the *dynamic* TTC is the hitting time

$$(1.1) \quad \tau(\mathbf{z}) \stackrel{\text{def}}{=} \inf\{\theta > 0 : \Xi(\theta; \mathbf{z}) = 0\},$$

where $\Xi(\cdot; \mathbf{z})$ evolves under OVM prediction dynamics started from $\mathbf{z}$, with a leader acceleration that is merely measurable and bounded, $\mathbf{a}_\ell \in L^\infty$. The OVM is the natural choice of prediction model: it is the base layer of the proposed dynamics with the TTC term removed [4, 3], so the controller anticipates the trajectory that the uncontrolled system would in fact follow. The resulting control law is genuinely predictive — but the map τ now appears in the vector field whose flow defines it, and the model is well posed only if τ is regular enough to serve as a coefficient. Three obstructions follow, and they organize the paper.

(i) Regularity of a hitting time. Hitting times are generically discontinuous; a trajectory tangent to the target set can be perturbed into one that misses it entirely. Regularity of exit- and minimum-time functions is classical [6, 10], where Lipschitz continuity is typically obtained from Petrov-type controllability conditions ensuring that some admissible control drives the state into the target at a definite rate; without such a condition the value function is continuous but need not be locally Lipschitz. Our setting is different in two respects. The prediction flow is *uncontrolled*, so the relevant nondegeneracy is not controllability but *transversality*; the predicted trajectory must cross $\{\Xi = 0\}$ with $\Lambda < 0$ rather than graze it. And the forcing is only L^∞ in time, placing the problem in the Carathéodory rather than the classical framework [20]—so that a pointwise transversality condition is not even well defined for a fixed representative, and every estimate must be phrased through integrated quantities. We prove that on the domain of transversal initial data, τ is locally Lipschitz and in fact C^1 , with all constants controlled by a single uniform transversality margin that degenerates precisely as tangency is approached.

(ii) No closed form, hence delay. Eliminating Λ reduces the prediction dynamics to a forced, damped nonlinear oscillator with measurable forcing; no closed-form solution is available, and τ additionally requires inverting the flow to locate a zero of the gap. Evaluating τ therefore means simulating an ODE at every control instant. The resulting latency is not merely an inefficiency; by the time the simulation completes the state has moved, the feedback loop is broken, and the collision guarantees that the controller was designed to provide no longer

apply. A real-time surrogate is thus mathematically necessary, not only practically convenient [15, 43].

(iii) *Approximation with guarantees.* Neural networks approximate controlled dynamics compactly and evaluate in constant time [34, 28, 12], and rigorous foundations exist for their use in safety-critical control [49, 50, 7, 29]. The physics-informed framework [48] embeds the governing equation in the loss. What is required here, however, is not merely a good fit but a *guarantee*: that a small residual implies a small error in the learned object. We obtain this by recasting τ as the value function of a minimum-time problem. The dynamic programming principle yields a first-order Hamilton–Jacobi equation whose unique viscosity solution [14, 6] is τ itself, and whose Hamiltonian is *affine* in the gradient; consequently the characteristics coincide with the OVM trajectories and do not cross. This is the structural reason an almost-everywhere classical solution here agrees with the viscosity solution, in contrast to the general Hamilton–Jacobi setting, where gradient ambiguity admits spurious almost-everywhere solutions.

Related work on learning Hamilton–Jacobi solutions. Hamilton–Jacobi reachability provides a set-based characterisation of safety through the value function of a differential game [41, 18], and learned approximations have made high-dimensional instances tractable [5]. These treat the dynamics as given and the value function as the object to be learned; the regularity of that object is inherited rather than established. Convergence theory for physics-informed approximation has developed in parallel [52, 47], largely for linear elliptic and parabolic problems. Closest to the present work is [33], which bounds solution error by residual for stationary Lyapunov and Hamilton–Jacobi–Bellman equations by integrating the residual along closed-loop trajectories — the same mechanism we use. The settings differ in ways that matter: there the value function is an infinite-horizon cost to an equilibrium, and a *pointwise relative* residual bound is required, since a merely uniform bound may integrate to infinity along trajectories; here the horizon is the finite hitting time (1.1) with a free terminal boundary, which removes that obstruction and permits an L^2 hypothesis, but requires establishing regularity of the hitting time itself. To our knowledge the TTC map has not previously been treated as a value function, and no regularity theory for it exists; the traffic literature computes TTC kinematically throughout.

Sobolev regularity of the target is what converts approximation theory into explicit rates [64]; the $W^{1,\infty}$ and C^1 results of Section 4 play exactly that role here, and the C^1 property is what makes a vanishing strong-form residual attainable at all within a smooth hypothesis class.

Contributions. The main contributions of this paper are as follows:

- **A dynamically-informed TTC and its closed-loop model.** We introduce a feedback law in which the time-to-collision is computed as the hitting time of the collision set under the system’s own prediction dynamics, rather than under a frozen-state assumption. The resulting controller is genuinely predictive, and resolves the circularity of classical TTC feedback — at the cost of a new one: the map τ appears in the vector field whose flow defines it.
- **Well-posedness and collision avoidance.** We establish existence and uniqueness for the coupled system by a Picard-type fixed-point argument in a Carathéodory set-

ting, continuation up to the predictive collision time, and a maximum principle for the velocities. Collision is then *excluded* by a contradiction argument: near the boundary the barrier term integrates to infinity, contradicting the a priori velocity bound. The singularity that complicates the analysis is precisely the mechanism that forbids collision.

- **Regularity of the hitting-time map.** We prove that τ is locally Lipschitz, $\tau \in \mathcal{W}^{1,\infty}$, and in fact $\tau \in C^1$, on compact subsets of the transversal domain, with an explicit gradient obtained by implicit differentiation. Two features are structural: every constant is governed by a single uniform transversality margin γ_D , which degenerates as tangential collisions are approached and accounts for the failure of global Lipschitz continuity; and because the forcing is merely L^∞ in time, pointwise transversality is ill-defined, so the estimates are organized throughout around integrated quantities. This regularity is not covered by existing exit-time theory, where Lipschitz continuity is obtained from controllability conditions unavailable for an uncontrolled prediction flow.
- **Two-sided convergence of the learned value function.** Recasting τ as the value function of a minimum-time problem, we prove that the L^2 error of a physics-informed approximation is bounded by the square root of the total loss, via a co-area argument on the endpoint map — requiring a positive lower bound on its 2-Jacobian — together with a change-of-variables estimate along the flow. At zero residual the conclusion sharpens to *pointwise* identification with the viscosity solution; the C^1 regularity above is exactly what makes a vanishing strong-form residual attainable in a smooth hypothesis class.
- **Delay-free architecture and validation.** We propose a two-stage physics-informed architecture $\mathcal{N}_{\text{sol}}$ for approximating the full system trajectory without delay, designed around the two features that make the problem hard: the near-collision singularity and the oscillatory dynamics. Local stability of the non-delayed system is established, and the framework is validated on a six-vehicle platoon.

2. Mathematical Model of Dynamic-based TTC Control. In this section, we present the general proposed control framework, which consists of the proposed AV dynamical model with dynamic-based TTC, a novel approach in computing TTC.

2.1. Motivation. While the OVM effectively captures key traffic phenomena, controllers that depend solely on inter-vehicle spacing can exhibit overly aggressive behavior in practice. This issue arises because spacing alone carries no information about the rate of closure between vehicles: as spacing decreases, the control response grows exaggerated even when the actual collision risk is low — for instance, when the leading vehicle is moving significantly faster than the ego vehicle (see [Figure 2](#)). The OVFL model [\[36\]](#), a well-studied OVM-based model for AVs — [\[4, 37, 38\]](#) — addresses collision avoidance theoretically through boundary-layer analysis [\[43\]](#), yet can still exhibit this aggressive behavior in practice due to its spacing-only dependence. Incorporating relative velocity into the control resolves this issue by providing information about future spacing evolution. When both spacing and relative velocity are considered, the controller naturally operates as a predictive controller, capable of anticipating future vehicle interactions rather than reacting to instantaneous states alone. This predictive

structure enhances both safety and efficiency, [25, 35, 30, 1, 26, 32], and motivates the development of the nonlinear microscopic model with TTC-based predictive control proposed in this paper; [Appendix A](#) illustrates the road map of this paper.

With this motivation, we propose a nonlinear feedback control in the following form:

$$(2.1) \quad \begin{cases} \dot{x}_\ell = v_\ell \\ \dot{v}_\ell = a_\ell \\ \dot{x}_n = v_n \\ \dot{v}_n = \alpha \{ \mathcal{V}(x_{n-1}(t) - x_n(t)) - v_n(t) \} + \beta \frac{(v_{n-1}(t) - v_n(t))}{(\tau_n(t, x_{n-1}, x_n, v_{n-1}, v_n))^2} \\ (x_\ell(0), x_n(0), v_\ell(0), v_n(0)) = (x_{\ell,0}, x_{n,0}, v_{\ell,0}, v_{n,0}), \end{cases}$$

for $n \in \{1, \dots, N\}$ or in compact form, we define $\varphi_n := (x_{n-1}, x_n, v_{n-1}, v_n)$ and the dynamical model [Equation \(2.1\)](#) can be written in the form of

$$(2.2) \quad \begin{cases} \dot{\Phi} = F(t, \Phi) = F(t, \Phi, \tau(t, \Phi)), & \Phi = (\varphi_1, \varphi_2, \dots, \varphi_N), \quad \tau = (\tau_1, \dots, \tau_N) \\ \Phi(t_0) = \Phi_0. \end{cases}$$

The index $n \in \{1, \dots, N\}$, indicates the vehicle, and the position and velocity of n -th vehicle are denoted by x_n and v_n , respectively. The preceding vehicle of vehicle $n \in \{1, \dots, N\}$ is indexed by $n - 1$. The position, velocity, and acceleration of the leading vehicle are denoted by x_ℓ , v_ℓ , and a_ℓ , respectively. For notational consistency, the exogenous leader, which is the preceding vehicle of vehicle 1, is identified as vehicle 0, making $x_0 = x_\ell$ and $v_0 = v_\ell$. Here, τ_n is the time-to-collision (TTC) control strategy which will be defined later in this section—see [Equation \(3.5\)](#). The theoretical assumptions and approximation theory regarding the TTC function are detailed in [Sections 3 to 5](#), respectively. The proposed platoon may include all interacting AVs or AVs interacting with a human-driven vehicle (HV) leader (See the schematic in [Figure 1](#)).

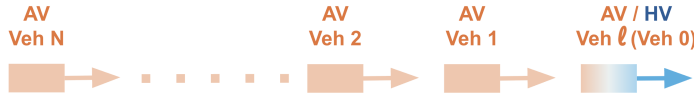


Figure 1: Platoon Schematic

The optimal velocity function $\mathcal{V} : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ is a smooth, bounded, monotonically increasing function, and satisfies the Lipschitz condition. In this paper, we consider

$$(2.3) \quad \mathcal{V}(x_{n-1} - x_n) \stackrel{\text{def}}{=} \tanh(x_{n-1} - x_n - 2) + \tanh(2),$$

see [\[43, 4\]](#) for properties of this function, and [\[39\]](#) for model's sensitivity analysis with respect to the coefficients α and β .

In addition, we assume a reasonable behaviour for the leading vehicle from a practical point of view; that is, the leader's acceleration is a prescribed measurable function of time,

$$(2.4) \quad a_\ell \in L^\infty(0, T_h), \quad \text{with} \quad 0 \leq v_{\ell,0} + \int_0^t a_\ell(s) ds \leq v_{\max} < \infty \quad \text{for all } t \in [0, T_h],$$

where $v_{\max}$ is the maximum permitted speed. In particular the leader does not react to the follower.

Thanks to the general choice of the acceleration function $\mathbf{a}_\ell$, for the analytical parts we only need to consider $N = 1$ —the interaction between the dynamics of two vehicles—and the generalization of results to any finite number of vehicles will follow immediately [39]. Therefore, from hereon in this paper, we focus our analysis on a general leader-follower control dynamics in the compact form

$$(2.5) \quad \begin{cases} \dot{\varphi} = F(t, \varphi), & t \in [0, T_h] \\ \varphi(t_o) = \varphi_o. \end{cases}$$

for $\varphi = (x_\ell, x, v_\ell, v)$.

Remark 2.1 (Acceleration Function). *In this research, we assume $\mathbf{a}_\ell$ can be approximated/learned. Such assumption is justified by the following remarks.*

- **Sequential Learning in vehicle platoons.** *First, in a platoon setting, the trajectory of each vehicle becomes the input for the subsequent follower. Specifically, once the trajectory (x_ℓ, v_ℓ) of a leading vehicle is learned or observed, it can be treated as a known signal for the following vehicle. In this sense, the learning framework can be applied recursively along the platoon: learning the dynamics of one vehicle effectively provides the input (i.e., $\mathbf{a}_\ell$) for the next.*
- **Interpretation as control input.** *Second, the leading acceleration $\mathbf{a}_\ell$ can be interpreted as a control input. From a control-theoretic standpoint, the leader's behavior is not arbitrary, but rather reflects a control strategy—either human or automated. Under this interpretation, assuming $\mathbf{a}_\ell$ is known corresponds to analyzing the follower dynamics under a fixed control policy. This is standard in reachability and optimal control, where the system is studied under a given control signal or as part of the predictive trajectory in feedback control. Moreover, this viewpoint allows for future extensions in which acceleration is designed or optimized to achieve specific objectives such as safety, efficiency, or stability.*
- **Parametric approximation.** *The leading acceleration can be modeled parametrically. For example,*

$$\mathbf{a}_\ell(t) \approx \sum_{k=0}^n c_k t^k,$$

or even more generally, representing the $\mathbf{a}_\ell$ using proper basis (polynomials, NN, functions, and so on). The coefficients can then be estimated from observed trajectory data in a reliable way.

Remark 2.2 (Length of Vehicles). *The length of vehicles is neglected for the analytical part of this paper—the vehicles are modeled (as particles)—since considering a constant length and safety headway only shifts the analytical results by a constant without altering their fundamental structures [36].*

2.2. The Dynamic-Based TTC and its Approximation. In this paper, the dynamic-based TTC mapping $\tau : [0, T_h] \times \mathbb{R}^2 \times \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}_{\geq 0}$ is assumed to evolve according to the

OVM-dynamics. More precisely, given any state $(\bar{t}, \bar{\varphi})$ of system Equation (2.5), the OVM-dynamical model Equation (2.6) predicts the time to collision of system Equation (2.2) starting from such current state of the system. The choice of OVM for predicting the time to collision can be justified by the fact that the OVM-dynamics is a simple characterization of dynamic evolution of acceleration/deceleration as a function of distance between consecutive vehicles. In addition, the OVM is essentially part of the dynamical model (2.1), excluding the β -control term that potentially is used in AV's dynamics to prevent the collision. This dynamical model is proven to be a proper representation of critical properties of the system. It should also be noted that the choice of OVM dynamics is not restrictive in any sense and can simply be replaced by any other dynamical model as long as it satisfied a certain properties—e.g. Lipschitz solution.

To continue with the motivation of the proposed dynamical model, it would be helpful to further compare its behavior with the existing microscopic traffic models—OVM, OVFL—considering different initial conditions. Let us recall

$$(2.6) \quad \text{OVM} : \dot{v}(t) = \alpha \{ \mathcal{V}(x_\ell(t) - x(t)) - v(t) \}$$

$$(2.7) \quad \text{OVFL} : \dot{v}(t) = \alpha \{ \mathcal{V}(x_\ell(t) - x(t)) - v(t) \} + \beta \frac{v_\ell(t) - v(t)}{(x_\ell(t) - x(t))^2}.$$

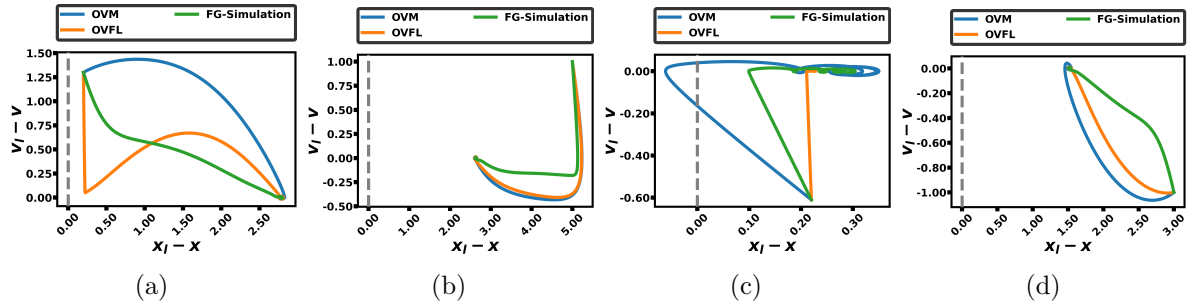


Figure 2: Comparison between the trajectories of OVFL, OVM, and the proposed dynamics with dynamic-based TTC (FG-simulation; see Def. B.1 for the definition). Parameters: $\alpha = 2$, $\beta = 3$, $x = 0$, Simulation Time = 50, $a_\ell = 0.01 \sin(0.5t)$ Initial Conditions: (a) $x_\ell = 0.2$, $v = 0.3$, $v_\ell = 1.6$; (b) $x_\ell = 5$, $v = 0.5$, $v_\ell = 1.5$; (c) $x_\ell = 0.22$, $v = 0.62$, $v_\ell = 0.01$ (d) $x_\ell = 3$, $v = 1.9$, $v_\ell = 0.1$

Figure 2 highlights the properties and the differences among these models. In particular, the OVM and OVFL models can lead to physically undesirable behaviors—as explained in Subsection 2.1. The proposed TTC-based control dynamical model effectively addresses these issues, ensuring smoother and more favorable responses; compare with the top and bottom left cases in these illustrations.

3. Analytical Properties. In this section we collect the required analytical properties of the proposed feedback control dynamical model Equation (2.5). More precisely, in this section we prove the well-posedness of the dynamical system under the specified assumptions.

In addition, we show that such feedback control system provides a safety-aware, collision free dynamical model. We start with a few key definitions:

Domain We consider the domain

$$(3.1) \quad \mathbb{S} \stackrel{\text{def}}{=} \{(x_1, x_2, v_1, v_2) \in \mathbb{R}^2 \times \mathbb{R}_{\geq 0}^2 : x_1 - x_2 > 0\}.$$

Additionally, we introduce the domain

$$(3.2) \quad \mathbb{S}_{T_h} \stackrel{\text{def}}{=} (0, T_h) \times \mathbb{S}.$$

Collision time of the dynamical model Equation (2.5) Let us fix the initial condition

$(t_o, \varphi_o) \in \mathbb{S}_{T_h}$. We define

$$(3.3) \quad \mathcal{T}(t_o, \varphi_o) \stackrel{\text{def}}{=} \inf\{t \geq t_o : x_\ell(t; t_o, \varphi_o) - x(t; t_o, \varphi_o) = 0, v_\ell(t; t_o, v_o) - v(t; t_o, v_o) < 0\},$$

as the collision time of the dynamical model of Equation (2.5). Strictly negative relative velocity in the Equation (3.3) is named *transversality* condition of velocity. This in particular means we consider a state $x_\ell - x = 0$ a collision state if the relative velocity is negative.

In fact, the validity of this assumption is by considering a safe headway, and hence unless the relative speed is not strictly negative at the boundary location, such a headway will not be violated.

Predictive collision time and transversal property Given the state $(\bar{t}, \bar{\varphi}) \in \mathbb{S}_{T_h}$ of the system Equation (2.5), *predictive collision time* calculates the collision time predicted by the OVM dynamics starting from such given state. In particular, given any initial condition $(\bar{t}, \bar{\varphi}) \in \mathbb{S}_{T_h}$

$$(3.4) \quad \tilde{\mathcal{T}}(\bar{t}, \bar{\varphi}) \stackrel{\text{def}}{=} \inf\{t \in [\bar{t}, T_h] : \xi_\ell(t; \bar{t}, \bar{\varphi}) - \xi(t; \bar{t}, \bar{\varphi}) = 0, \zeta_\ell(t; \bar{t}, \bar{\varphi}) - \zeta(t; \bar{t}, \bar{\varphi}) < 0\},$$

with $\inf(\emptyset) = T_h$ and where $(\xi_\ell, \xi, \zeta_\ell, \zeta)$ represents the solution of the OVM-dynamic, Equation (2.6). More precisely, the $\tilde{\mathcal{T}}$ is considered to be the predictive collision time, if in the OVM-based dynamics the predictive distance is zero and in addition the predictive relative velocity is transversal.

Time to collision Time to collision can now be defined simply by

$$(3.5) \quad \tau(\bar{t}, \bar{\varphi}) \stackrel{\text{def}}{=} \tilde{\mathcal{T}}(\bar{t}, \bar{\varphi}) - \bar{t}.$$

Safe region The goal is to define a region over which τ remains away from the zero. This is in particular important in proving properties of the function F.

To this end, we define

$$(3.6) \quad \hat{\mathbb{S}}_{T_h} \stackrel{\text{def}}{=} \{(t, \varphi) \in \mathbb{S}_{T_h} : \tau(t, \varphi) > 0\}.$$

Lemma 3.1. Suppose $(\bar{t}, \bar{\varphi}) \in \text{int}[\mathbb{S}_{T_h}]$. Then, $\hat{\mathbb{S}}_{T_h} \neq \emptyset$.

Proof. The proof of this lemma is an immediate consequence of the well-posedness of the OVM dynamical model [11, 39]. In particular, starting from a non-boundary point, by continuity of the solution of the OVM-dynamics, $\tau(\bar{t}, \bar{\varphi}) > 0$. ■

Definition 3.2 (Weak Solution of Equation (2.5)). Let $T_h > 0$. A vector-valued function $(x_\ell, x, v_\ell, v) \in \mathcal{W}_{\text{loc}}^{1,\infty}((0, T_h); \mathbb{R}^2 \times \mathbb{R}_+^2)$ is a solution to the system (2.5) if

$$(3.7) \quad \begin{cases} x_\ell(t) = x_{\ell,0} + \int_0^t v_\ell(s) ds \\ v_\ell(t) = v_{\ell,0} + \int_0^t a_\ell(s) ds \\ x(t) = x_0 + \int_0^t v(s) ds \\ v(t) = v_0 + \int_0^t \mathcal{A}(t, x_\ell(s), x(s), v_\ell(s), v(s)) ds \end{cases}$$

where

$$(3.8) \quad \mathcal{A}(t, x, y, v, w) \stackrel{\text{def}}{=} \alpha \{ \mathcal{V}(x - y) - v \} + \beta \frac{v - w}{[\tau(t, \varphi)]^2}.$$

Before we start with some analytical results on the properties of the model, let us consider some examples on function $\mathcal{T}$ which provide some insight about the problem.

Example 1. (Constant speed)

$$(3.9) \quad \tilde{\mathcal{T}}(t, x, y, v, w) = \frac{x - y}{\bar{v}},$$

where $\bar{v}$ can be interpreted as a constant velocity profile such as maximum speed, or average speed of traffic from historic data collection. In other words, in this model the information about dynamic velocity is not used.

Example 2. (Motion-based TTC) In this approach, the velocity and the distance at any point of time is calculated employing the Newtonian rule of motion to predict the time to collision.

$$(3.10) \quad \tilde{\mathcal{T}}(t, x_\ell, x, v_\ell, v) \stackrel{\text{def}}{=} \begin{cases} \frac{x_\ell(t) - x(t)}{v(t) - v_\ell(t)}, & v_\ell(t) - v(t) < 0 \\ \infty, & \text{else} \end{cases}$$

Example 3. (Non-local speed profile TTC)

Let $\lambda > 0$ and $\eta_\lambda \in \mathcal{C}_c^\infty((-\lambda, 0); [0, 1])$ be the kernel with $\int_{\mathbb{R}} \eta_\lambda(s) ds = 1$. We define

$$(3.11) \quad \tilde{\mathcal{T}}(t, x_\ell, x, v_\ell, v) \stackrel{\text{def}}{=} \begin{cases} \frac{x_\ell(t) - x(t)}{\int_{t-\lambda}^t \eta_\lambda(t-s)(v(s) - v_\ell(s)) ds}, & v(t) > v_\ell(t) \\ \infty, & \text{otherwise} \end{cases}$$

In this case, the weighted average of the speed presents the speed profile.

Regularities of vector field F. Considering the dynamical model of Equation (2.5) along with the assumption Equation (2.4), we have the following regularities:

Measurability

$$(3.12) \quad t \mapsto F(t, \varphi) \quad \text{is measurable for all } \varphi,$$

Boundedness

$$(3.13) \quad F \in L_{\text{loc}}^\infty(\hat{\mathbb{S}}_{T_h}).$$

Local Lipschitz property For any time $t \in [t_o, T_h]$ and $\varphi_1, \varphi_2 \in \mathcal{U} \Subset \mathbb{S}$ compact, such that $(t, \varphi_i) \in \mathbb{S}_{T_h}$, we have

$$(3.14) \quad \|F(t, \varphi_1) - F(t, \varphi_2)\| \leq C_F \|\varphi_1 - \varphi_2\|, \quad \text{for any } t \in [t_o, T_h],$$

where $C_F = C_F(\mathcal{U})$ is local Lipschitz constant of F .

Next, we show that the solution to the dynamical model (2.5) exists uniquely on $[0, \mathcal{T})$, i.e. up to the collision time. To this end, we first show that the dynamical model Equation (2.5) is locally well-posed.

Lemma 3.3 (Local Existence and Regularity). *Let function F be defined as in (2.5) with $\alpha, \beta > 0$ and regularity Equations (3.12)–(3.14) and assuming that $(t_o, \varphi_o) \in \hat{\mathbb{S}}_{T_h}$. Then, there exists $\eta > 0$ such that a unique solution φ of (2.5) in the sense of Def. 3.2 exists on $[t_o, t_o + \eta]$.*

Proof. We start with Picard-type iterative approach to construct the solution by introducing a proper kernel, customized for our proposed dynamical model by employing some of the key properties and regularities of the solution as it reaches the collision boundaries. Let us fix $\delta > 0$ such that $\mathcal{B}_{\varphi_o}(\delta) \stackrel{\text{def}}{=} \text{Ball}(\varphi_o, \delta) \Subset \mathbb{S}$. For any $t \in [t_o, \mathcal{T})$, we define

$$(3.15) \quad \begin{aligned} R(t; \delta) &\stackrel{\text{def}}{=} \int_{t_o}^t \sup_{\varphi \in \text{cl}[\mathcal{B}_{\varphi_o}(\delta)]} \|F(s, \varphi)\|_{\ell^2} ds, \\ \lambda(t; \delta) &\stackrel{\text{def}}{=} \sup_{\substack{\varphi \neq \psi: \\ \varphi, \psi \in \text{cl}[\mathcal{B}_{\varphi_o}(\delta)]}} \frac{\|F(t, \varphi) - F(t, \psi)\|_{\ell^2}}{\|\varphi - \psi\|_{\ell^2}}. \end{aligned}$$

where by Equations (3.12)–(3.14) both functions are well-defined. So, we can define

$$(3.16) \quad \eta = \eta(\delta) \stackrel{\text{def}}{=} \sup\{t \in [t_o, T_h] : R(t) \leq \delta, \quad \bar{\lambda}(t) < \infty\}.$$

where,

$$(3.17) \quad \bar{\lambda}(t) \stackrel{\text{def}}{=} \int_{t_o}^t \lambda(s) ds.$$

Let us consider the Banach space $\mathcal{X} \stackrel{\text{def}}{=} \mathcal{C}([t_o, t_o + \eta]; \mathbb{R}^4)$ with the uniform norm topology $\mathcal{T}_{\|\cdot\|_{L^\infty}}$. Then, in a canonical way, we may consider a kernel $\mathcal{K} : \mathcal{X} \rightarrow \mathcal{X}$ by

$$(3.18) \quad \mathcal{K}(\varphi)(t) \stackrel{\text{def}}{=} \varphi_o + \int_{t_o}^t F(s, \varphi(s)) ds, \quad \mathcal{K}^m \stackrel{\text{def}}{=} \mathcal{K}^{m-1} \circ \mathcal{K}$$

Self-mapping property of $\mathcal{K}$. Let us define

$$\mathcal{Y} \stackrel{\text{def}}{=} \{\varphi \in \mathcal{X} : \|\varphi - \varphi_o\|_{L^\infty} \leq \delta\}.$$

For any $\varphi \in \mathcal{Y}$ under uniform topology $\mathcal{T}_{\|\cdot\|_{L^\infty}}$, (3.16) and (3.18) imply that

$$(3.19) \quad \|\mathcal{K}(\varphi)(t) - \varphi_o\|_{\ell^2} \leq \int_{t_o}^t \|F(s, \varphi(s))\|_{\ell^2} ds \leq R(t) \leq \delta, \quad t \in [t_o, t_o + \eta]$$

and taking supremum over $t \in [t_o, t_o + \eta]$ provides $\mathcal{K}(\mathcal{Y}) \subset \mathcal{Y}$.

Contraction property of $\mathcal{K}$. For any $\varphi, \psi \in (\mathcal{Y}, \mathcal{T}_{\|\cdot\|_{L^\infty}})$ and $t \in [t_o, t_o + \eta]$, we have

$$\begin{aligned}
 \|\mathcal{K}(\varphi)(t) - \mathcal{K}(\psi)(t)\|_{\ell^2} &\leq \int_{t_o}^t \|\mathbf{F}(s, \varphi(s)) - \mathbf{F}(s, \psi(s))\|_{\ell^2} ds \\
 &\leq \int_{t_o}^t \lambda(s) \|\varphi(s) - \psi(s)\|_{\ell^2} ds \\
 (3.20) \quad &\leq \int_{t_o}^t \bar{\lambda}'(s) \bar{\lambda}(s) \sup_{t_o \leq r \leq s} \|\varphi(r) - \psi(r)\|_{\ell^2} ds \\
 &\leq \frac{1}{2} (\bar{\lambda}(t))^2 \sup_{t_o \leq r \leq t} \|\varphi(r) - \psi(r)\|_{\ell^2}
 \end{aligned}$$

where the second inequality uses the self-mapping and consequently the Lipschitz property in Equation (3.15) and third inequalities is by employing the Equation (3.18) as well as Lipschitz property. Using induction argument, one can show that

$$(3.21) \quad \|\mathcal{K}^m(\varphi)(t) - \mathcal{K}^m(\psi)(t)\|_{\ell^2} \leq \frac{\bar{\lambda}(t)^{m+1}}{(m+1)!} \sup_{t_o \leq r \leq t} \|\varphi(r) - \psi(r)\|_{\ell^2}$$

Taking supremum over $t \in [t_o, t_o + \eta]$ on both sides of (3.21), Weissinger's contraction principle implies that the kernel $\mathcal{K}$ has a unique fixed point

$$\varphi^* \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \mathcal{K}^n(\varphi),$$

where the limit is taken with respect to $(\mathcal{X}, \mathcal{T}_{\|\cdot\|_{L^\infty}})$. Therefore, by Equation (3.18) a unique solution of dynamics of Equation (2.5) in the sense of Def. 3.2 over $[t_o, t_o + \eta]$ exists. ■

Theorem 3.4 (Well-posedness up to collision time). *Let $\mathcal{T} = \mathcal{T}(t_o, \varphi_o)$ be as in Equation (3.3). Then, the solution of (2.5) under the regularity conditions Equations (3.12)–(3.14) exists uniquely on $[t_o, \mathcal{T})$ in the sense of Def. 3.2.*

Proof. In Lem. 3.3 we have proven the existence and uniqueness of the solution over $[t_o, t_o + \eta(\delta)]$. We can generalize the result by considering a sequence $(\delta_n)_{n \in \mathbb{N}}$ such that

$$\text{Ball} \left(\varphi(t_o + \sum_{k=1}^{n-1} \eta(\delta_k)), \delta_n \right) \in \mathbb{S}.$$

Now let us define

$$(3.22) \quad \bar{\eta} \stackrel{\text{def}}{=} \sum_{k \geq 1} \eta(\delta_k).$$

Let $[t_o, T_{\max} \stackrel{\text{def}}{=} t_o + \bar{\eta})$ be the maximal interval of existence and we start by assuming—on the contrary to our claim—that

$$(3.23) \quad T_{\max} < \mathcal{T}.$$

We need to characterize the behavior of the solution $t \mapsto \varphi(t)$ as $t \rightarrow T_{\max}$.

By extensibility theorem of dynamic systems, under the condition of Equation (3.23), as $t \rightarrow T_{\max}$, the following cases can happen.

Case 1. The solution is bounded. Let $(t_n)_{n \in \mathbb{N}}$ be any sequence such that $t_n \rightarrow T_{\max}$ as $n \rightarrow \infty$. By condition Equation (2.4) and Lem. 3.7, v_ℓ and v satisfy the maximal principle property. Consequently, the solutions x_ℓ and x cannot also grow unbounded in finite time. In other words, the solution φ remains bounded along every subsequence.

Case 2. Approaching the boundary of domain $\mathbb{S}$. Considering the boundedness of the solution along any sequence (Case 1 above), under the maximal interval of existence assumption $[t_o, T_{\max})$, along any sequence $(r_n)_{n \in \mathbb{N}}$ with $r_n \rightarrow T_{\max}$, we must have

$$(3.24) \quad \text{dist}(\varphi(r_n), \partial \mathbb{S}) \rightarrow 0, \quad n \rightarrow \infty,$$

or equivalently for any compact set $K \Subset \mathbb{S}$, there exists a r_K such that $\varphi(r_K) \notin K$. At this point, we split the boundary of the domain. Let us assume there exists a sequence $(s_n)_{n \in \mathbb{N}}$ with $s_n \rightarrow T_{\max}$, such that

$$(3.25) \quad \text{dist}(\varphi(s_n), \partial^c \mathbb{S}) \rightarrow 0, \quad n \rightarrow \infty,$$

where, the collision-boundary of the domain $\mathbb{S}$ is defined by

$$\partial^c \mathbb{S} \stackrel{\text{def}}{=} \{(x, y, v, w) \in \mathbb{R}^2 \times \mathbb{R}_{\geq 0}^2 : x - y = 0\}.$$

In other words, assume by Equation (3.24) and considering the boundedness of solution, and consequently invoking Bolzano-Weierstrass theorem, there exists a subsequence—still denoted by $(s_n)_{n \in \mathbb{N}}$ —along which $x_\ell(s_n) - x(s_n) \rightarrow 0$ as $n \rightarrow \infty$. However, this in particular implies that $s_n \rightarrow \mathcal{T}$ as $n \rightarrow \infty$. This contradicts the assumption of Equation (3.23).

Therefore, Equation (3.24) forces that along any sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow T_{\max}$ we must have

$$\text{dist}(\varphi(t_n), \partial^v \mathbb{S}) \rightarrow 0, \quad n \rightarrow \infty,$$

where considering the definition Equation (3.1) of $\mathbb{S}$, we define

$$(3.26) \quad \partial^v \mathbb{S} \stackrel{\text{def}}{=} [\{v = 0\} \cup \{w = 0\}] \setminus \{x - y = 0\}$$

which is the non-collision part of the boundary of admissible domain $\mathbb{S}$.

Keeping the boundedness of the solution discussed in Case 1 and consequently Bolzano-Weierstrass theorem, for any sequence $(t_n)_{n \in \mathbb{N}}$, we can choose a subsequence—still denoted by $(t_n)_{n \in \mathbb{N}}$ —such that $v_\ell(t_n) \rightarrow 0$ and/or $v(t_n) \rightarrow 0$ while $x_\ell(t_n) - x(t_n) \rightarrow c > 0$ as $n \rightarrow \infty$. We claim, there exists a lower bound $\underline{\delta} = \underline{\delta}(\bar{\eta}) > 0$ such that along any sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow T_{\max}$

$$(3.27) \quad \liminf_n |x_\ell(t_n) - x(t_n)| \geq \underline{\delta}.$$

To see this, we assume Equation (3.27) can be violated. Under such assumption, therefore, one can construct a diagonal sequence $(\bar{\varphi}_k)_{k \in \mathbb{N}} \subset \partial^v \mathbb{S}$ of accumulation points with corresponding $\underline{\delta}_k \rightarrow 0$ as $k \rightarrow \infty$. However, this falls under the previous category and following Equation (3.25), this contradicts Equation (3.23). In other words,

This in particular implies that there exists a $\delta' > 0$, such that $\tau(t_n, \varphi(t_n)) \geq \delta'$ along any sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow T_{\max}$. Therefore, for any $s, t \in [t_o, T_{\max})$

$$(3.28) \quad \|\varphi(t) - \varphi(s)\|_{\ell^2} \leq \int_{\min\{s,t\}}^{\max\{s,t\}} \|F(r, \varphi(r))\|_{\ell^2} dr \leq \sup_{[t_o, t_o + \bar{\eta}] \times [-k, k]} \|F(t, u)\|_{\ell^2} |t - s|,$$

where the last inequality is by $\|\varphi\|_{L^\infty([t_o, t_o + \bar{\eta}]; \mathbb{R}^4)} \leq k$ and Equation (3.13).

Now let us fix the sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow T_{\max}$ as $n \rightarrow \infty$. Then by (3.28), $(\varphi(t_n))_{n \in \mathbb{N}} \subset [-k, k]$ is a Cauchy sequence and hence there exists a $\bar{\varphi}$ such that $\varphi(t_n) \rightarrow \bar{\varphi}$ as $n \rightarrow \infty$. We consider $\bar{\varphi}$ as the extension of the solution φ to $T_{\max}$ (i.e. $\varphi(T_{\max}) \stackrel{\text{def}}{=} \bar{\varphi}$) along the sequence $(t_n)_{n \in \mathbb{N}}$. In addition, we need to show that such extension can be carried out continuously along any sequence. The next classical result in continuation theory of dynamical systems justifies this concept rigorously.

Lemma 3.5. *Let $(r_n)_{n \in \mathbb{N}}$ be any sequence with $r_n \rightarrow T_{\max}$ as $n \rightarrow \infty$. Then, $\lim_{n \rightarrow \infty} \varphi(r_n) = \bar{\varphi}$; i.e. the extension is continuous.* ■

Proof of Lem. 3.5. Let us fix $\varepsilon > 0$. Thanks to Equation (3.27) and consequently boundedness of F , Equation (3.28) implies that for sufficiently small $\varsigma > 0$, $|t - s| < \varsigma$ implies that

$$(3.29) \quad \|\varphi(t) - \varphi(s)\|_{\ell^2} \leq \varepsilon.$$

In addition, we can write

$$\|\varphi(r_n) - \bar{\varphi}\|_{\ell^2} \leq \|\varphi(r_n) - \varphi(t_n)\|_{\ell^2} + \|\varphi(t_n) - \bar{\varphi}\|_{\ell^2}$$

By the construction, for sufficiently large n , $|t_n - r_n| < \varsigma/2$, and we will have that

$$\|\varphi(r_n) - \bar{\varphi}\|_{\ell^2} \leq \varepsilon.$$

This proves the claim.

We continue with the proof of Thm. 3.4. By Lem. 3.5, φ is well-defined solution over $[t_o, T_{\max})$ and a continuous function over $[t_o, T_{\max}]$. Therefore, $t \mapsto \int_{t_o}^{t_o + \bar{\eta}} F(s, \varphi(s)) ds$ is continuous over $[t_o, t_o + \bar{\eta}]$. By letting $t \rightarrow \bar{\eta}$, we have

$$(3.30) \quad \varphi(t_o + \bar{\eta}) = \varphi_o + \int_{t_o}^{t_o + \bar{\eta}} F(s, \varphi(s)) ds.$$

In other words, $t \mapsto \varphi(t)$ is a well-defined solution over $[t_o, t_o + \bar{\eta}]$. But considering the condition (3.27), $\varphi(T_{\max}) \in \text{int}[\mathbb{S}]$ and therefore following the proof of Lem. 3.3 starting from the initial time $\bar{\eta}$, the solution φ can be extended beyond $t_o + \bar{\eta}$ which contradicts the maximal interval of existence assumption.

Therefore, collecting all the cases together solution φ is well-posed over $[t_o, \mathcal{T})$ as claimed and this completes the proof. ■

Before we state the main result of this section, we construct the following barrier domain. We will employ the consequent results to discuss the behavior of the dynamics near the collision boundaries and eventually to the collision analysis.

Theorem 3.6 (Barrier Domain). Suppose $\mathcal{T} < \mathbf{T}_h$, i.e. the collision happens before the time horizon. Let $t \mapsto \varphi(t; t_o, \varphi_o)$ denotes the solution of (2.5) in the sense of the Def. 3.2 starting from the initial condition (t_o, φ_o) . Then, there exists a scalar $\delta > 0$ and $\check{t} > 0$ such that

- (i) $(x_\ell(t) - x(t), v_\ell(t) - v(t)) \in [0, \delta] \times [-v_{\max}, 0]$ for $t \in [\check{t}, \mathcal{T})$, and
 - (ii) $x_\ell(t) - x(t)$ is strictly decreasing on $[\check{t}, \mathcal{T})$,
- where

$$(3.31) \quad \check{t} = \check{t}(\delta) \stackrel{\text{def}}{=} \sup\{t \in (t_o, \mathbf{T}_h) : x_\ell(t) - x(t) > \delta\}, \quad \sup(\emptyset) = \infty.$$

Proof. For the remaining results, it would be conceptually more convenient to work with the dynamics of distance and relative velocity—referred to as difference dynamics from hereon. More precisely, for $t \in [t_o, \mathcal{T})$, we define

$$(3.32) \quad \begin{cases} X(t) \stackrel{\text{def}}{=} x_\ell(t) - x(t) \\ V(t) \stackrel{\text{def}}{=} v_\ell(t) - v(t) \end{cases}, \quad \dot{X}(t) = V(t).$$

Using the representation of Equation (3.32), the dynamics of acceleration reads:

$$(3.33) \quad \dot{V}(t) = \mathbf{a}_\ell(t) - \alpha \{\mathcal{V}(X(t)) + V(t) - v_\ell(t)\} - \beta \frac{V(t)}{[\tau(t, \Phi(t))]^2}.$$

where $\Phi \stackrel{\text{def}}{=} (X, V)$. The key observation in the proof of this theorem is that if there is *no asymptotic behavior/perturbation* in the trajectories of the solutions as $t \rightarrow \mathcal{T}$, then the statement of the theorem follows immediately. Therefore, it is sufficient to study *an asymptotic behavior* in trajectories approaching the collision point. Figure 3b illustrates an example of such behavior.

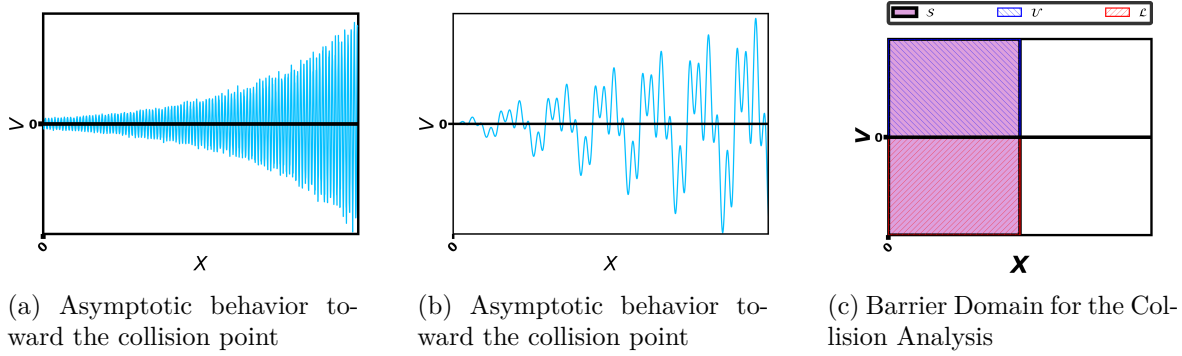


Figure 3: Phase-plane visualization for collision analysis

In particular, from a mathematical point of view, the asymptotic behavior of the trajectories asserts that there must exist a *strictly decreasing (nested)* sequence of intervals

$$(3.34) \quad \{\mathcal{J}_m \stackrel{\text{def}}{=} [t_m, t_{m+1}) : m \in \mathbb{N}\},$$

such that $t_m \rightarrow \mathcal{T}$ as $m \rightarrow \infty$, $\sum_m |\mathcal{J}_m| = \mathcal{T} - t_1$, and

$$(3.35) \quad V(t) |_{\mathcal{J}_m} \geq 0, \quad V(t) |_{\mathcal{J}_{m+1}} < 0.$$

We claim that under the condition (2.4) the asymptotic behavior of the trajectories and in particular a sequence characterized by (3.34)-(3.35) does not exist.

The proof is by contradiction. In particular, we show if the sequence (3.34) exists and satisfies the condition (3.35), it will violate the statement of the theorem.

It suffices to consider the case that the asymptotic behavior in $V(t)$ happens between the two regions

$$(3.36) \quad \mathcal{U} = \mathbb{S} \cap \mathcal{Y}^-, \quad \mathcal{L} = \mathbb{S} \cap \mathcal{Y}^+.$$

where

$$(3.37) \quad \begin{aligned} \mathcal{Y}^+ &\stackrel{\text{def}}{=} \{(x, y, v, w) \in \mathbb{R}^2 \times \mathbb{R}_{\geq 0}^2 : v - w < 0\} \\ \mathcal{Y}^- &\stackrel{\text{def}}{=} \{(x, y, v, w) \in \mathbb{R}^2 \times \mathbb{R}_{\geq 0}^2 : v - w \geq 0\}. \end{aligned}$$

In any other case, either the problem will be reduced to this case, or the statement of the theorem will be immediate. To continue the proof, we need a few more results. The visual representation of these two regions is illustrated in Figure 3c.

Lemma 3.7 (Maximum principle). *Let $v(\cdot)$ be the solution of the dynamical model (2.5) with the initial values $v_{\ell, \circ}, v_{\circ} \in [0, v_{\max}]$ and the parameters $\alpha, \beta > 0$. Then, we have that*

$$(3.38) \quad 0 \leq v(t) \leq v_{\max}, \quad t \in [t_{\circ}, \mathcal{T}).$$

Proof. We prove the result by contradiction. More precisely, we assume that there exists a $t \in (t_{\circ}, \mathcal{T})$ such that $v(t) = v_{\max}$ and $v(t+) > v(t)$ (i.e. violates the maximum principle). But then, by continuity of the solution, there exists $\mathfrak{T} \in (t, \mathcal{T})$ such that on $(t, \mathfrak{T})$, $v(t) > v_{\max}$.

On the other hand using the dynamical model of Equation (2.1), we can write that

$$(3.39) \quad v(\mathfrak{T}) - v(t) = \int_t^{\mathfrak{T}} \left[\alpha \{ \mathcal{V}(x_{\ell}(s) - x(s)) - v(s) \} + \beta \frac{v_{\ell}(s) - v(s)}{[\tau(s, \varphi(s))]^2} \right] ds.$$

On the left hand side, we have $v(\mathfrak{T}) - v(t) > 0$ by the assumption. On the right hand side, by the definition of the optimal velocity function the first term is non-positive on $(t, \mathfrak{T})$. Similarly, by condition (2.4), the control term on the right hand side is non-positive on the interval $(t, \mathfrak{T})$ which is a contradiction.

Following the same argument, one can show by the method of contradiction that the $v \geq 0$ on $[t_{\circ}, \mathcal{T})$. This completes the proof. ■

Lemma 3.8. *Under the assumption $\mathcal{T} < \mathbf{T}_h$ for the dynamical model (2.5) with $\alpha, \beta > 0$, we have that*

$$(3.40) \quad \lim_{t \rightarrow \mathcal{T}} X(t) = 0.$$

In other words, the only possible state as $t \rightarrow \mathcal{T}$ is collision.

Proof. The uniqueness and continuity of trajectory over $[t_o, \mathcal{T})$ —[Thm. 3.4](#)—implies that the blow-up of the solution cannot happen along an isolated trajectory. Therefore, the maximum principle, [Lem. 3.7](#), suggests that along any sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow \mathcal{T}$ we must have

$$(3.41) \quad \liminf_n \|\Phi(t_n)\| < \infty.$$

Therefore, along any sequence $(t_n)_{n \in \mathbb{N}}$ we should have

$$\text{dist}(\Phi(t_n), \partial \mathbb{S}_d) \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

where,

$$(3.42) \quad \mathbb{S}_d \stackrel{\text{def}}{=} \{(X, V) \in \mathbb{R}^2 : X > 0\}.$$

Therefore, considering [Equation \(3.41\)](#), any sequence $(t_n)_{n \in \mathbb{N}}$ with $t_n \rightarrow \mathcal{T}$, has a convergent subsequence with the accumulation point $(0, \bar{V}) \in \partial \mathbb{S}_d$ where $\bar{V} \in [-v_{\max}, v_{\max}]$. This proves the claim [Equation \(3.40\)](#). ■

We continue with the proof of [Thm. 3.6](#) by considering the following possible cases below.

Non-zero accumulation points. Recalling the conditions presented in [Equations \(3.34\)–\(3.35\)](#) and following the discussion of [Lem. 3.8](#), suppose there are sequences $(t_n)_{n \in \mathbb{N}}$ and $(s_n)_{n \in \mathbb{N}}$ with $t_n, s_n \rightarrow \mathcal{T}$ as $n \rightarrow \infty$ and with accumulation points $\bar{V} > 0$ and $\bar{V} < 0$, respectively. In this case, there exists a constant $c > 0$ such that $V(t_n) - V(s_n) \geq c$ for sufficiently large n . Therefore,

$$(3.43) \quad |V|_{\text{TV}((t_o, \mathcal{T}); \mathbb{R})} \geq \sum_{n \in \mathbb{N}} |V(t_n) - V(s_n)|.$$

In other words, the total variation of the solution V grows unbounded which violates the well-posedness of the solution in terms of [Def. 3.2](#).

Zero accumulation point. Eliminating the possibility studied in the first case above, there are two possibilities to explore both mainly with $V \rightarrow 0$ along some sequence. The first possibility is to investigate the oscillation presented by [Equations \(3.34\)–\(3.35\)](#) for sequences $(t_n)_{n \in \mathbb{N}}, (s_n)_{n \in \mathbb{N}}$ approaching $\mathcal{T}$ as $n \rightarrow \infty$ with $V(t_n) > 0, V(s_n) < 0$ for all $n \in \mathbb{N}$ and $V(t_n), V(s_n) \rightarrow 0$ as $n \rightarrow \infty$. But by the definition of collision time [Equation \(3.3\)](#)—and more specifically the transversality assumption—this contradicts $\mathcal{T} < T_h$.

Next, let us assume that we have the sequence $(t_{n,k})_{n,k \in \mathbb{N}}$ such that for each $k \in \mathbb{N}$, $X(t_{n,k}) \rightarrow 0$ as $n \rightarrow \infty$, $V(t_{n,k}) \rightarrow \bar{V}_k$, and $\bar{V}_k \rightarrow 0$ as $k \rightarrow \infty$. In other words, we consider the case where there are several subsequences along which the limit points $\bar{V}_k$'s vanish. In this case, one may diagonally create a subsequence $(t_m) \subset (t_{n,k})$ for which $V(t_m) \rightarrow 0$ as $m \rightarrow \infty$. But this case is precluded by total variation blowup similar to [Equation \(3.43\)](#). ■

Theorem 3.9. *Let $t \mapsto (X(t), V(t))$ be the trajectory of the dynamical model (2.5) with associated difference dynamics of (3.32) and (3.33). Then, there exists a $\lambda > 0$ such that the set*

$$(3.44) \quad \mathcal{C}_\lambda \stackrel{\text{def}}{=} \{(X, V) \in \mathbb{R}_{\geq 0} \times \mathbb{R} : V < 0, X \in [0, \lambda]\}$$

is not reachable along the trajectory $(X(t), V(t))$. In other words, the solution exists over $[0, T_h]$.

Proof. We first note that set $\mathcal{C}_\lambda$ is increasing nested over $\mathcal{L} = \mathbb{S} \cap \mathcal{Y}^+$ in the sense that:

$$(3.45) \quad \mathcal{C}_{\lambda_1} \subset \mathcal{C}_{\lambda_2}, \quad \text{if } \lambda_1 < \lambda_2.$$

and shrinks to the collision boundary as $\lambda \downarrow 0$:

$$(3.46) \quad \bigcap_{\lambda > 0} \mathcal{C}_\lambda = \{(X, V) \in \mathbb{R}_{\geq 0} \times \mathbb{R} : V < 0, X = 0\}.$$

In particular, if $\lambda_n \downarrow 0$ and $(X(t_n), V(t_n)) \in \mathcal{C}_{\lambda_n}$, then $0 \leq X(t_n) \leq \lambda_n \rightarrow 0$.

Assumption 3.10. Let us on the contrary assume that for any $\lambda > 0$, there exists a $t \in [t_o, \mathcal{T})$ such that $(X(t), V(t)) \in \text{int}[\mathcal{C}_\lambda]$.

Without loss of generality, we only need to consider $\mathcal{T} < T_h$ (otherwise the claim follows). In this case, by [Thm. 3.6](#) there exists a $\bar{\lambda}$ such that the function $X(\cdot)$ is decreasing over the interval $[\check{t}(\bar{\lambda}), \mathcal{T})$. Under [Asm. 3.10](#), in particular, we consider a sequence $(\lambda_n)_{n \in \mathbb{N}}$ with $\lambda_n < \bar{\lambda}$ for all $n \in \mathbb{N}$, and $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$. We define the associated $(t_n)_{n \in \mathbb{N}}$ of $(\lambda_n)_{n \in \mathbb{N}}$ as in [Asm. 3.10](#). Hence $X(t_n) \rightarrow 0$, and by [Lem. 3.8](#) together with the strict monotonicity of X on $[\check{t}(\bar{\lambda}), \mathcal{T})$ furnished by [Thm. 3.6](#), this forces $t_n \rightarrow \mathcal{T}$. Therefore, by the standing assumption [Asm. 3.10](#)

$$(3.47) \quad (X(t_n), V(t_n)) \in \text{int}[\mathcal{C}_{\lambda_n}].$$

In this case the dynamics of [\(3.33\)](#) reads:

$$(3.48) \quad V(t) = V(\check{t}) - \int_{\check{t}}^t \mathbf{a}_\ell(s) ds - \alpha \int_{\check{t}}^t \left\{ \mathcal{V}(X(s)) + V(s) - v_\ell(s) \right\} ds - \beta \int_{\check{t}}^t \frac{V(s)}{[\tau(s, \Phi(s))]^2} ds,$$

where $\check{t} = \check{t}(\lambda_n)$ for any $n \in \mathbb{N}$. Considering the maximum principle ([Lem. 3.7](#)), the Assumption of [Equation \(2.4\)](#), the assumption $\mathcal{T} < T_h$ (and consequently, the fact that following the proof of [Lem. 3.8](#), $(X(t_n), V(t_n)) \rightarrow (0, \bar{V})$ as $n \rightarrow \infty$), the first three terms in [Equation \(3.48\)](#) are bounded. In addition by the latter fact, and in particular by [Thm. 3.6](#), there exists an $N \in \mathbb{N}$ such that by [Equations \(3.45\)–\(3.47\)](#), $X(t_N) < \lambda_N$ is sufficiently small and strictly decreasing on $[\check{t}(\lambda_N), \mathcal{T})$.

Let us investigate the behavior of the dynamical model under these conditions. Let (X, V) and (Ξ, Λ) present the trajectory of the main and OVM dynamics, respectively. Let us assume first that $V < 0$ close to the collision boundaries (i.e. X is strictly decreasing). Then, starting from $X(t) > 0$ for some $t \in [\check{t}(\lambda_N), \mathcal{T})$ and considering the dynamics of Λ and $\dot{V}$, we have that

$$(3.49) \quad \begin{aligned} X(t + \delta) &= X(t) + \int_t^{t+\delta} V(s) ds \\ \Xi(t + \delta) &= X(t) + \int_t^{t+\delta} \Lambda(s) ds \\ &\leq X(t) + \int_t^{t+\delta} V(s) ds = X(t + \delta; t). \end{aligned}$$

The last inequality is by the fact that the singularity term $\int_t^t \frac{V}{\tau^2} < 0$ for the dynamical model [Equation \(2.5\)](#) and does not exist for the OVM-dynamics. Therefore,

$$(3.50) \quad X(t + \delta) \geq \Xi(t + \delta; t).$$

In addition, by definition $\Xi(t + \tau(t); t) = 0$. Therefore,

$$(3.51) \quad 0 \leq X(t + \tau(t)) = X(t) + \int_t^{t+\tau(t)} V(s) ds$$

By [Thm. 3.6](#) and the transversality assumption, $V < 0$ over $[\check{t}(\lambda_N), \mathcal{T})$. In other words, there exists $\underline{u}, \bar{u} > 0$ such that, $V \in [-\underline{u}, -\bar{u}]$ over $[t, t + \tau(t)]$. [Equation \(3.51\)](#) then implies that

$$(3.52) \quad 0 \leq X(t + \tau(t)) \leq X(t) - \bar{u} \tau(t).$$

More accurately, [Equation \(3.52\)](#) implies that up to a constant, $\tau \sim X$ near the collision boundary which in turn implies that $\int \frac{V}{\tau^2} \sim \int \frac{dX}{X^2}$ which blow up near the collision boundaries. But this contradicts the maximum principle [Lem. 3.7](#).

Collecting all together, there exists a $\lambda > 0$ such that for any $\mathbf{t} \in [\check{t}(\lambda), \mathcal{T})$, $(X(\mathbf{t}), V(\mathbf{t})) \notin \text{int}[\mathcal{C}_\lambda]$. In addition, due to the continuity of the solution and the fact that $\lambda > 0$, we have $X(\mathbf{t}) > 0$, which implies that $\mathcal{T} \geq \mathbf{T}_h$. ■

In fact, the following result is the immediate consequence of [Thm. 3.9](#).

Corollary 3.11. *The solution of dynamical model (2.5) is well-posed over $[0, \mathbf{T}_h]$.*

4. Analytical properties of OVM dynamics. The main goal of this section is to investigate some analytical properties of the TTC-mapping τ . Such properties turn out to be essential in establishing the results of next sections (see [Section 5](#)).

In particular, we consider the OVM-dynamical model

$$(4.1) \quad \begin{cases} \dot{\Xi} = \Lambda \\ \dot{\Lambda} = \mathbf{a}_\ell - \alpha \{ \mathcal{V}(\Xi) + \Lambda - \zeta_\ell \} \\ (\Xi(t_o), \Lambda(t_o)) = (\Xi_o, \Lambda_o) \end{cases}$$

over $[t_o, \tilde{\mathcal{T}})$ in the Carathéodory sense, with the acceleration function is defined as in [Equation \(2.4\)](#).

We start by deriving some stability results on the solution of OVM-dynamical model. We define denote the solution of the OVM-dynamical model [Equation \(4.1\)](#) by

$$(4.2) \quad \begin{aligned} \Xi(\cdot) &= \Xi(\cdot; t_o, \Xi_o, \Lambda_o) \stackrel{\text{def}}{=} \xi_\ell(\cdot; t_o, \Xi_o, \Lambda_o) - \xi(\cdot; t_o, \Xi_o, \Lambda_o), \\ \Lambda(\cdot) &= \Lambda(\cdot; t_o, \Xi_o, \Lambda_o) \stackrel{\text{def}}{=} \zeta_\ell(\cdot; t_o, \Xi_o, \Lambda_o) - \zeta(\cdot; t_o, \Xi_o, \Lambda_o), \end{aligned}$$

where ξ_ℓ and ξ are the position and ζ_ℓ and ζ are the velocity of the leading and following vehicles, respectively. The collision time under the OVM-dynamics starting from t_o

$$(4.3) \quad \tilde{\mathcal{T}} = \tilde{\mathcal{T}}(t_o, \Xi_o, \Lambda_o) \stackrel{\text{def}}{=} \inf \{ t > t_o : \Xi(t; t_o, \Xi_o, \Lambda_o) \leq 0 \}, \quad \inf(\emptyset) = \mathbf{T}_h.$$

Next, we will introduce some of the most essential properties of the OVM-dynamics which will be the critical for the rest of the results in the paper.

The OVM dynamics are well posed on the whole state space, independently of whether a collision is predicted; we therefore work on the full set of initial conditions

$$(4.4) \quad \mathbb{I} \stackrel{\text{def}}{=} [0, \mathbf{T}_h] \times \mathbb{R}_{>0} \times \mathbb{R} = \{(t_o, \Xi_o, \Lambda_o) : t_o \in [0, \mathbf{T}_h], \Xi_o > 0, \Lambda_o \in \mathbb{R}\}.$$

Recall from (4.3) that $\tilde{\mathcal{T}} \leq \mathbf{T}_h$ everywhere, with $\tilde{\mathcal{T}} = \mathbf{T}_h$ exactly on the no-collision data; thus $[t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o))$ is the maximal pre-collision interval on which the OVM solution is defined, for every $(t_o, \mathbf{x}_o) \in \mathbb{I}$.

Definition 4.1 (Pre-collision reachable tube of the OVM dynamics). Write $\mathbf{x}(\cdot; t_o, \mathbf{x}_o) = (\Xi(\cdot; t_o, \mathbf{x}_o), \Lambda(\cdot; t_o, \mathbf{x}_o))$ for the solution of (4.1) on $[t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o))$. The pre-collision reachable tube is

$$(4.5) \quad \mathbf{R} \stackrel{\text{def}}{=} \{(s, \Xi(s; t_o, \mathbf{x}_o), \Lambda(s; t_o, \mathbf{x}_o)) \in [0, \mathbf{T}_h] \times \mathbb{R}_{>0} \times \mathbb{R} : (t_o, \mathbf{x}_o) \in \mathbb{I}, s \in [t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o))\}.$$

We will be using this domain to uniformly bound all required fields; see [Remark 4.3](#).

Lemma 4.2 (Measurability of R). The reachable tube $\mathbf{R}$ of (4.5) is Borel measurable (even σ -compact) subset of $[0, \mathbf{T}_h] \times \mathbb{R}_{>0} \times \mathbb{R}$.

Proof. We show $\mathbf{R}$ is Lebesgue measurable. The continuity used below is that of the flow, which follows from the existence and continuous-dependence theory of [Lem. 3.3–Lem. 3.8](#), established prior to [Thm. 4.4](#).

Measurability of the hitting time. The map $(t, t_o, \mathbf{x}_o) \mapsto \Xi(t; t_o, \mathbf{x}_o)$ is jointly continuous on its domain (continuous dependence of Carathéodory solutions on time and initial data, [Lem. 3.3](#)). Hence, for the hitting time (4.3),

$$\tilde{\mathcal{T}}(t_o, \mathbf{x}_o) = \inf\{t > t_o : \Xi(t; t_o, \mathbf{x}_o) \leq 0\} \wedge \mathbf{T}_h,$$

the sublevel sets $\{\tilde{\mathcal{T}} < c\}$ are open in $(t_o, \mathbf{x}_o)$ for each c (if Ξ dips to 0 before c at one initial condition, it does so on a neighbourhood, by continuity), so $\tilde{\mathcal{T}}$ is upper semicontinuous, in particular Borel measurable.

Measurability of the tube. Consequently

$$\mathbf{D} \stackrel{\text{def}}{=} \{(t_o, \mathbf{x}_o, s) : (t_o, \mathbf{x}_o) \in \mathbb{I}, t_o \leq s < \tilde{\mathcal{T}}(t_o, \mathbf{x}_o)\}$$

is a Borel subset of $\mathbb{I} \times [0, \mathbf{T}_h]$, and $\mathbf{R}$ is its image under the continuous map $F(t_o, \mathbf{x}_o, s) \stackrel{\text{def}}{=} (s, \mathbf{x}(s; t_o, \mathbf{x}_o))$. A continuous image of a Borel set is analytic, hence Lebesgue (universally) measurable; this suffices for the essential supremum in (4.7) to be well defined.

σ -compactness (sharper). If in addition one uses that $\tilde{\mathcal{T}}$ is continuous on each compact $\mathbf{K}_n \subset \mathbb{I}$ of a fixed exhaustion—available a posteriori from [Thm. 4.13](#), and not needed for the measurability claim—then, reparametrising $[t_o, \tilde{\mathcal{T}}]$ to $[0, 1]$, each $\mathbf{R}(\mathbf{K}_n)$ is a continuous image of the compact $\mathbf{K}_n \times [0, 1]$, hence compact, and $\mathbf{R}$ is σ -compact. ■

Remark 4.3 (Uniform a priori bound and Lipschitz property of the OVM field on $\mathbb{R}$). *The field*

$$(4.6) \quad \mathbf{G} : \mathbb{R} \rightarrow \mathbb{R}^2, \quad \mathbf{G}(s, \Xi, \Lambda) = \left(\Lambda, \mathbf{a}_\ell(s) - \alpha(\mathcal{V}(\Xi) + \Lambda - \zeta_\ell(s)) \right),$$

of (4.1) is bounded on the reachable tube (4.5), uniformly over all initial conditions in $\mathbb{I}$:

$$(4.7) \quad \|\mathbf{G}\|_{L^\infty(\mathbb{R}; \ell^2)} \leq M, \quad M \stackrel{\text{def}}{=} \left(v_{\max}^2 + (\|\mathbf{a}_\ell\|_{L^\infty(0, \mathbf{T}_h)} + \alpha(2 + 2v_{\max}))^2 \right)^{1/2} < \infty,$$

with $M = M(\alpha, \|\mathbf{a}_\ell\|_{L^\infty(0, \mathbf{T}_h)}, v_{\max})$. The essential supremum is over $(s, \Xi, \Lambda) \in \mathbb{R}$; the “essential” qualifier attaches only to the time slot s , through the L^∞ -in-time forcing $\mathbf{a}_\ell, \zeta_\ell$, whereas the state dependence is bounded everywhere. Indeed:

- (i) the first component is the OVM relative velocity $\Lambda = \zeta_\ell - \zeta$, and $|\Lambda| \leq v_{\max}$ since the OVM speeds obey the global constraint $0 \leq \zeta, \zeta_\ell \leq v_{\max}$ of (2.4) and by maximal principle property of the OVM-dynamics;
- (ii) for the second component, for a.e. s and every $(s, \Xi, \Lambda) \in \mathbb{R}$,

$$\begin{aligned} |\mathbf{a}_\ell(s) - \alpha(\mathcal{V}(\Xi) - \Lambda + \zeta_\ell(s))| &\leq \|\mathbf{a}_\ell\|_{L^\infty(0, \mathbf{T}_h)} + \alpha(\|\mathcal{V}\|_{L^\infty(\mathbb{R})} + |\Lambda| + \|\zeta_\ell\|_{L^\infty(0, \mathbf{T}_h)}) \\ &\leq \|\mathbf{a}_\ell\|_{L^\infty(0, \mathbf{T}_h)} + \alpha(2 + 2v_{\max}), \end{aligned}$$

using $\|\mathcal{V}\|_{L^\infty(\mathbb{R})} \leq 2$; in particular no singularity arises at $\{\Xi = 0\}$, $|\Lambda| \leq v_{\max}$, and $0 \leq \zeta_\ell \leq v_{\max}$.

Consequently, along any trajectory $\mathbf{x}(\cdot; t_o, \mathbf{x}_o)$, $(t_o, \mathbf{x}_o) \in \mathbb{I}$, the Carathéodory identity $\dot{\mathbf{x}}(s) = \mathbf{G}(s, \mathbf{x}(s))$ holds for a.e. $s \in [t_o, \tilde{\mathcal{T}})$, whence $|\dot{\mathbf{x}}(s)| \leq M$ a.e. and

$$(4.8) \quad \left| \int_a^b \mathbf{G}(r, \mathbf{x}(r)) dr \right|_{\ell^2} \leq M |b - a| \quad \text{for all } [a, b] \subseteq [t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o)].$$

In particular $\mathbf{x}(\cdot; t_o, \mathbf{x}_o) \in \mathcal{W}^{1, \infty}((t_o, \tilde{\mathcal{T}}); \mathbb{R}^2)$, M -Lipschitz in time, uniformly over $\mathbb{I}$.

Theorem 4.4 (OVM dynamical model: well-posedness and Lipschitz stability). *For every $(t_o, \mathbf{x}_o) \in \mathbb{I}$, as in (4.4), the OVM-type dynamical model Equation (4.1) with acceleration $\mathbf{a}_\ell$ as in (2.4), admits, in the Carathéodory sense of Def. 3.2, a unique solution*

$$\mathbf{x}(\cdot; t_o, \mathbf{x}_o) \in \mathcal{W}^{1, \infty}((t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o)); \mathbb{R}^2)$$

on its maximal pre-collision interval. If moreover a collision is predicted, $\tilde{\mathcal{T}}(t_o, \mathbf{x}_o) < \mathbf{T}_h$, the solution meets the boundary $\{\Xi = 0\}$ transversally; i.e. $\Lambda(\tilde{\mathcal{T}}; t_o, \mathbf{x}_o) < 0$.

The solution is Lipschitz stable, uniformly over all initial conditions in $\mathbb{I}$; i.e. there is a constant

$$(4.9) \quad C_F = C_F(\alpha, \|\mathbf{a}_\ell\|_{L^\infty(0, \mathbf{T}_h)}, v_{\max}, \mathbf{T}_h),$$

independent of the initial condition, such that the following hold whenever the indicated intervals are nonempty.

(a) Stability in the initial state. For every $t_o \in [0, \mathbf{T}_h]$ and $(t_o, \mathbf{x}_o), (t_o, \tilde{\mathbf{x}}_o) \in \mathbb{I}$,

$$(4.10) \quad \|\mathbf{x}(\cdot; t_o, \mathbf{x}_o) - \tilde{\mathbf{x}}(\cdot; t_o, \tilde{\mathbf{x}}_o)\|_{L^\infty((t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o) \wedge \tilde{\mathcal{T}}(t_o, \tilde{\mathbf{x}}_o)); \ell^2)} \leq C_F \|\mathbf{x}_o - \tilde{\mathbf{x}}_o\|_{\ell^2}.$$

(b) Stability in the initial time. For every $0 \leq s_o < t_o \leq \mathbf{T}_h$ with $(t_o, \mathbf{x}_o), (s_o, \mathbf{x}_o) \in \mathbb{I}$,

$$(4.11) \quad \|\mathbf{x}(\cdot; t_o, \mathbf{x}_o) - \tilde{\mathbf{x}}(\cdot; s_o, \mathbf{x}_o)\|_{L^\infty((t_o \vee s_o, \tilde{\mathcal{T}}(s_o, \mathbf{x}_o) \wedge \tilde{\mathcal{T}}(t_o, \mathbf{x}_o)); \ell^2)} \leq C_F |t_o - s_o|.$$

(c) Joint stability. For every $0 \leq s_o \leq t_o \leq \mathbf{T}_h$ with $(t_o, \mathbf{x}_o), (s_o, \tilde{\mathbf{x}}_o) \in \mathbb{I}$,

$$(4.12) \quad \|\mathbf{x}(\cdot; t_o, \mathbf{x}_o) - \tilde{\mathbf{x}}(\cdot; s_o, \tilde{\mathbf{x}}_o)\|_{L^\infty((t_o \vee s_o, \tilde{\mathcal{T}}(s_o, \tilde{\mathbf{x}}_o) \wedge \tilde{\mathcal{T}}(t_o, \mathbf{x}_o)); \ell^2)} \leq C_F (\|\mathbf{x}_o - \tilde{\mathbf{x}}_o\|_{\ell^2} + |t_o - s_o|).$$

Proof. Well-posedness and transversality. Existence and uniqueness of a Carathéodory solution on $[t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o))$ follow as in the proofs of [Lem. 3.3–Lem. 3.8](#); see also [\[39, 11\]](#). The regularity $\mathbf{x}(\cdot; t_o, \mathbf{x}_o) \in \mathcal{W}^{1,\infty}$ is shown in [Remark 4.3](#). When $\tilde{\mathcal{T}} < \mathbf{T}_h$, transversality $\Lambda(\tilde{\mathcal{T}}) \neq 0$ holds by the definition [\(4.3\)](#) of the predictive collision time, which selects the first boundary hit with strictly negative relative velocity. All stability constants below trace back to the uniform field bound [\(4.7\)](#); in particular none depends on the initial condition.

Proof of (a): stability in the initial state. Fix t_o and let $\mathbf{x} = (\Xi, \Lambda)$, $\tilde{\mathbf{x}} = (\tilde{\Xi}, \tilde{\Lambda})$ solve [\(4.1\)](#) from $\mathbf{x}_o$, $\tilde{\mathbf{x}}_o$ at t_o , on the common interval

$$J \stackrel{\text{def}}{=} (t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o) \wedge \tilde{\mathcal{T}}(t_o, \tilde{\mathbf{x}}_o)).$$

In integral (Carathéodory) form, for $t \in J$,

$$\begin{aligned} \Xi(t) - \tilde{\Xi}(t) &= (\Xi_o - \tilde{\Xi}_o) + \int_{t_o}^t (\Lambda - \tilde{\Lambda}) ds, \\ \Lambda(t) - \tilde{\Lambda}(t) &= (\Lambda_o - \tilde{\Lambda}_o) - \alpha \int_{t_o}^t (\Lambda - \tilde{\Lambda}) ds - \alpha \int_{t_o}^t (\mathcal{V}(\Xi) - \mathcal{V}(\tilde{\Xi})) ds - \alpha \int_{t_o}^t (\zeta_\ell(s) - \tilde{\zeta}_\ell(s)) ds, \end{aligned}$$

the difference $\Lambda - \tilde{\Lambda}$ entering the first line through $\dot{\Xi} = \Lambda$ and the second through the $-\alpha\Lambda$ term of $\dot{\Lambda}$.

There is a pointwise Lipschitz bounds on the forcing, which is valid for a.e. $s \in J$. In particular, $\mathcal{V} \in C^\infty(\mathbb{R})$ with $\|\mathcal{V}'\|_{L^\infty(\mathbb{R})} \leq 1$, so $|\mathcal{V}(\Xi(s)) - \mathcal{V}(\tilde{\Xi}(s))| \leq |\Xi(s) - \tilde{\Xi}(s)| \leq \|\mathbf{x}(s) - \tilde{\mathbf{x}}(s)\|_{\ell^2}$;

Squared estimate and Gronwall. Squaring each component, applying

$$\left(\sum_{k=1}^m a_k\right)^2 \leq m \sum_{k=1}^m a_k^2, \quad \text{and} \quad \left(\int_{t_o}^t g ds\right)^2 \leq (t - t_o) \int_{t_o}^t g^2 ds \leq \mathbf{T}_h \int_{t_o}^t g^2 ds$$

with (i)–(iii), we get for a.e. $t \in J$

$$\|\mathbf{x}(t) - \tilde{\mathbf{x}}(t)\|_{\ell^2}^2 \leq C_1(1 + \mathbf{T}_h^2) \|\mathbf{x}_o - \tilde{\mathbf{x}}_o\|_{\ell^2}^2 + C_2 \mathbf{T}_h \int_{t_o}^t \|\mathbf{x}(s) - \tilde{\mathbf{x}}(s)\|_{\ell^2}^2 ds,$$

C_1, C_2 depending only on α . With $\phi(t) \stackrel{\text{def}}{=} \text{ess sup}_{t_0 \leq r \leq t} \|\mathbf{x}(r) - \tilde{\mathbf{x}}(r)\|_{\ell^2}^2$ the right side is nondecreasing, so Gronwall gives $\phi(t) \leq C_1(1 + \mathbf{T}_h^2)e^{C_2\mathbf{T}_h^2}\|\mathbf{x}_o - \tilde{\mathbf{x}}_o\|_{\ell^2}^2$. Taking square roots yields (4.10) with $C_F = (C_1(1 + \mathbf{T}_h^2))^{1/2}e^{C_2\mathbf{T}_h^2/2}$, depending only on $\alpha, \mathbf{T}_h$ —hence, through L_a and (2.4), on $\alpha, \|\mathbf{a}_\ell\|_{L^\infty}, v_{\max}, \mathbf{T}_h$, and independent of $(t_o, \mathbf{x}_o)$.

Proof of (b): stability in the initial time. Let $0 \leq s_o < t_o \leq \mathbf{T}_h$ and write $\mathbf{x}(\cdot) \stackrel{\text{def}}{=} \mathbf{x}(\cdot; t_o, \mathbf{x}_o)$, $\tilde{\mathbf{x}}(\cdot) \stackrel{\text{def}}{=} \tilde{\mathbf{x}}(\cdot; s_o, \mathbf{x}_o)$: two solutions from the *same* datum $\mathbf{x}_o$ but at *different* initial times. Because (4.1) is non-autonomous, the two are *not* time-translates; we route through the common time t_o .

Startup gap at t_o . On $[s_o, t_o]$ only $\tilde{\mathbf{x}}$ has evolved, from $\tilde{\mathbf{x}}(s_o) = \mathbf{x}_o$. By the integrated field bound (4.8) applied to $\tilde{\mathbf{x}}$,

$$(4.13) \quad \|\tilde{\mathbf{x}}(t_o) - \mathbf{x}_o\|_{\ell^2} = \left| \int_{s_o}^{t_o} \mathbf{G}(r, \tilde{\mathbf{x}}(r)) dr \right|_{\ell^2} \leq M |t_o - s_o|.$$

Since $\mathbf{x}(t_o) = \mathbf{x}_o$, the trajectories are $M|t_o - s_o|$ apart at t_o .

Propagation for $t \geq t_o$. For $t \geq t_o$ both solve (4.1) with the same initial time t_o and forcing, differing only in their states at t_o , namely $\mathbf{x}_o$ and $\tilde{\mathbf{x}}(t_o)$. The initial-state estimate (4.10), applied on $(t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o) \wedge \tilde{\mathcal{T}}(s_o, \mathbf{x}_o))$ with initial states $\mathbf{x}_o, \tilde{\mathbf{x}}(t_o)$, gives

$$(4.14) \quad \|\mathbf{x} - \tilde{\mathbf{x}}\|_{L^\infty((t_o, \tilde{\mathcal{T}}(t_o, \mathbf{x}_o) \wedge \tilde{\mathcal{T}}(s_o, \mathbf{x}_o)); \ell^2)} \leq C_F \|\mathbf{x}_o - \tilde{\mathbf{x}}(t_o)\|_{\ell^2} \leq C_F M |t_o - s_o|,$$

the last step by (4.13). On (s_o, t_o) only $\tilde{\mathbf{x}}$ is defined, and the norm in (4.11) is over $(t_o \vee s_o, \cdot) = (t_o, \cdot)$, so that interval does not enter. Hence (4.11) follows from (4.14) with C_F replaced by $C_F(1 \vee M)$, still of the asserted form.

Proof of (c): joint stability. Insert the intermediate solution $\mathbf{x}(\cdot; t_o, \tilde{\mathbf{x}}_o)$ and use the triangle inequality in the $L^\infty(\cdot; \ell^2)$ norm over the common interval:

$$\begin{aligned} \|\mathbf{x}(\cdot; t_o, \mathbf{x}_o) - \tilde{\mathbf{x}}(\cdot; s_o, \tilde{\mathbf{x}}_o)\| &\leq \underbrace{\|\mathbf{x}(\cdot; t_o, \mathbf{x}_o) - \mathbf{x}(\cdot; t_o, \tilde{\mathbf{x}}_o)\|}_{\leq C_F \|\mathbf{x}_o - \tilde{\mathbf{x}}_o\|_{\ell^2} \text{ (by (a))}} + \underbrace{\|\mathbf{x}(\cdot; t_o, \tilde{\mathbf{x}}_o) - \tilde{\mathbf{x}}(\cdot; s_o, \tilde{\mathbf{x}}_o)\|}_{\leq C_F |t_o - s_o| \text{ (by (b))}}. \end{aligned}$$

Summing yields (4.12) with the same constant C_F (up to the factor $1 \vee M$), completing the proof. ■

Using Thm. 4.4, we formalize the function τ and we discuss some of its analytical properties which would be crucial in designing the proper NN-architecture.

4.1. Analytical Properties of the TTC Map. We start our discussion by introducing a domain

$$(4.15) \quad \mathbb{D} \stackrel{\text{def}}{=} \{(t_o, \mathbf{x}_o) \in [0, \mathbf{T}_h) \times \mathbb{R}_{>0} \times \mathbb{R} : \tilde{\mathcal{T}}(t_o, \mathbf{x}_o) < \mathbf{T}_h, \Lambda(\tilde{\mathcal{T}}(t_o, \mathbf{x}_o); t_o, \mathbf{x}_o) < 0\},$$

the states from which the OVM dynamics predict a (transversal) collision strictly before $\mathbf{T}_h$, with the collision not yet reached (i.e. $\Xi > 0$ on $[t_o, \tilde{\mathcal{T}})$ —defined in Equation (4.3)). Along any such trajectory the state remains in $\mathbb{D}$ up to its predictive collision time.

For simplicity we denote an initial condition by $\mathbf{z} = (t_o, \mathbf{x}_o) \in \mathbb{D}$, and write

$$(4.16) \quad \Xi(\theta; \mathbf{z}), \quad \Lambda(\theta; \mathbf{z}), \quad \theta \in [0, \tau(\mathbf{z})],$$

for the predictive gap and relative velocity along the OVM trajectory started at $\mathbf{z}$, evaluated at elapsed time θ ; these are the two components of $\mathbf{x}(t_o + \theta; \mathbf{z})$ solving (4.1). In this notation the TTC map is the first zero of the gap,

$$(4.17) \quad \tau: \begin{cases} \mathbb{D} & \rightarrow \mathbb{R}_{\geq 0} \\ \mathbf{z} & \mapsto \inf\{\theta > 0 : \Xi(\theta; \mathbf{z}) = 0\}, \end{cases}$$

which is well defined by the definition of $\mathbb{D}$ and well-posedness of the OVM dynamical model (Thm. 4.4). Finally we fix the two constants used throughout:

$$(4.18) \quad M \stackrel{\text{def}}{=} \|\mathbf{G}\|_{L^\infty(\mathbb{R}; \ell^2)}, \quad C_F = C_F(\alpha, \|\mathbf{a}_\ell\|_{L^\infty}, v_{\max}, \mathbf{T}_h),$$

the uniform field bound of Remark 4.3 and the stability constant of Thm. 4.4.

We first collect the terminal-window estimates on which the Lipschitz property of τ rests. All are stated on a fixed compact $\mathbf{D} \Subset \mathbb{D}$ and depend only on the *integrated* behaviour of the relative velocity Λ , hence remain valid for the rough L^∞ -in-time forcing.

Lemma 4.5 (Continuity of τ). *The TTC map τ of (4.17) is continuous on $\mathbb{D}$. In particular, on any compact $\mathbf{D} \Subset \mathbb{D}$ it attains a positive minimum,*

$$(4.19) \quad \tau_{\min}(\mathbf{D}) \stackrel{\text{def}}{=} \min_{\mathbf{z} \in \mathbf{D}} \tau(\mathbf{z}) > 0.$$

Proof. Fix $\mathbf{z}^0 \in \mathbb{D}$, write $\nu^0 \stackrel{\text{def}}{=} \tau(\mathbf{z}^0)$, and set

$$\gamma \stackrel{\text{def}}{=} -\Lambda(\nu^0; \mathbf{z}^0) > 0,$$

which is well defined and strictly positive by the transversality condition in (4.15). Throughout, M is the field bound of (4.18), so that by (4.6) both components of the flow are M -Lipschitz in time,

$$(4.20) \quad |\Xi(\theta_1; \mathbf{z}) - \Xi(\theta_2; \mathbf{z})| \leq M|\theta_1 - \theta_2|, \quad |\Lambda(\theta_1; \mathbf{z}) - \Lambda(\theta_2; \mathbf{z})| \leq M|\theta_1 - \theta_2|,$$

and by Thm. 4.4 the flow is C_F -Lipschitz in the initial datum, uniformly in θ :

$$(4.21) \quad |\Xi(\theta; \mathbf{z}) - \Xi(\theta; \mathbf{z}^0)| + |\Lambda(\theta; \mathbf{z}) - \Lambda(\theta; \mathbf{z}^0)| \leq C_F \|\mathbf{z} - \mathbf{z}^0\|.$$

Let $\epsilon > 0$ be given; shrinking it if necessary we assume

$$(4.22) \quad \epsilon \leq \min\left\{\frac{\nu^0}{2}, \frac{\gamma}{2M}, \frac{\gamma^2}{128M^2}\right\}.$$

Step 1: lower bound, $\tau(\mathbf{z}) > \nu^0 - \epsilon$. By (4.17) the gap of the reference trajectory is strictly positive before collision, so

$$(4.23) \quad m_\epsilon \stackrel{\text{def}}{=} \min_{\theta \in [0, \nu^0 - \epsilon]} \Xi(\theta; \mathbf{z}^0) > 0,$$

the minimum being attained since $\Xi(\cdot; \mathbf{z}^0)$ is continuous on the compact interval $[0, \nu^0 - \epsilon]$, which lies strictly inside the pre-collision interval $[0, \nu^0]$. If $\|\mathbf{z} - \mathbf{z}^0\| < m_\epsilon/(2C_F)$, then (4.21) gives

$$\Xi(\theta; \mathbf{z}) \geq \Xi(\theta; \mathbf{z}^0) - \frac{m_\epsilon}{2} \geq \frac{m_\epsilon}{2} > 0, \quad \theta \in [0, \nu^0 - \epsilon],$$

so the gap of $\mathbf{z}$ has no zero on $[0, \nu^0 - \epsilon]$ and therefore $\tau(\mathbf{z}) > \nu^0 - \epsilon$. Note that only times in $[0, \nu^0 - \epsilon]$ are used, all strictly before the collision of either trajectory.

Step 2: the reference state just before collision. Put $t_1 \stackrel{\text{def}}{=} \nu^0 - \epsilon \in (0, \nu^0)$. Using $\Xi(\nu^0; \mathbf{z}^0) = 0$ and (4.20),

$$(4.24) \quad \Xi(t_1; \mathbf{z}^0) = |\Xi(t_1; \mathbf{z}^0) - \Xi(\nu^0; \mathbf{z}^0)| \leq M\epsilon,$$

and, again by (4.20) together with $\Lambda(\nu^0; \mathbf{z}^0) = -\gamma$ and $\epsilon \leq \gamma/(2M)$,

$$(4.25) \quad \Lambda(t_1; \mathbf{z}^0) \leq -\gamma + M\epsilon \leq -\frac{\gamma}{2}.$$

Step 3: upper bound, $\tau(\mathbf{z}) \leq \nu^0 + \frac{16M}{\gamma}\epsilon$. Choose $\varrho > 0$ with $C_F\varrho \leq \min\{M\epsilon, \gamma/4\}$ and let $\|\mathbf{z} - \mathbf{z}^0\| < \varrho$. Then (4.21), (4.24) and (4.25) give, at the single time t_1 ,

$$(4.26) \quad \Xi(t_1; \mathbf{z}) \leq 2M\epsilon, \quad \Lambda(t_1; \mathbf{z}) \leq -\frac{\gamma}{2} + \frac{\gamma}{4} = -\frac{\gamma}{4}.$$

By the Lipschitz-in-time bound (4.20) applied to $\Lambda(\cdot; \mathbf{z})$,

$$(4.27) \quad \Lambda(\theta; \mathbf{z}) \leq -\frac{\gamma}{8} \quad \text{for } \theta \in [t_1, t_1 + \ell], \quad \ell \stackrel{\text{def}}{=} \frac{\gamma}{8M},$$

since Λ can increase by at most $M\ell = \gamma/8$ over that interval. Consequently, as long as the trajectory has not collided,

$$\Xi(\theta; \mathbf{z}) \leq \Xi(t_1; \mathbf{z}) - \frac{\gamma}{8}(\theta - t_1), \quad \theta \in [t_1, t_1 + \ell],$$

by integrating (4.27). The right-hand side vanishes at

$$\theta^* \stackrel{\text{def}}{=} t_1 + \frac{8\Xi(t_1; \mathbf{z})}{\gamma} \leq t_1 + \frac{16M\epsilon}{\gamma} \leq t_1 + \ell,$$

the last inequality by $\epsilon \leq \gamma^2/(128M^2)$ from (4.22). Hence $\Xi(\cdot; \mathbf{z})$ must vanish at some time in $(t_1, \theta^*]$, so

$$\tau(\mathbf{z}) \leq \theta^* \leq \nu^0 - \epsilon + \frac{16M\epsilon}{\gamma} \leq \nu^0 + \frac{16M}{\gamma}\epsilon.$$

Again every evaluation occurs at a time not exceeding the collision time of the trajectory in question.

Conclusion. Combining Steps 1 and 3, for $\|\mathbf{z} - \mathbf{z}^0\| < \min\{m_\epsilon/(2C_F), \varrho\}$,

$$|\tau(\mathbf{z}) - \tau(\mathbf{z}^0)| \leq C\epsilon, \quad C \stackrel{\text{def}}{=} \max\left\{1, \frac{16M}{\gamma}\right\},$$

and since $\epsilon > 0$ was arbitrary, τ is continuous at $\mathbf{z}^0$. Finally $\tau > 0$ on $\mathbb{D}$, so continuity and compactness give (4.19). ■

Lemma 4.6 (Uniform transversality margin on compacts). *Define the terminal approach rate*

$$(4.28) \quad g: \mathbb{D} \rightarrow \mathbb{R}, \quad g(\mathbf{z}) \stackrel{\text{def}}{=} -\Lambda(\tau(\mathbf{z}); \mathbf{z}),$$

the magnitude of the relative velocity at the collision instant. Then g is continuous and strictly positive on $\mathbb{D}$, and for every compact $\mathbf{D} \Subset \mathbb{D}$ the transversality margin

$$(4.29) \quad \gamma_{\mathbf{D}} \stackrel{\text{def}}{=} \min_{\mathbf{z} \in \mathbf{D}} g(\mathbf{z}) = \min_{\mathbf{z} \in \mathbf{D}} (-\Lambda(\tau(\mathbf{z}); \mathbf{z}))$$

is attained and satisfies $\gamma_{\mathbf{D}} > 0$. Equivalently,

$$(4.30) \quad \Lambda(\tau(\mathbf{z}); \mathbf{z}) \leq -\gamma_{\mathbf{D}} < 0, \quad \mathbf{z} \in \mathbf{D}.$$

Proof. *Strict positivity, pointwise.* For every $\mathbf{z} \in \mathbb{D}$ the transversality condition in the definition (4.15) of $\mathbb{D}$ reads $\Lambda(\tau(\mathbf{z}); \mathbf{z}) < 0$; since $\tilde{T}(\mathbf{z}) = t_o + \tau(\mathbf{z})$, this is exactly $g(\mathbf{z}) > 0$.

Continuity. Write g as the composition $\mathbf{z} \mapsto (\tau(\mathbf{z}), \mathbf{z}) \mapsto -\Lambda(\tau(\mathbf{z}); \mathbf{z})$. The outer map $(\theta, \mathbf{z}) \mapsto \Lambda(\theta; \mathbf{z})$ is jointly continuous: it is M -Lipschitz in θ by Equation (4.6) and C_F -Lipschitz in $\mathbf{z}$ uniformly in θ by Thm. 4.4, whence for $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{D}$

$$(4.31) \quad |g(\mathbf{z}_1) - g(\mathbf{z}_2)| \leq M|\tau(\mathbf{z}_1) - \tau(\mathbf{z}_2)| + C_F \|\mathbf{z}_1 - \mathbf{z}_2\|.$$

The inner map $\mathbf{z} \mapsto \tau(\mathbf{z})$ is continuous by Lem. 4.5. Hence g is continuous on $\mathbb{D}$.

Uniformity on compacts. Let $\mathbf{D} \Subset \mathbb{D}$ be compact. By continuity therefore, there is $\mathbf{z}^* \in \mathbf{D}$ with $g(\mathbf{z}^*) = \min_{\mathbf{D}} g$. Since $\mathbf{z}^* \in \mathbf{D} \subset \mathbb{D}$, the first part gives $g(\mathbf{z}^*) > 0$, and therefore $\gamma_{\mathbf{D}} = g(\mathbf{z}^*) > 0$, which is (4.30). ■

Corollary 4.7 (Uniform continuity of τ on compacts). *For every compact $\mathbf{D} \Subset \mathbb{D}$, the map τ is uniformly continuous on $\mathbf{D}$; more precisely, the constant C in the proof of Lem. 4.5 may be taken uniform over $\mathbf{D}$,*

$$(4.32) \quad C = C(\mathbf{D}) \stackrel{\text{def}}{=} \max\left\{1, \frac{16M}{\gamma_{\mathbf{D}}}\right\},$$

with $\gamma_{\mathbf{D}}$ as in (4.29).

Proof. In the proof of Lem. 4.5 the only $\mathbf{z}^0$ -dependent quantity entering the final constant is $\gamma = \gamma(\mathbf{z}^0) = -\Lambda(\tau(\mathbf{z}^0); \mathbf{z}^0) = g(\mathbf{z}^0)$. By Lem. 4.6, $g \geq \gamma_{\mathbf{D}}$ on $\mathbf{D}$, so replacing γ by $\gamma_{\mathbf{D}}$ throughout—which only strengthens the smallness requirements (4.22)—yields the single constant (4.32), valid simultaneously for all $\mathbf{z}^0 \in \mathbf{D}$. ■

Lemma 4.8 (Uniform lower bound on the terminal monotone window). *Let $\mathbf{D} \Subset \mathbb{D}$ be compact and, for $\mathbf{z} \in \mathbf{D}$, let*

$$(4.33) \quad \check{t}_{\text{OVM}}(\mathbf{z}) \stackrel{\text{def}}{=} \inf\left\{\theta \in [0, \tau(\mathbf{z})) : \Lambda(s; \mathbf{z}) < 0 \text{ for a.e. } s \in [\theta, \tau(\mathbf{z}))\right\}$$

be the start of the maximal terminal window on which the gap $\Xi(\cdot; \mathbf{z})$ is strictly decreasing (the window is nonempty by Thm. 3.6 specialized to the OVM dynamics). Then there exists $\varsigma = \varsigma(\mathbf{D}) > 0$ such that

$$(4.34) \quad \eta(\mathbf{z}) \stackrel{\text{def}}{=} \tau(\mathbf{z}) - \check{t}_{\text{OVM}}(\mathbf{z}) \geq \varsigma, \quad \mathbf{z} \in \mathbf{D}.$$

Proof. Suppose, for contradiction, that (4.34) fails for every $\varsigma > 0$. Then there is a sequence $(z_n) \subset \mathbf{D}$ with

$$(4.35) \quad \eta_n \stackrel{\text{def}}{=} \tau(z_n) - \check{t}_{\text{OVM}}(z_n) \longrightarrow 0,$$

and, by compactness of $\mathbf{D}$, we may assume $z_n \rightarrow \bar{z} \in \mathbf{D}$. We distinguish two cases according to whether the terminal window starts at the initial time.

Case 1: $\check{t}_{\text{OVM}}(z_n) = 0$ along a subsequence. Then the whole trajectory lies in the monotone window and $\eta_n = \tau(z_n)$. By Lem. 4.5, $\tau(z_n) \geq \tau_{\min}(\mathbf{D}) > 0$ for every n , contradicting (4.35). Hence this case cannot occur along any subsequence, and we may assume $\check{t}_{\text{OVM}}(z_n) > 0$ for all n .

Case 2: $\check{t}_{\text{OVM}}(z_n) > 0$ for all n . Since $\check{t}_{\text{OVM}}(z_n)$ is defined in (4.33) as an *infimum*, no admissible window starts strictly earlier: for each n there exists

$$(4.36) \quad s_n \in \left(\check{t}_{\text{OVM}}(z_n) - \frac{1}{n}, \check{t}_{\text{OVM}}(z_n) \right) \cap [0, \tau(z_n)) \quad \text{with} \quad \Lambda(s_n; z_n) \geq 0.$$

Indeed, were $\Lambda(\cdot; z_n) < 0$ a.e. on $[\check{t}_{\text{OVM}}(z_n) - \frac{1}{n}, \tau(z_n))$, the value $\check{t}_{\text{OVM}}(z_n) - \frac{1}{n}$ would belong to the set in (4.33), contradicting the definition of the infimum.

We now pass to the limit in (4.36). By (4.35) and (4.36),

$$|s_n - \tau(z_n)| \leq |s_n - \check{t}_{\text{OVM}}(z_n)| + |\check{t}_{\text{OVM}}(z_n) - \tau(z_n)| \leq \frac{1}{n} + \eta_n \longrightarrow 0,$$

while $\tau(z_n) \rightarrow \tau(\bar{z})$ by Lem. 4.5. Hence $s_n \rightarrow \tau(\bar{z})$. By Thm. 4.4 the relative velocity $(\theta, z) \mapsto \Lambda(\theta; z)$ is jointly continuous, so passing to the limit in the inequality of (4.36),

$$0 \leq \lim_{n \rightarrow \infty} \Lambda(s_n; z_n) = \Lambda(\tau(\bar{z}); \bar{z}).$$

This contradicts the transversality condition of Lem. 4.6, which implies $\Lambda(\tau(\bar{z}); \bar{z}) < 0$ as $\bar{z} \in \mathbf{D} \subset \mathbb{D}$.

Both cases being impossible, (4.34) holds for some $\varsigma(\mathbf{D}) > 0$. ■

Lemma 4.9 (Boundary-layer profile). *Let $\mathbf{D} \Subset \mathbb{D}$ be compact, let $\gamma_{\mathbf{D}} > 0$ be the transversality margin of (4.29), and set*

$$(4.37) \quad h_0 = h_0(\mathbf{D}) \stackrel{\text{def}}{=} \frac{\gamma_{\mathbf{D}}}{2M},$$

with M as in (4.18). Then, for every $z \in \mathbf{D}$,

$$(4.38) \quad \Lambda(\theta; z) \leq -\frac{\gamma_{\mathbf{D}}}{2} \quad \text{for every } \theta \in [\tau(z) - h_0, \tau(z)].$$

Proof. Fix $z \in \mathbf{D}$. By the uniform transversality margin (4.30),

$$\Lambda(\tau(z); z) \leq -\gamma_{\mathbf{D}}.$$

By Remark 4.3 the relative velocity is M -Lipschitz in time, so for every $\theta \in [\tau(\mathbf{z}) - h_0, \tau(\mathbf{z})]$,

$$\Lambda(\theta; \mathbf{z}) \leq \Lambda(\tau(\mathbf{z}); \mathbf{z}) + M|\theta - \tau(\mathbf{z})| \leq -\gamma_D + Mh_0 = -\gamma_D + \frac{\gamma_D}{2} = -\frac{\gamma_D}{2},$$

by the choice (4.37) of h_0 . Since $\Lambda(\cdot; \mathbf{z})$ is continuous, the bound holds at every such θ , not merely almost everywhere. Both h_0 and $\gamma_D/2$ depend only on $\mathbf{D}$, so the estimate is uniform over $\mathbf{D}$. ■

Corollary 4.10 (Finite-difference transversality). *Under Lem. 4.9, for every $\mathbf{z} \in \mathbf{D}$ and every $h \in [0, h_0]$,*

$$(4.39) \quad \Xi(\tau(\mathbf{z}) - h; \mathbf{z}) \geq \frac{\gamma_D}{2} h.$$

Proof. By Remark 4.3 the map $\theta \mapsto \Xi(\theta; \mathbf{z})$ is Lipschitz with $\frac{d}{d\theta}\Xi = \Lambda$ a.e., hence absolutely continuous, and $\Xi(\tau(\mathbf{z}); \mathbf{z}) = 0$ by (4.17). The fundamental theorem of calculus therefore gives, for $h \in [0, h_0]$,

$$\Xi(\tau(\mathbf{z}) - h; \mathbf{z}) = \Xi(\tau(\mathbf{z}) - h; \mathbf{z}) - \Xi(\tau(\mathbf{z}); \mathbf{z}) = - \int_{\tau(\mathbf{z})-h}^{\tau(\mathbf{z})} \Lambda(\theta; \mathbf{z}) d\theta \geq \frac{\gamma_D}{2} h,$$

using (4.38) on the whole interval of integration. The relative velocity Λ appears only under the integral sign: no pointwise value of Λ —hence none of $\mathbf{a}_\ell$ or ζ_ℓ —is used, and the left-hand side is a value of the Lipschitz-continuous gap Ξ . ■

Remark 4.11 (The rough forcing is neutralized). *Cor. 4.10 replaces the pointwise transversality $\Lambda(\tau(\mathbf{z}); \mathbf{z}) < 0$ —a value of Λ at a single instant, undefined for an L^∞ -in-time field—by an inequality for the continuous gap Ξ at times strictly before collision, obtained by integrating Λ . Modifying the leader acceleration $\mathbf{a}_\ell$, and hence the leader velocity ζ_ℓ , on an $\mathcal{L}^1$ -null set of times leaves the Carathéodory integral Ξ unchanged; consequently (4.39), and with it the Lipschitz bound (4.41) below, is independent of the representative chosen for the forcing.*

Lemma 4.12 (Finite-difference Lipschitz bound on τ). *Fix a compact $\mathbf{D} \Subset \mathbb{D}$. There is a constant*

$$(4.40) \quad L_\tau(\mathbf{D}) \stackrel{\text{def}}{=} \frac{2\sqrt{2} \max\{C_F, M\}}{\gamma_D}$$

such that, for all $\mathbf{z}_1, \mathbf{z}_2 \in \mathbf{D}$ with $\|\mathbf{z}_1 - \mathbf{z}_2\|$ sufficiently small,

$$(4.41) \quad |\tau(\mathbf{z}_1) - \tau(\mathbf{z}_2)| \leq L_\tau(\mathbf{D}) \|\mathbf{z}_1 - \mathbf{z}_2\|,$$

with M, C_F as in (4.18) and γ_D as in (4.29).

Proof. Write $\nu_i \stackrel{\text{def}}{=} \tau(\mathbf{z}_i)$ and $\mathbf{z}_i = (t_i, \mathbf{x}_i)$; without loss of generality $\nu_1 \leq \nu_2$, and set $h \stackrel{\text{def}}{=} \nu_2 - \nu_1 \geq 0$.

Step 0 (admissibility of h). By Lem. 4.5 the map τ is continuous on $\mathbb{D}$, so $h = |\tau(\mathbf{z}_1) - \tau(\mathbf{z}_2)| \rightarrow 0$ as $\|\mathbf{z}_1 - \mathbf{z}_2\| \rightarrow 0$. Hence there is $\varrho_0 = \varrho_0(\mathbf{D}) > 0$ such that $\|\mathbf{z}_1 - \mathbf{z}_2\| < \varrho_0$ implies $h \leq h_0$, with h_0 as in (4.37); we assume this throughout. By (4.17), $\Xi(\nu_i; \mathbf{z}_i) = 0$ for $i = 1, 2$.

Step 1 (transversality). Since $\nu_1 = \nu_2 - h$ with $h \in [0, h_0]$, Cor. 4.10 applied to $\mathbf{z}_2$ gives

$$\Xi(\nu_1; \mathbf{z}_2) \geq \frac{\gamma_D}{2} h.$$

Step 2 (flow stability). Trajectory 1 has collided at ν_1 while trajectory 2 has not, by Step 1. Comparing the two gaps at the common elapsed time ν_1 and using Thm. 4.4—Lipschitz dependence on the initial state with constant C_F , and Lipschitz dependence in time with constant M

$$|\Xi(\nu_1; \mathbf{z}_1) - \Xi(\nu_1; \mathbf{z}_2)| \leq C_F (\|\mathbf{x}_1 - \mathbf{x}_2\|_{\ell^2}) + M|t_1 - t_2| \leq \max\{C_F, M\} (\|\mathbf{x}_1 - \mathbf{x}_2\|_{\ell^2} + |t_1 - t_2|),$$

and, by the elementary inequality $a + b \leq \sqrt{2} (a^2 + b^2)^{1/2}$ applied to $a = \|\mathbf{x}_1 - \mathbf{x}_2\|_{\ell^2}$, $b = |t_1 - t_2|$,

$$|\Xi(\nu_1; \mathbf{z}_1) - \Xi(\nu_1; \mathbf{z}_2)| \leq \sqrt{2} \max\{C_F, M\} \|\mathbf{z}_1 - \mathbf{z}_2\|.$$

Step 3 (conclusion). Combining Steps 1 and 2 with $\Xi(\nu_1; \mathbf{z}_1) = 0$,

$$\frac{\gamma_D}{2} h \leq \Xi(\nu_1; \mathbf{z}_2) = |\Xi(\nu_1; \mathbf{z}_2) - \Xi(\nu_1; \mathbf{z}_1)| \leq \sqrt{2} \max\{C_F, M\} \|\mathbf{z}_1 - \mathbf{z}_2\|,$$

that is, $h \leq L_\tau(\mathbf{D}) \|\mathbf{z}_1 - \mathbf{z}_2\|$ with $L_\tau(\mathbf{D})$ as in (4.40), which is (4.41). Only the continuous gap Ξ and Cor. 4.10 are used; no time derivative of Ξ and no pointwise value of the forcing appears. $\blacksquare$

Theorem 4.13. *Consider the TTC map of Equation (4.17) calculated based on the OVM dynamics. For any compact set $\mathbf{D} \in \mathbb{D}$, $\tau \in \mathcal{W}^{1,\infty}(\mathbf{D}; \mathbb{R}_{\geq 0})$.*

Proof. The proof has two parts: a local Lipschitz bound on small balls with a constant independent of the centre, followed by a chaining argument that upgrades it to a global bound on $\mathbf{D}$.

Local Lipschitz. For $\mathbf{z}_1, \mathbf{z}_2 \in \mathbf{D}$ with $\|\mathbf{z}_1 - \mathbf{z}_2\|$ sufficiently small, the finite-difference bound Lem. 4.12 gives directly

$$|\tau(\mathbf{z}_1) - \tau(\mathbf{z}_2)| \leq L_\tau(\mathbf{D}) \|\mathbf{z}_1 - \mathbf{z}_2\|.$$

Hence τ is Lipschitz on the relatively open ball $\text{Ball}(\mathbf{z}, \varsigma') \cap \mathbf{D}$ for each $\mathbf{z} \in \mathbf{D}$, where $\varsigma' = \varsigma'(\mathbf{D}) > 0$ may be taken uniform in $\mathbf{z}$: the window length h_0 of (4.37) and the transversality margin γ_D of (4.29) are uniform on $\mathbf{D}$, and by Lem. 4.5 so is the radius $\varrho_0(\mathbf{D})$ of Step 0 of Lem. 4.12. That is,

$$(4.42) \quad \tau \in \mathcal{W}^{1,\infty}(\text{Ball}(\mathbf{z}, \varsigma') \cap \mathbf{D}; \mathbb{R}_{\geq 0}), \quad \mathbf{z} \in \mathbf{D},$$

with the single Lipschitz constant $L_\tau(\mathbf{D})$ of (4.40), independent of the center $\mathbf{z}$.

Global Lipschitz extension by chaining. Partition of unity is a standard tool to glue the compact sets with local Lipschitz properties as in Equation (4.42) together. However, we adopt the following direct chaining proof since it provided a significantly tighter bound on the Lipschitz constant of the map τ .

The local bound (4.42) controls $|\tau(\mathbf{z}) - \tau(\tilde{\mathbf{z}})|$ only for $\mathbf{z}, \tilde{\mathbf{z}}$ in a common ball. We remove this restriction by chaining, the uniformity of $L_\tau(\mathbf{D})$ in the center being exactly what makes this possible.

Assume first that $\mathbf{D}$ is connected. Being compact, $\mathbf{D}$ is *totally bounded*, so finitely many balls $\{\text{Ball}(\mathbf{z}_i, \varsigma'/2)\}_{i=1}^N$ cover $\mathbf{D}$, with $N = N(\mathbf{D}, \varsigma')$. Let $\mathbf{z}, \tilde{\mathbf{z}} \in \mathbf{D}$.

If $\|\mathbf{z} - \tilde{\mathbf{z}}\| < \varsigma'$, the two points lie in a common ball of radius ς' and (4.42) applies directly.

If $\|\mathbf{z} - \tilde{\mathbf{z}}\| \geq \varsigma'$, connectedness of $\mathbf{D}$ provides a chain $\mathbf{z} = \mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_m = \tilde{\mathbf{z}}$ of points of $\mathbf{D}$ with consecutive points in a common ball of the cover, so that $\|\mathbf{w}_k - \mathbf{w}_{k-1}\| < \varsigma'$ for each k ; since the cover is finite and the chain may be taken to visit each ball at most once, $m \leq N$. Applying (4.42) on each link and using the triangle inequality,

$$|\tau(\mathbf{z}) - \tau(\tilde{\mathbf{z}})| \leq \sum_{k=1}^m |\tau(\mathbf{w}_k) - \tau(\mathbf{w}_{k-1})| \leq L_\tau(\mathbf{D}) \sum_{k=1}^m \|\mathbf{w}_k - \mathbf{w}_{k-1}\| \leq L_\tau(\mathbf{D}) N \varsigma'.$$

Since $\|\mathbf{z} - \tilde{\mathbf{z}}\| \geq \varsigma'$, we can write

$$|\tau(\mathbf{z}) - \tau(\tilde{\mathbf{z}})| \leq L_\tau(\mathbf{D}) N \|\mathbf{z} - \tilde{\mathbf{z}}\|.$$

Combining the two cases,

$$(4.43) \quad |\tau(\mathbf{z}) - \tau(\tilde{\mathbf{z}})| \leq \hat{L}_\tau(\mathbf{D}) \|\mathbf{z} - \tilde{\mathbf{z}}\|, \quad \hat{L}_\tau(\mathbf{D}) \stackrel{\text{def}}{=} N L_\tau(\mathbf{D}), \quad \mathbf{z}, \tilde{\mathbf{z}} \in \mathbf{D}.$$

If $\mathbf{D}$ is not connected, it has finitely many connected components $\mathbf{D}_1, \dots, \mathbf{D}_p$ (finitely many, since each contains a ball of the finite cover), and $\text{dist}_{\min} \stackrel{\text{def}}{=} \min_{a \neq b} \text{dist}(\mathbf{D}_a, \mathbf{D}_b) > 0$ by compactness. Estimate (4.43) applies within each component. For $\mathbf{z} \in \mathbf{D}_a$ and $\tilde{\mathbf{z}} \in \mathbf{D}_b$ with $a \neq b$, we have $\|\mathbf{z} - \tilde{\mathbf{z}}\| \geq \text{dist}_{\min}$, while τ is bounded on $\mathbf{D}$ by Lem. 4.5 and compactness, say $\tau \leq \tau_{\max}(\mathbf{D})$; hence

$$|\tau(\mathbf{z}) - \tau(\tilde{\mathbf{z}})| \leq 2\tau_{\max}(\mathbf{D}) \leq \frac{2\tau_{\max}(\mathbf{D})}{\text{dist}_{\min}} \|\mathbf{z} - \tilde{\mathbf{z}}\|.$$

Combining the two situations gives (4.43) on all of $\mathbf{D}$ with $\hat{L}_\tau(\mathbf{D})$ replaced by

$$\max\{\hat{L}_\tau(\mathbf{D}), 2\tau_{\max}(\mathbf{D})/\text{dist}_{\min}\}.$$

In either case τ is Lipschitz on $\mathbf{D}$ and, being bounded there by Lem. 4.5, satisfies $\tau \in \mathcal{W}^{1,\infty}(\mathbf{D}; \mathbb{R}_{\geq 0})$, which completes the proof. ■

5. Collision Boundary Invariance Set: A Hamilton-Jacobi-Reachability framework. In this section, we propose a deep learning scheme that approximates the collision invariance set. We situate collision invariance analysis within a Hamilton-Jacobi(HJ) reachability framework, which provides a global, set-based characterization of collision inevitability under bounded uncertainty and admissible dynamics via HJ partial differential equations.

HJB is a proper framework. In Section 4, we established the well-posedness and local Lipschitz continuity of the TTC map directly from the OVM dynamics.

We analytically explained that the behavior of the dynamical model is notably different near the collision boundaries and to approximate such behavior, we need a proper characterization of the dynamical model through introducing a proper framework which turns out to be

HJ-reachability approach. In particular, the HJ-reachability framework provides an intrinsic characterization of the TTC function.

Consequently, when the TTC map is represented as the value function of a minimum-time reachability problem and characterized through the HJB equation in this section, the PDE formulation is not merely formal. Rather, it is consistent with the previously established analytical properties of the TTC mapping. In particular, the HJB solution inherits the well-posedness and regularity properties already derived directly from the underlying dynamics.

Furthermore, our analytical result of the TTC-mapping highlights the fact that the viscosity solution of HJB corresponds to a physically meaningful TTC-mapping. In other words, the HJB solution is not an abstract PDE object; it is exactly the TTC-mapping characterization when the collision is projected before T_h ; i.e. on set $\mathbb{D}$. A detailed schematics of the correspondence between OVM dynamics, HJB framework, viscosity solution and its approximation by NN is illustrated in Figure 6.

5.1. Mathematical formalization: reachability characterization. We start by defining the set

$$(5.1) \quad \mathcal{Q} \stackrel{\text{def}}{=} \mathbb{R}_{\leq 0} \times \mathbb{R},$$

and we define the value function as the minimum hitting time of $\mathcal{Q}$. In particular,

$$(5.2) \quad V(s, \mathbf{y}) \stackrel{\text{def}}{=} \inf\{T \in [0, T_h - s] \mid \mathbf{x}(s + T; s, \mathbf{y}) \in \mathcal{Q}\}, \quad \inf(\emptyset) = 0.$$

In particular, the *reachable* set can be defined by

$$\mathbb{L}(\mathcal{Q}) = \{(s, \mathbf{y}) \mid V(s, \mathbf{y}) < T_h - s\},$$

i.e. the level sets of the value function V . If $\mathbf{y} \in \partial\mathcal{Q}$, then $V(s, \mathbf{y}) = 0$ which in fact defines the boundary condition.

Lemma 5.1 (Optimality Principle). *Let us fix the initial condition $(s, \mathbf{y}) \in \mathbb{D}$. For any $0 \leq \kappa \leq \nu \stackrel{\text{def}}{=} V(s, \mathbf{y})$, we define*

$$\mathbf{x}_\kappa \stackrel{\text{def}}{=} \mathbf{x}(s + \kappa; s, \mathbf{y}).$$

Then, the optimality principle

$$(5.3) \quad \nu = \kappa + V(s + \kappa, \mathbf{x}_\kappa)$$

holds.

Proof. Let

$$\Phi(\theta; s, \mathbf{y}) \stackrel{\text{def}}{=} \mathbf{x}(s + \theta; s, \mathbf{y}), \quad \theta \in [0, T_h - s],$$

denotes the flow associated with the OVM dynamical system starting from $\mathbf{y}$ at time s . By uniqueness of solutions, the flow satisfies evolution (Champan-Kolmogorov or also known as semigroup) property

$$\Phi(T + \kappa; s, \mathbf{y}) = \Phi(T; s + \kappa, \mathbf{x}_\kappa),$$

whenever

$$0 \leq \kappa \leq T_h - s, \quad 0 \leq T \leq T_h - s - \kappa,$$

which is valid for the non-autonomous field $G(t, \cdot)$. By $(s, \mathbf{y}) \in \mathbb{D}$, the OVM trajectory reaches the target set $\mathcal{Q}$ before the horizon. Hence

$$\nu = V(s, \mathbf{y}) = \inf\{T \in [0, T_h - s) \mid \Phi(T; s, \mathbf{y}) \in \mathcal{Q}\},$$

and the infimum is attained at the first hitting time. In particular,

$$\Phi(\nu; s, \mathbf{y}) \in \mathcal{Q}, \quad \Phi(T; s, \mathbf{y}) \notin \mathcal{Q} \quad \text{for every } 0 \leq T < \nu.$$

Now fix $0 \leq \kappa \leq \nu$. We claim that the remaining hitting time from the shifted state $(s + \kappa, \mathbf{x}_\kappa)$ is exactly $\nu - \kappa$. Indeed, by the semigroup property,

$$\Phi(\nu - \kappa; s + \kappa, \mathbf{x}_\kappa) = \Phi(\nu; s, \mathbf{y}) \in \mathcal{Q}.$$

In other words, starting from $(s + \kappa, \mathbf{x}_\kappa)$, OVM-flow is at the boundary at $\nu - \kappa$. On the other hand by Equation (5.2), starting from this point, the first collision time should be $V(s + \kappa, \mathbf{x}_\kappa)$; i.e.

$$(5.4) \quad \nu - \kappa \geq V(s + \kappa, \mathbf{x}_\kappa),$$

which would be satisfied as equality if $\kappa = \nu$.

Next, we show that Equation (5.4) holds in equality form for any $\kappa \in [0, \nu]$. To do so, we suppose, by contradiction, that for some $\kappa < \nu$, $V(s + \kappa, \mathbf{x}_\kappa) < \nu - \kappa$. Equivalently, we suppose there exists some $0 \leq T < \nu - \kappa$, such that

$$\Phi(T; s + \kappa, \mathbf{x}_\kappa) \in \mathcal{Q}.$$

Using the semigroup property again, we get

$$\Phi(T + \kappa; s, \mathbf{y}) = \Phi(T; s + \kappa, \mathbf{x}_\kappa) \in \mathcal{Q}.$$

But this implies the collision happens after $T + \kappa < \nu$ which contradicts the minimality of ν as the first hitting time from $(s, \mathbf{y})$.

Combining these results, we conclude

$$V(s + \kappa, \mathbf{x}_\kappa) = \nu - \kappa.$$

This completes the proof. ■

From optimality principle to HJ-PDE. Let us fix $(s, \mathbf{y}) \in \mathbb{D}$ and a $\kappa > 0$ such that $(s + \kappa, \Phi(\kappa; s, \mathbf{y})) \in \text{cl}[\mathbb{D}]$. By the optimality principle Lem. 5.1, we have

$$V(s + \kappa, \Phi(\kappa; s, \mathbf{y})) = V(s, \mathbf{y}) - \kappa.$$

Thus the value function decreases linearly with unit rate along OVM characteristics. Formally, assuming that V is differentiable at this point and $\dot{\Phi}(0; s, \mathbf{y}) = G(s, \mathbf{y})$, then differentiating this identity at $\kappa = 0$, we will have

$$\partial_s V(s, \mathbf{y}) + \nabla V(s, \mathbf{y}) \cdot G(s, \mathbf{y}) = -1.$$

It should be noted that even if $\mathbf{V}$ is differentiable, this identity holds a.e. as $\mathbf{G}$ has a time dependence on a rough forcing $\mathbf{a}_\ell$ which is L^∞ and hence the equation is understood in the viscosity sense. Therefore considering the boundary and terminal values, we will be interested in the HJB-type equations in the form of:

$$(5.5) \quad \begin{aligned} \partial_t \mathbf{V}(s, \mathbf{y}) + \nabla \mathbf{V}(s, \mathbf{y}) \cdot \mathbf{G}(s, \mathbf{y}) &= -1, & (s, \mathbf{y}) \in \mathbb{D} \\ \mathbf{V}(s, \mathbf{y}) &= 0, & (s, \mathbf{y}) \in (0, T_h] \times \mathcal{Q} \\ \mathbf{V}(T_h, \mathbf{y}) &= 0, & \mathbf{y} \in \mathbb{R}_{>0} \times \mathbb{R}. \end{aligned}$$

which is the classic HJB (also known as transport) equations.

Remark 5.2 (Characterization of TTC-mapping by HJB equations). *The terminal condition $\mathbf{V}(T_h, \cdot)$ does not affect the solution on the predictive domain $\mathbb{D}$. Indeed, by the definition Equation (4.15) of $\mathbb{D}$, every characteristic reaches $\mathcal{Q}$ before the terminal time T_h .*

Furthermore, while on the domain $\mathbb{D}$ the value function $\mathbf{V}$ is completely determined by the target boundary condition on $\partial\mathcal{Q}$, we define such boundary condition to be absorbing over entire $\mathcal{Q}$. This would be helpful in analytical results of the paper.

Remark 5.3 (Regularity of the value function as HJB solution). *The value function $\mathbf{V} = \tau$ is the unique viscosity solution of the HJB equation (5.5) on $\mathbb{D}$; this follows from the dynamic programming principle (Lem. 5.1) by the standard argument testing $\mathbf{V}$ against smooth functions touching it from above and below, using continuity of $\mathbf{V}$ and the evolution property of the OVM flow. Viscosity is the selective notion here—it singles out the physically correct solution and yields uniqueness via the comparison principle.*

On any compact $\mathbf{D} \in \mathbb{D}$, the viscosity solution is moreover regular. In particular, Thm. 4.13 gives $\mathbf{V} \in \mathcal{W}^{1,\infty}(\mathbf{D})$, so by Rademacher’s theorem $\mathbf{V}$ is differentiable at a.e. point of $\mathbf{D}$, and at every such point the HJB equation (5.5) holds classically,

$$\partial_s \mathbf{V} + \mathbf{G} \cdot \nabla_{\mathbf{y}} \mathbf{V} + 1 = 0 \quad \text{for a.e. } (s, \mathbf{y}) \in \mathbf{D}.$$

This a.e. classical identity is the sense in which the trained residual $\mathcal{R}[\mathbf{V}_\theta]$ is driven to zero (Equation (5.12)); it is a regularity refinement of the viscosity property on compacts. It should be noted that for general Hamilton–Jacobi equations an a.e. classical solution need not be the viscosity solution. $H(\mathbf{z}, p) = p \cdot \mathbf{G}(\mathbf{z}) - 1$ is affine in the gradient $p = \nabla \mathbf{V}$ (though with variable, non-autonomous coefficients through $\mathbf{G}(\mathbf{z})$), affineness in p excludes the gradient-direction ambiguity that makes a.e. solutions non-unique for genuinely nonlinear Hamiltonians, and the induced characteristic field $\partial_p H = \mathbf{G}$ is independent of p , so the characteristics are exactly the (non-crossing) OVM trajectories, along which the terminal condition determines $\mathbf{V}$ uniquely (see Prop. 5.23).

While the HJB equation provides the reachability interpretation and theoretical characterization of TTC on $\mathbb{D}$, the computational framework for learning is formalized through two different approaches, weak and strong fixed point properties which is formalized in the next result.

Definition 5.4 (Weak and Strong Fixed point). *A function $W \in L^\infty(\mathcal{cl}[\mathbb{D}])$ is a strong fixed point on $\mathbb{D}$ if for every $(s, \mathbf{y}) \in \mathbb{D}$ and every $\kappa \geq 0$ with $(s + \kappa, \Phi(\kappa; s, \mathbf{y})) \in \mathcal{cl}[\mathbb{D}]$, the*

optimality condition is satisfied:

$$(5.6) \quad \kappa + W(s + \kappa, \Phi(\kappa; s, \mathbf{y})) = W(s, \mathbf{y}).$$

Equivalently, the map $\kappa \mapsto \kappa + W(s + \kappa, \Phi(\kappa; s, \mathbf{y}))$ is invariance along every OVM trajectory over $\mathcal{CI}[\mathbb{D}]$.

5.2. Learning HJB-Equation: strong fixed-point approach. In this section, we show that by modifying the lost function, we can target the strong fixed point solution which is unique and in addition two-sided estimate can be established for the proximal. We start by defining a new loss function.

Definition 5.5 (Augmented PINN training loss). Let $\mathbf{D} \Subset \mathbb{D}$ be compact and $\mathbf{V}_\theta \in C^1(\mathbb{D}) \cap C(\bar{\mathbb{D}})$. Let us modify the definition of the tube $\mathbf{R}$ (defined as in Equation (4.5)) by restricting the initial condition to the compact set $\mathbf{D}$:

$$(5.7) \quad \mathbf{D}^+ \stackrel{\text{def}}{=} \{(s + \kappa, \Phi(\kappa; s, \mathbf{y})) : (s, \mathbf{y}) \in \mathbf{D}, \quad \kappa \in [0, V(s, \mathbf{y}))\} \subset \mathbb{D}.$$

In fact, $\mathbf{D}^+$ is not compactly embedded in $\mathbb{D}$, but its closure touches $\partial\mathcal{Q}$. Therefore, we can define the hitting-point map:

$$(5.8) \quad \Psi : \mathbf{D}^+ \rightarrow \partial\mathcal{Q}, \quad \Psi(s, \mathbf{y}) \stackrel{\text{def}}{=} (s + V(s, \mathbf{y}), \Phi(V(s, \mathbf{y}); s, \mathbf{y})).$$

From here on, we define

$$(5.9) \quad \mathcal{Q}_\mathbf{D} \stackrel{\text{def}}{=} \Psi(\mathbf{D}^+) \subset \partial\mathcal{Q},$$

and we note that $\mathcal{Q}_\mathbf{D} \cap \mathbb{D} = \emptyset$ since $\mathcal{Q}$ and $\mathbb{D}$ are disjoint by definition.

The loss components are:

$$(5.10) \quad J_{\text{HJB}}(\theta) \stackrel{\text{def}}{=} \int_{\mathbf{D}^+} |\partial_s \mathbf{V}_\theta + \mathbf{G} \cdot \nabla_{\mathbf{y}} \mathbf{V}_\theta + 1|^2 d\mu(s, \mathbf{y}),$$

$$(5.11) \quad J_{\partial\mathcal{Q}}(\theta) \stackrel{\text{def}}{=} \int_{\mathcal{Q}_\mathbf{D}} |\mathbf{V}_\theta(s, \mathbf{y})|^2 d\sigma(s, \mathbf{y}),$$

Here $\mu = \mathcal{H}^3 \llcorner_{\mathbf{D}^+}$ and $\sigma = \mathcal{H}^2 \llcorner_{\mathcal{Q}_\mathbf{D}}$ denote, respectively, the Lebesgue (volume) measure on $\mathbf{D} \subset \mathbb{R}^3$ and the surface (Hausdorff) measure on the two-dimensional collision boundary $\mathcal{Q}_\mathbf{D} \subset \partial\mathcal{Q}$; all convergence results below are stated for these continuum objectives. In implementation, the integrals are approximated by Monte-Carlo quadrature over N collocation points drawn from a sampling measure equivalent to μ (resp. σ), so that the empirical losses $\hat{J}_N \rightarrow J$ as $N \rightarrow \infty$; we carry out the analysis at the continuum level, where a vanishing objective forces the residual to vanish almost everywhere.

The augmented training loss is

$$(5.12) \quad J_{\text{total}}(\theta) \stackrel{\text{def}}{=} \lambda_{\text{HJB}} J_{\text{HJB}}(\theta) + \lambda_{\partial\mathcal{Q}} J_{\partial\mathcal{Q}}(\theta),$$

with weights $\lambda_{\text{HJB}}, \lambda_{\partial\mathcal{Q}} > 0$.

The loss component J_{HJB} enforces the strong fixed point condition, which by [Remark 5.3](#) is equivalent to the HJB equation and uniquely selects $\mathbf{V}$. $J_{\partial\mathcal{Q}}$ enforces the boundary condition $\mathbf{V}|_{\mathcal{Q}} = 0$. Together, $J_{\text{HJB}} + J_{\partial\mathcal{Q}} = 0$ uniquely characterizes $\mathbf{V}$.

Theorem 5.6 (Two-sided convergence via PINN). *Let $\mathbb{D}$ be defined as in [Equation \(4.15\)](#) and $\mathbf{D} \Subset \mathbb{D}$ be a compact set. Let in addition, $\mathbf{V}_\theta \in C^1(\mathbb{D}) \cap C(\bar{\mathbb{D}})$. Define the HJB residual and boundary residual:*

$$(5.13) \quad r_{\text{HJB}}(s, \mathbf{y}) \stackrel{\text{def}}{=} \partial_s \mathbf{V}_\theta(s, \mathbf{y}) + \mathbf{G}(s, \mathbf{y}) \cdot \nabla_{\mathbf{y}} \mathbf{V}_\theta(s, \mathbf{y}) + 1, \quad r_{\partial\mathcal{Q}} \stackrel{\text{def}}{=} \mathbf{V}_\theta|_{\mathcal{Q}_D}.$$

If $J_{\text{HJB}}(\theta) + J_{\partial\mathcal{Q}}(\theta) \leq \delta$, then:

$$(5.14) \quad \|\mathbf{V}_\theta - \mathbf{V}\|_{L^2(\mathbf{D})} \leq C_{\mathbf{D}} \sqrt{\delta},$$

where $C_{\mathbf{D}}$ depends only on $\mathbf{D}$, T_h , and the OVM flow Φ .

Proof. Before we proceed with the details of the proof of this theorem, let us establish the high level intuition. In particular, the L^2 stability estimate in this theorem is based on three theoretical facts:

- *Uniqueness of viscosity solutions (comparison principle).* The HJB equation (5.5) with BC $\mathbf{V}|_{\mathcal{Q}} = 0$ has a unique continuous viscosity solution on $\mathbb{D}$, namely $\mathbf{V} = \tau$ (Crandall-Lions [13]). The theorem states a *stability estimate*: C^1 functions with small PDE residual are forced close to $\mathbf{V}$.
- *L^2 stability via characteristics.* For first-order transport equations, the method of characteristics provides a natural L^2 stability estimate; the error propagates along OVM characteristics and the co-area formula converts the pointwise error bound into an $L^2(\mathbf{D})$ bound.
- *Density of C^1 in L^2 .* The approximation class $\{\mathbf{V}_\theta \in C^1(\mathbb{D})\}$ is dense in $L^2(\mathbf{D})$ (standard Sobolev density). This ensures the approximation problem is well-posed; the infimum of $\|\mathbf{V}_\theta - \mathbf{V}\|_{L^2(\mathbf{D})}$ over smooth networks is zero. While this is very insightful, we will not use this fact in our proof below. We rather consider this as justification of the proof.

Now we establish the proof. Let $e \stackrel{\text{def}}{=} \mathbf{V}_\theta - \mathbf{V}$. We derive a pointwise bound on e via the method of characteristics, then integrate over $\mathbf{D}$.

Error dynamics along characteristics. Fix $(s, \mathbf{y}) \in \mathbf{D}$ and let $\nu \stackrel{\text{def}}{=} \mathbf{V}(s, \mathbf{y})$. Differentiate e along the OVM characteristic $\kappa \mapsto (s + \kappa, \Phi(\kappa; s, \mathbf{y}))$:

$$\begin{aligned} \frac{d}{d\kappa} e(s + \kappa, \Phi(\kappa; s, \mathbf{y})) &= \partial_s \mathbf{V}_\theta + \mathbf{G} \cdot \nabla \mathbf{V}_\theta - \underbrace{\frac{d}{d\kappa} \left(\mathbf{V}(s + \kappa, \Phi(\kappa; s, \mathbf{y})) \right)}_{=-1 \text{ by DPP (Eq. (5.3))}} \\ &= r_{\text{HJB}}(s + \kappa, \Phi(\kappa; s, \mathbf{y})). \end{aligned}$$

Integration along the characteristic. Noting that $\mathbf{V}_\theta \in C^1(\mathbb{D}) \cap C(\bar{\mathbb{D}})$, by integrating from $\kappa = 0$ to $\kappa = \nu$ (when the characteristic reaches $\mathcal{Q}$):

$$(5.15) \quad \begin{aligned} e(s, \mathbf{y}) &= e(s + \nu, \Phi(\nu; s, \mathbf{y})) - \int_0^\nu r_{\text{HJB}}(s + \kappa, \Phi(\kappa; s, \mathbf{y})) d\kappa \\ &= \underbrace{\mathbf{V}_\theta(s + \nu, \Phi(\nu; s, \mathbf{y}))}_{=r_{\partial\mathcal{Q}}} - \int_0^\nu r_{\text{HJB}}(s + \kappa, \Phi(\kappa; s, \mathbf{y})) d\kappa. \end{aligned}$$

Here we used $\mathbf{V}|_{\partial\mathcal{Q}} = 0$.

Pointwise bound. By the triangle inequality and Cauchy–Schwarz applied to (5.15):

$$(5.16) \quad |e(s, \mathbf{y})|^2 \leq 2 |\mathbf{V}_\theta(s + \nu, \Phi(\nu; s, \mathbf{y}))|^2 + 2\nu \int_0^\nu |r_{\text{HJB}}(s + \kappa, \Phi(\kappa; s, \mathbf{y}))|^2 d\kappa.$$

L^2 -bound on D . Integrate (5.16) over $(s, \mathbf{y}) \in D$ with respect to μ

$$(5.17) \quad \|e\|_{L^2(D)}^2 \leq 2 I_1 + 2T_h I_2,$$

where,

$$(5.18) \quad I_1 \stackrel{\text{def}}{=} \int_D |\mathbf{V}_\theta(s + \mathbf{V}(s, \mathbf{y}), \Phi(\mathbf{V}(s, \mathbf{y}); s, \mathbf{y}))|^2 d\mu(s, \mathbf{y}),$$

$$(5.19) \quad I_2 \stackrel{\text{def}}{=} \int_D \int_0^\nu |r_{\text{HJB}}(s + \kappa, \Phi(\kappa; s, \mathbf{y}))|^2 d\kappa d\mu(s, \mathbf{y}),$$

where in I_1 , the argument $(s + \nu, \Phi(\nu; s, \mathbf{y})) \in \mathcal{Q}$ for every $(s, \mathbf{y}) \in D$, so I_1 is an integral of $\mathbf{V}_\theta$ evaluated on $\mathcal{Q}$.

To show proper bounds on Equations (5.18)–(5.19), we need to employ Federer co-area formalism. In particular, this framework helps us to shift from integration over D to integration along the trajectory of the OVM-dynamics where the analytical properties of dynamical model can help bounding the integral terms (see Equation (5.58)).

Let us briefly collect some results established in Section 4 for consistency of notations and accessibility.

Lemma 5.7 (Regularity of the OVM flow). *Let $\mathbf{a}_\ell \in L^\infty([0, T_h])$ and let $\Phi(\kappa; s, \mathbf{y})$ denote the OVM flow with initial condition $(s, \mathbf{y}) \in \mathbb{D}$.*

The following regularity properties hold within this interval:

- (i) **Lipschitz in κ .** *Since $\|\mathbf{G}\|_{L^\infty(\mathbf{R})} \leq M < \infty$ (see Equation (4.7)) where $\mathbf{R}$ is defined as in Equation (4.5), for any $\kappa_1, \kappa_2 \in [0, \mathbf{V}(s, \mathbf{y})]$:*

$$(5.20) \quad |\Phi(\kappa_1; s, \mathbf{y}) - \Phi(\kappa_2; s, \mathbf{y})| \leq M |\kappa_1 - \kappa_2|.$$

- (ii) **Differentiable a.e. in κ .** *For any $(s, \mathbf{y}) \in D$, by Rademacher’s theorem applied to the Lipschitz map $\kappa \mapsto \Phi(\kappa; s, \mathbf{y})$, the ODE*

$$(5.21) \quad \frac{d}{d\kappa} \Phi(\kappa; s, \mathbf{y}) = \mathbf{G}(s + \kappa, \Phi(\kappa; s, \mathbf{y})) \quad \text{a.e. in } \kappa \in [0, \mathbf{V}(s, \mathbf{y})]$$

holds in the Carathéodory sense. At $\kappa = \mathbf{V}(s, \mathbf{y})$, only the left derivative exists. In particular, the left derivative at $\kappa = \nu$ will be in the form of

$$\lim_{\kappa \rightarrow \nu^-} \frac{\Xi(\kappa; s, \mathbf{y}) - \Xi(\nu; s, \mathbf{y})}{\kappa - \nu} = \lim_{\kappa \rightarrow \nu^-} \Lambda(\kappa; s, \mathbf{y}) = \Lambda(\nu; s, \mathbf{y}),$$

- (iii) **Lipschitz in initial datum.** *For two initial conditions $(s_1, \mathbf{y}_1), (s_2, \mathbf{y}_2) \in D$, both flows are defined only up to their respective hitting times $\mathbf{V}(s_1, \mathbf{y}_1)$ and $\mathbf{V}(s_2, \mathbf{y}_2)$. By Gronwall’s inequality, for any $\kappa \in [0, \min(\mathbf{V}(s_1, \mathbf{y}_1), \mathbf{V}(s_2, \mathbf{y}_2))]$:*

$$(5.22) \quad |\Phi(\kappa; s_1, \mathbf{y}_1) - \Phi(\kappa; s_2, \mathbf{y}_2)| \leq C_F (|\mathbf{y}_1 - \mathbf{y}_2| + |s_1 - s_2|),$$

where C_F is defined as in Equation (4.9).

Lemma 5.8 (Endpoint map and its image). *For $\mathbf{D} \in \mathbb{D}$ compact, the endpoint map Ψ is defined as in Equation (5.8). Then Ψ is Lipschitz on $\mathbf{D}$ jointly in $(s, \mathbf{y})$. In particular, for any $z_1 = (s_1, \mathbf{y}_1), z_2 = (s_2, \mathbf{y}_2) \in \mathbf{D}$,*

$$(5.23) \quad |\Psi(z_1) - \Psi(z_2)| \leq C_\Psi(\mathbf{D}) |z_1 - z_2|,$$

where $C_\Psi(\mathbf{D})$ depends on $M = \|\mathbf{G}\|_{L^\infty(\mathbb{R})}$, on $L_\tau(\mathbf{D})$ (Lipschitz constant of $\mathbf{V}$, Thm. 4.13), and on the OVM flow stability constant C_F .

Proof of Lem. 5.8. Write $\nu_i \stackrel{\text{def}}{=} \mathbf{V}(z_i)$. For the time component,

$$|(s_1 + \nu_1) - (s_2 + \nu_2)| \leq |s_1 - s_2| + |\mathbf{V}(z_1) - \mathbf{V}(z_2)| \leq (1 + L_\tau(\mathbf{D})) |z_1 - z_2|,$$

using $\mathbf{V} \in \mathcal{W}^{1,\infty}(\mathbf{D})$ (Thm. 4.13). For the space component, add and subtract $\Phi(\nu_2; s_1, \mathbf{y}_1)$:

$$|\Phi(\nu_1; s_1, \mathbf{y}_1) - \Phi(\nu_2; s_2, \mathbf{y}_2)| \leq \underbrace{|\Phi(\nu_1; s_1, \mathbf{y}_1) - \Phi(\nu_2; s_1, \mathbf{y}_1)|}_{\leq M |\nu_1 - \nu_2| \text{ (Eq.(5.20)(i))}} + \underbrace{|\Phi(\nu_2; s_1, \mathbf{y}_1) - \Phi(\nu_2; s_2, \mathbf{y}_2)|}_{\leq C_F(|s_1 - s_2| + |\mathbf{y}_1 - \mathbf{y}_2|) \text{ (Eq.(5.22))}}.$$

Since $|\nu_1 - \nu_2| \leq L_\tau(\mathbf{D}) |z_1 - z_2|$, both components are bounded by a constant multiple of $|z_1 - z_2|$, giving (5.23) with $C_\Psi(\mathbf{D}) = 1 + L_\tau(\mathbf{D}) + M L_\tau(\mathbf{D}) + C_F$. $\blacksquare$

In co-area formula that we will be using to bound the I_1 and I_2 integrals (see Equation (5.58)) the Jacobian term $J_2\Psi$ will show up in the denominator. In the following section, we show that $J_2\Psi$ is lower bounded by a positive constant over any compact set.

Theorem 5.9 ($J_2\Psi \geq c_D > 0$). *Let $\mathbf{D} \in \mathbb{D}$ be compact. The 2-dimensional Jacobian of the endpoint map $\Psi : \mathbf{D} \subset \mathbb{R}^3 \rightarrow \mathcal{Q}_D \subset \mathbb{R}^2$ satisfies:*

$$(5.24) \quad J_2\Psi(s, \mathbf{y}) \stackrel{\text{def}}{=} \{\det(D\Psi(s, \mathbf{y})D\Psi(s, \mathbf{y})^\top)\}^{\frac{1}{2}} \geq c_D > 0 \quad \text{a.e. on } \mathbf{D},$$

where $c_D > 0$ depends on $\mathbf{D}$ but not on θ , and $c_D \rightarrow 0$ as $\mathbf{D} \rightarrow \partial\mathbb{D}$.

Proof. We proceed in two steps. First, we show that $\text{rank}(D\Psi) \leq 2$ and in then we show that in fact this result holds in equality form and we bound the error integrals.

Step 1: $\tilde{\mathbf{G}} \in \ker(D\Psi)$ for a.e. $(s, \mathbf{y}) \in \mathbf{D}$.

Using local coordinates (s, Ξ, Λ) on $\mathbf{D}$ and (s_*, Λ_*) on $\partial\mathcal{Q}$ (since $\Xi = 0$ on $\partial\mathcal{Q}$), we write $\tilde{\Psi} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ for the coordinate representation of Ψ .

Let us define $\mathbf{V}_{\min}$ by

$$(5.25) \quad \mathbf{V}_{\min}(\mathbf{D}) \stackrel{\text{def}}{=} \min_{(s, \mathbf{y}) \in \mathbf{D}} \mathbf{V}(s, \mathbf{y}) > 0.$$

Since $\mathbb{D}$ is open and $\mathbf{D} \in \mathbb{D}$ is compact, we have $d_D \stackrel{\text{def}}{=} \text{dist}(\mathbf{D}, \partial\mathbb{D}) > 0$. Fix

$$(5.26) \quad 0 < \delta < \min\left\{\frac{1}{2}\mathbf{V}_{\min}(\mathbf{D}), \frac{d_D}{2M}\right\},$$

with $M = \|\mathbf{G}\|_{L^\infty}$ as in (4.18), and following Equation (5.7) set

$$(5.27) \quad \mathbf{D}^{\pm, \delta} \stackrel{\text{def}}{=} \{(s + \kappa, \Phi(\kappa; s, \mathbf{y})) : (s, \mathbf{y}) \in \mathbf{D}, \kappa \in [-\delta, \delta]\},$$

the two-sided tube swept out by the OVM characteristics through $\mathbf{D}$ over elapsed times $|\kappa| \leq \delta$.

The tube is compact, being the image of the compact $\mathbf{D} \times [-\delta, \delta]$ under the continuous map $((s, \mathbf{y}), \kappa) \mapsto (s + \kappa, \Phi(\kappa; s, \mathbf{y}))$. It lies in $\mathbb{D}$; in particular, integrating (4.6) gives the displacement bound

$$|(s + \kappa, \Phi(\kappa; s, \mathbf{y})) - (s, \mathbf{y})| \leq (1 + M)|\kappa| \leq \frac{d_{\mathbf{D}}}{2} \cdot \frac{1+M}{M}$$

for $|\kappa| \leq \delta$, so after shrinking δ if necessary each point of the tube lies within $d_{\mathbf{D}}$ of $\mathbf{D}$ and hence in $\mathbb{D}$. Moreover, by the DPP every point of $\mathbf{D}^{\pm, \delta}$ has remaining hitting time at least $V_{\min}(\mathbf{D}) - \delta > 0$, so the tube is bounded away from $\partial \mathcal{Q}$ as well. Hence

$$(5.28) \quad \mathbf{D}^{\pm, \delta} \Subset \mathbb{D},$$

By Thm. 4.13 together with Lem. 5.7(iii), Ψ (and hence $\tilde{\Psi}$) is Lipschitz on $\mathbf{D}^{\pm, \delta}$ with a constant $C_{\Psi}(\mathbf{D}^{\pm, \delta}) < \infty$ (see Equation (5.23)) that is *uniform* in $(s, \mathbf{y}) \in \mathbf{D}$ and therefore, $D\tilde{\Psi}$ exists a.e. by Rademacher's theorem.

Fix $(s, \mathbf{y}) \in \mathbf{D}$ and we define the end-point mapping along the OVM-dynamics by

$$(5.29) \quad h(\epsilon) \stackrel{\text{def}}{=} \tilde{\Psi}(s + \epsilon, \Phi(\epsilon; s, \mathbf{y})), \quad \epsilon \in [0, \delta].$$

Of course, this map is well-defined as from Equation (5.8) and the definition δ , it follows that $(s + \epsilon, \Phi(\epsilon; s, \mathbf{y})) \in \mathbf{D}^{+, \delta} \Subset \mathbb{D}$.

By the DPP (Lem. 5.1), all points on the OVM characteristic through $(s, \mathbf{y})$ have the same collision point on $\partial \mathcal{Q}$; i.e.

$$(5.30) \quad h(\epsilon) = \tilde{\Psi}(s, \mathbf{y}) \quad \forall \epsilon \in [0, \delta],$$

so h is *constant* on its interval of definition.

In particular, the following properties follow immediately:

- (a) $\epsilon \mapsto (s + \epsilon, \Phi(\epsilon; s, \mathbf{y}))$ is Lipschitz in ϵ with constant $1 + M$ (Lem. 5.7(i)) which maps $[0, \delta]$ into $\mathbf{D}^{\pm, \delta}$.
- (b) $\tilde{\Psi}$ is Lipschitz on $\mathbf{D}^{\pm, \delta}$ with constant $C_{\Psi}(\mathbf{D}^{\pm, \delta})$ (Lem. 5.8).
- (c) The composition h of two Lipschitz maps is Lipschitz in $\epsilon \in [0, \delta]$ with constant $(1 + M) \cdot C_{\Psi}(\mathbf{D}^{\pm, \delta})$.

By Rademacher's theorem applied to the Lipschitz map h , $h'(\epsilon)$ exists a.e. for $\epsilon \in [0, \delta]$. Since h is constant (Equation (5.30)):

$$(5.31) \quad h'(\epsilon) = 0 \quad \text{a.e. for } \epsilon \in [0, \delta].$$

The chain rule for Lipschitz compositions (valid a.e., since both maps are Lipschitz) gives:

$$(5.32) \quad h'(\epsilon) = D\tilde{\Psi}(s + \epsilon, \Phi(\epsilon; s, \mathbf{y})) \cdot \underbrace{\frac{d}{d\epsilon}(s + \epsilon, \Phi(\epsilon; s, \mathbf{y}))}_{\tilde{\mathbf{G}}(\epsilon, \Phi(\epsilon; s, \mathbf{y})) = (1, \mathbf{G}(\epsilon, \Phi(\epsilon; s, \mathbf{y})))} = 0 \quad \text{a.e.}$$

The constant $(1 + M) C_\Psi(\mathbf{D}^{\pm, \delta})$ and the interval length δ depend only on $\mathbf{D}$ and $\|\mathbf{G}\|_{L^\infty}$, not on the base point $(s, \mathbf{y}) \in \mathbf{D}$. If we had enough regularity to pass $\epsilon \rightarrow 0$, the claim of this step would have followed directly. However, due to lack of temporal continuity in $\mathbf{G}$, such limit might not be well defined.

It will be convenient to name the flow map appearing in (5.27):

$$(5.33) \quad \mathcal{F} : \mathbf{D}^{\pm, \delta} \times [-\delta, \delta] \longrightarrow \mathbb{D}, \quad \mathcal{F}((s, \mathbf{y}), \epsilon) \stackrel{\text{def}}{=} (s + \epsilon, \Phi(\epsilon; s, \mathbf{y})),$$

so that $\mathbf{D}^{\pm, \delta} = \bigcup_{|\kappa| \leq \delta} \mathcal{F}_\kappa(\mathbf{D})$; see Equation (5.34).

By Lem. 5.7(i) and (iii), $\mathcal{F}$ is Lipschitz; by uniqueness of solutions to the Carathéodory system, it is injective in $(s, \mathbf{y})$ for each fixed ϵ . In other words, the map

$$(5.34) \quad \mathcal{F}_\epsilon : \mathbf{D}^{\pm, \delta} \rightarrow \mathbb{D}, \quad \mathcal{F}_\epsilon(s, \mathbf{y}) = (s + \epsilon, \Phi(\epsilon; s, \mathbf{y})), \quad \epsilon \in [-\delta, \delta],$$

has Lipschitz inverse given by the backward flow whose Lipschitz constant is defined by

$$(5.35) \quad C_{\mathcal{F}}(\mathbf{D}^{\pm, \delta}) \stackrel{\text{def}}{=} \sqrt{2} \max\{1, C_F, M\}$$

(apply Thm. 4.4 to the time-reversed flow, whose field $-\mathbf{G}$ obeys the same bounds). Hence, for each fixed $\epsilon \in [0, \delta]$, the map $(s, \mathbf{y}) \mapsto \mathcal{F}((s, \mathbf{y}), \epsilon)$ is bi-Lipschitz onto its image and therefore maps null sets to null sets and conull sets to conull sets.

Now set

$$(5.36) \quad N \stackrel{\text{def}}{=} \left\{ (s, \mathbf{y}) \in \mathbf{D} : D\tilde{\Psi}(s, \mathbf{y}) \text{ exists and } D\tilde{\Psi}(s, \mathbf{y}) \cdot \tilde{\mathbf{G}}(s, \mathbf{y}) \neq 0 \right\}.$$

First we note that, the set N is Borel. By Rademacher's theorem $\tilde{\Psi}$ is differentiable a.e. on $\mathbf{D}^{\pm, \delta}$, its differentiability set $D_{\tilde{\Psi}}$ is Borel (a countable intersection over rational tolerances of countable unions over rational radii of closed conditions), and on $D_{\tilde{\Psi}}$ the derivative $D\tilde{\Psi}$ is a pointwise limit of continuous difference quotients, hence Borel measurable; since $\tilde{\mathbf{G}}$ is Carathéodory, the map $x \mapsto D\tilde{\Psi}(x) \cdot \tilde{\mathbf{G}}(x)$ is Borel and N is the preimage of $\mathbb{R}^2 \setminus \{0\}$ under it, intersected with $\mathbf{D}$.

We would like to show that $\mathcal{L}^3(N) = 0$. To do so, we consider the incidence set

$$(5.37) \quad A \stackrel{\text{def}}{=} \mathcal{F}^{-1}(N) = \left\{ ((s, \mathbf{y}), \epsilon) \in \mathbf{D}^{\pm, \delta} \times [-\delta, \delta] : \mathcal{F}((s, \mathbf{y}), \epsilon) \in N \right\},$$

measurable because N is Borel (the differentiability set is Borel) and $\mathcal{F}$ is continuous.

Slicing in ϵ . Fix $(s, \mathbf{y}) \in \mathbf{D}^{\pm, \delta}$. By (5.32) the integrand vanishes for a.e. ϵ , i.e. $\mathcal{F}((s, \mathbf{y}), \epsilon) \notin N$ for a.e. ϵ , so the corresponding slice of A is $\mathcal{L}^1$ -null:

$$(5.38) \quad \mathcal{L}^1 \left(\underbrace{\left\{ \epsilon \in [-\delta, \delta] : \mathcal{F}((s, \mathbf{y}), \epsilon) \in N \right\}}_{\stackrel{\text{def}}{=} A_{(s, \mathbf{y})}} \right) = 0.$$

Every characteristic therefore meets N in a null set of times, and Fubini gives

$$(5.39) \quad (\mathcal{L}^3 \otimes \mathcal{L}^1)(A) = \int_{\mathbf{D}^{\pm, \delta}} \mathcal{L}^1(A_{(s, \mathbf{y})}) d(s, \mathbf{y}) = 0.$$

Slicing in $(s, \mathbf{y})$. Applying Fubini to (5.39) in the opposite order, the ϵ -slices of A are $\mathcal{F}_\epsilon^{-1}(N) \cap \mathbf{D}^{\pm, \delta}$, whence

$$(5.40) \quad \mathcal{L}^3\left(\mathcal{F}_\epsilon^{-1}(N) \cap \mathbf{D}^{\pm, \delta}\right) = 0 \quad \text{for a.e. } \epsilon \in [-\delta, \delta].$$

Fix one such ϵ_0 . Since $\mathcal{F}_{\epsilon_0}$ is bi-Lipschitz onto its image, it maps null sets to null sets, so (5.40) gives

$$(5.41) \quad \mathcal{L}^3\left(N \cap \mathcal{F}_{\epsilon_0}(\mathbf{D}^{\pm, \delta})\right) = 0$$

One admissible shift already covers $\mathbf{D}$. By the definition of $\mathbf{D}^{\pm, \delta}$ and the evolution property of the flow, $\mathcal{F}_{\epsilon_0} \circ \mathcal{F}_\epsilon = \mathcal{F}_{\epsilon_0 + \epsilon}$, so

$$(5.42) \quad \mathcal{F}_{\epsilon_0}(\mathbf{D}^{\pm, \delta}) = \bigcup_{\epsilon \in [-\delta, \delta]} \mathcal{F}_{\epsilon_0 + \epsilon}(\mathbf{D}) = \bigcup_{\epsilon' \in [\epsilon_0 - \delta, \epsilon_0 + \delta]} \mathcal{F}_{\epsilon'}(\mathbf{D}) \supseteq \mathcal{F}_0(\mathbf{D}) = \mathbf{D},$$

the inclusion holding because $0 \in [\epsilon_0 - \delta, \epsilon_0 + \delta]$ whenever $|\epsilon_0| \leq \delta$. Combining (5.41) with (5.42),

$$(5.43) \quad \mathcal{L}^3(N \cap \mathbf{D}) = 0.$$

By Equation (5.36), therefore, we immediately have

$$(5.44) \quad D\tilde{\Psi}(s, \mathbf{y}) \cdot \tilde{\mathbf{G}}(s, \mathbf{y}) = 0 \quad \text{for a.e. } (s, \mathbf{y}) \in \mathbf{D}.$$

Furthermore, $\tilde{\mathbf{G}} \neq 0$ (its first component equals 1),

$$(5.45) \quad \dim(\ker(D\tilde{\Psi}(s, \mathbf{y}))) \geq 1 \Rightarrow \text{rank } D\tilde{\Psi}(s, \mathbf{y}) \leq 2 \quad \text{a.e. on } \mathbf{D}.$$

Assumption 5.10 (C^1 state dependence; additive rough forcing). *The OVM field of Equation (4.1) decomposes as*

$$\mathbf{G}(s, \Xi, \Lambda) = (\Lambda, f_{\text{OVM}}(\Xi, \Lambda) + b(s)), \quad f_{\text{OVM}}(\Xi, \Lambda) \stackrel{\text{def}}{=} -\alpha(\mathcal{V}(\Xi) + \Lambda), \quad b(s) \stackrel{\text{def}}{=} \mathbf{a}_\ell(s) + \alpha \zeta_\ell(s),$$

where $f_{\text{OVM}} \in C^1(\mathbb{R}_{>0} \times \mathbb{R})$ with $\|Df_{\text{OVM}}\|_{L^\infty(\mathbf{K})} < \infty$ on every compact $\mathbf{K} \Subset \mathbb{D}$, and the forcing $b \in L^\infty((0, T_h); \mathbb{R})$ is a function of time alone.

Now, we show Asm. 5.10 always hold true.

Remark 5.11 (Verification for the OVM model; the key structural fact). *For the optimal-velocity function $\mathcal{V}(\Xi) = \tanh(\Xi - 2) + \tanh 2$, the state Jacobian of the OVM field Equation (4.1) is*

$$(5.46) \quad \partial_{\mathbf{x}} \mathbf{G}(s, \Xi, \Lambda) = \begin{pmatrix} 0 & 1 \\ -\alpha \mathcal{V}'(\Xi) & -\alpha \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\alpha \text{sech}^2(\Xi - 2) & -\alpha \end{pmatrix},$$

which is continuous in (Ξ, Λ) and bounded ($0 < \text{sech}^2 \leq 1$), and is independent of the entire time forcing $b(s) = \mathbf{a}_\ell(s) + \alpha \zeta_\ell(s)$; both the leader acceleration $\mathbf{a}_\ell$ and the leader velocity ζ_ℓ enter Equation (4.1) additively in time and are annihilated by $\partial_{\mathbf{x}} = \partial_{(\Xi, \Lambda)}$. Thus Asm. 5.10 holds, and the rough forcing $b \in L^\infty$ —which prevents Φ from being C^1 in the elapsed time κ —is invisible to the linearization in the initial state. The flow will therefore be shown C^1 in $\mathbf{y}$ (Lem. 5.20) while remaining only Lipschitz in κ (Lem. 5.7(i)).

Lemma 5.12 (Reachable tube; uniform constants). *Let $\mathbf{D} \in \mathbb{D}$ be compact and set $\kappa_*(\mathbf{D}) \stackrel{\text{def}}{=} \frac{1}{2}V_{\min}(\mathbf{D})$. Define the reachable tube*

$$(5.47) \quad \mathcal{R}(\mathbf{D}) \stackrel{\text{def}}{=} \left\{ (s + \kappa, \Phi(\kappa; s, \mathbf{y})) : (s, \mathbf{y}) \in \mathbf{D}, 0 \leq \kappa \leq \kappa_*(\mathbf{D}) \right\}.$$

Then, (i) $\mathcal{R}(\mathbf{D})$ is compact; (ii) $\mathcal{R}(\mathbf{D}) \in \mathbb{D}$; and (iii) there is $\rho = \rho(\mathbf{D}) > 0$ such that the closed ρ -neighbourhood $\mathcal{R}_\rho(\mathbf{D}) \stackrel{\text{def}}{=} \{z : \text{dist}(z, \mathcal{R}(\mathbf{D})) \leq \rho\}$ still satisfies $\mathcal{R}_\rho(\mathbf{D}) \in \mathbb{D}$. Consequently

$$(5.48) \quad G_1(\mathbf{D}) \stackrel{\text{def}}{=} \|\partial_{\mathbf{y}} \mathbf{G}\|_{L^\infty(\mathcal{R}_\rho(\mathbf{D}))} < \infty, \quad C_{\text{Gr}}(\mathbf{D}) \stackrel{\text{def}}{=} e^{G_1(\mathbf{D})} V_{\min}(\mathbf{D}),$$

and $\partial_{\mathbf{y}} \mathbf{G}$ is uniformly continuous on $\mathcal{R}_\rho(\mathbf{D})$, with modulus of continuity $\omega = \omega_{\mathbf{D}}$ satisfying $\omega(\rho') \rightarrow 0$ as $\rho' \rightarrow 0$.

Proof. Part (i). $\mathcal{R}(\mathbf{D})$ is the image of the compact set $\mathbf{D} \times [0, \kappa_*(\mathbf{D})]$ under the mapping $(s, \mathbf{y}, \kappa) \mapsto (s + \kappa, \Phi(\kappa; s, \mathbf{y}))$, which is continuous by [Lem. 5.7\(i\),\(iii\)](#) (Lipschitz in κ with constant M , Lipschitz in $\mathbf{y}$); continuous images of compact sets are compact.

Part (ii). Let $(s, \mathbf{y}) \in \mathbf{D}$ and $\kappa \leq \kappa_*(\mathbf{D})$. By the evolution property of the flow ([Lem. 5.1](#)), $\mathbf{V}(s + \kappa, \Phi(\kappa; s, \mathbf{y})) = \mathbf{V}(s, \mathbf{y}) - \kappa$. Since $\mathbf{V}(s, \mathbf{y}) \geq V_{\min}(\mathbf{D})$ on $\mathbf{D}$ and $\kappa \leq \frac{1}{2}V_{\min}(\mathbf{D})$, every point of $\mathcal{R}(\mathbf{D})$ has hitting time at least $\frac{1}{2}V_{\min}(\mathbf{D}) > 0$; in particular $\Xi > 0$ there, no trajectory has reached collision, and $\mathcal{R}(\mathbf{D})$ is bounded away from the collision boundary of $\mathbb{D}$ ([Equation \(4.15\)](#)). Together with (i) this gives $\mathcal{R}(\mathbf{D}) \in \mathbb{D}$.

Part (iii). By (i)–(ii), $\mathcal{R}(\mathbf{D})$ is compact and $\text{dist}(\mathcal{R}(\mathbf{D}), \partial\mathbb{D}) > 0$; let us consider $\rho \stackrel{\text{def}}{=} \frac{1}{2} \text{dist}(\mathcal{R}(\mathbf{D}), \partial\mathbb{D})$. Then $\mathcal{R}_\rho(\mathbf{D})$ is compact and contained in $\mathbb{D}$. Finiteness in (5.48) and uniform continuity of $\partial_{\mathbf{y}} \mathbf{G}$ now follow from [Asm. 5.10](#) (continuity of $\partial_{\mathbf{y}} \mathbf{G}$, bounded on compacts of $\mathbb{D}$) and compactness of $\mathcal{R}_\rho(\mathbf{D})$. ■

Lemma 5.13 (Finite-difference lower bound for the state flow). *Under [Asm. 5.10](#), for every $\kappa \in [0, V_{\min}(\mathbf{D})]$ and every s with $(s, \mathbf{y}_1), (s, \mathbf{y}_2) \in \mathbf{D}$,*

$$(5.49) \quad |\Phi(\kappa; s, \mathbf{y}_1) - \Phi(\kappa; s, \mathbf{y}_2)| \geq C_{\text{Gr}}(\mathbf{D})^{-1} |\mathbf{y}_1 - \mathbf{y}_2|.$$

Proof. Fix s and κ , and set

$$w \stackrel{\text{def}}{=} \Phi(\kappa; s, \mathbf{y}_1) - \Phi(\kappa; s, \mathbf{y}_2), \quad u \stackrel{\text{def}}{=} \mathbf{y}_1 - \mathbf{y}_2.$$

The states $\Phi(\kappa; s, \mathbf{y}_i)$ are attained at the absolute time $s + \kappa$, so the backward flow returning them to time s is

$$(5.50) \quad \Theta_{s, \kappa} : \zeta \mapsto \Phi(-\kappa; s + \kappa, \zeta) = \mathbf{x}(s; s + \kappa, \zeta),$$

where the initial-time slot carries $s + \kappa$ because that is the time at which ζ is given. By uniqueness of solutions of the Carathéodory system, $\Theta_{s, \kappa}(\Phi(\kappa; s, \mathbf{y}_i)) = \mathbf{y}_i$ for $i = 1, 2$, whence

$$\Theta_{s, \kappa}(\Phi(\kappa; s, \mathbf{y}_1)) - \Theta_{s, \kappa}(\Phi(\kappa; s, \mathbf{y}_2)) = u.$$

We show that the map $\Theta_{s,\kappa}$ is Lipschitz with constant $C_{\text{Gr}}(\mathbf{D})$. To this end, applying Gronwall's inequality to the difference of two solutions of the time-reversed system, whose state Jacobian $-\partial_{\mathbf{y}}\mathbf{G}$ is forcing-free by (5.46), continuous in the state and bounded by $G_1(\mathbf{D})$ on the reachable tube, gives $\text{Lip}(\Theta_{s,\kappa}) \leq e^{G_1(\mathbf{D})\kappa} \leq C_{\text{Gr}}(\mathbf{D})$ for $\kappa \leq \mathbf{V}_{\min}(\mathbf{D})$. Hence $|u| \leq C_{\text{Gr}}(\mathbf{D})|w|$, which is (5.49). $\blacksquare$

Remark 5.14 (Reverse direction). *A mean-value inequality converts an upper bound on a derivative into an upper bound on a difference; it does not run backwards. The lower bound (5.49) is obtained instead as an upper bound on the difference for the inverse (backward) flow. Both directions of (5.68) rest solely on the forcing-free coefficient $\partial_{\mathbf{y}}\mathbf{G}$ of (5.46).*

Lemma 5.15 (Transverse bi-Lipschitz estimate for Ψ). *Let $\mathbf{D} \in \mathbb{D}$ be compact and let W_0 be the coordinate plane $W_0 = \{(w_0, \mathbf{w}) \in \mathbb{R}^3 : w_0 = 0\} = \{0\} \times \mathbb{R}^2$. Then, with*

$$(5.51) \quad \lambda_{\mathbf{D}} \stackrel{\text{def}}{=} \frac{C_{\text{Gr}}(\mathbf{D})^{-1}}{2 \max\{1, M\}} > 0, \quad \Lambda_{\mathbf{D}} \stackrel{\text{def}}{=} C_{\Psi}(\mathbf{D}) < \infty,$$

one has, for every $z_1 = (s_1, \mathbf{y}_1) \in \mathbf{D}$ and every $z_2 = (s_1, \mathbf{y}_2) \in \mathbf{D}$ lying in the same time slice (that is, $z_1 - z_2 \in W_0$),

$$(5.52) \quad \lambda_{\mathbf{D}} |P_{W_0}[z_1 - z_2]| \leq |\Psi(z_1) - \Psi(z_2)| \leq \Lambda_{\mathbf{D}} |z_1 - z_2|.$$

Proof. The upper bound is the Lipschitz continuity of Ψ (Lem. 5.8). For the lower bound, write $\nu_i \stackrel{\text{def}}{=} \mathbf{V}(z_i)$, and set

$$t \stackrel{\text{def}}{=} |\mathbf{y}_1 - \mathbf{y}_2| = |P_{W_0}[z_1 - z_2]|, \quad \eta \stackrel{\text{def}}{=} |\nu_1 - \nu_2|,$$

noting that $z_1 - z_2 \in W_0$ gives $s_1 = s_2$, so that $t = |z_1 - z_2|$. In its time and space components,

$$\Psi(z_i) = (s_1 + \nu_i, \Phi(\nu_i; s_1, \mathbf{y}_i)),$$

and since any product norm on $\mathbb{R} \times \mathbb{R}^2$ dominates each block,

$$(5.53) \quad |\Psi(z_1) - \Psi(z_2)| \geq \eta, \quad |\Psi(z_1) - \Psi(z_2)| \geq |\Phi(\nu_1; s_1, \mathbf{y}_1) - \Phi(\nu_2; s_1, \mathbf{y}_2)|.$$

The spatial block. Insert the aligned reference point $\Phi(\nu_1; s_1, \mathbf{y}_2)$, which differs from $\Phi(\nu_2; s_1, \mathbf{y}_2)$ only in the elapsed time. Since $s_1 = s_2$, no change of initial time is involved and the only transport error is the elapsed-time shift, which by (4.7) is at most $M\eta$:

$$(5.54) \quad |\Phi(\nu_1; s_1, \mathbf{y}_1) - \Phi(\nu_2; s_1, \mathbf{y}_2)| \geq |\Phi(\nu_1; s_1, \mathbf{y}_1) - \Phi(\nu_1; s_1, \mathbf{y}_2)| - M\eta \geq C_{\text{Gr}}(\mathbf{D})^{-1} t - M\eta,$$

the last step by Lem. 5.13, applicable since $\nu_1 \leq \mathbf{V}_{\min}(\mathbf{D})$ and both initial states lie in $\mathbf{D}$ at the common time s_1 .

Case split on η . Put $\eta_* \stackrel{\text{def}}{=} \frac{C_{\text{Gr}}(\mathbf{D})^{-1}}{2M} t$.

- If $\eta \geq \eta_*$, the first bound in (5.53) gives $|\Psi(z_1) - \Psi(z_2)| \geq \eta \geq \frac{C_{\text{Gr}}(\mathbf{D})^{-1}}{2M} t$.

- If $\eta < \eta_*$, then $M\eta < \frac{1}{2}C_{\text{Gr}}(\mathbf{D})^{-1}t$, and the second bound in (5.53) together with (5.54) gives $|\Psi(z_1) - \Psi(z_2)| \geq \frac{1}{2}C_{\text{Gr}}(\mathbf{D})^{-1}t$. In either case $|\Psi(z_1) - \Psi(z_2)| \geq \lambda_{\mathbf{D}} t$ with $\lambda_{\mathbf{D}}$ as in (5.51), which is the lower bound in (5.52). ■

Step 2: Positivity of the endpoint Jacobian $J_2\Psi \geq c_{\mathbf{D}} > 0$.

By Step 1 ((5.44)–(5.45)), $\text{span}\{\tilde{\mathbf{G}}(z)\} \subseteq \ker(D\tilde{\Psi}(z))$ and $\text{rank}(D\tilde{\Psi}(z)) \leq 2$ for a.e. $z \in \mathbf{D}$. We upgrade this to a strict lower bound on $J_2\Psi$ using the transverse bi-Lipschitz estimate Lem. 5.15, thereby avoiding any pointwise-in-time differentiation of Ψ .

Fix $z \in \mathbf{D}$ at which $D\tilde{\Psi}(z)$ exists (a.e., by Rademacher) and let $u \in W_0$ be a unit vector, so that $z - tu$ remains in the time slice of z . Applying (5.52) with $z_1 = z$, $z_2 = z - tu$ and letting $t \rightarrow 0^+$,

$$(5.55) \quad |D\tilde{\Psi}(z)u| = \lim_{t \rightarrow 0^+} \frac{|\Psi(z) - \Psi(z - tu)|}{t} \geq \lambda_{\mathbf{D}} |P_{W_0}u| = \lambda_{\mathbf{D}},$$

the limit existing because $D\tilde{\Psi}(z)$ does, and the constant $\lambda_{\mathbf{D}}$ of (5.51) being independent of the direction u . By homogeneity,

$$(5.56) \quad |D\tilde{\Psi}(z)u| \geq \lambda_{\mathbf{D}} |u| \quad \text{for all } u \in W_0,$$

Hence, $D\tilde{\Psi}(z)$ is bounded below, and so injective, on the two-dimensional plane W_0 (i.e. $\ker(D\tilde{\Psi}(z)) \upharpoonright_{W_0}$ is trivial). Now, we show that the kernel is exactly the flow line. First, $\dim \ker \geq 1$ since $\tilde{\mathbf{G}}(z) \in \ker(D\tilde{\Psi}(z))$ by (5.44); Secondly, if $\dim \ker \geq 2$ then, as $\ker$ and W_0 are subspaces of $\mathbb{R}^3$ of dimensions ≥ 2 and 2, and using the Grassmann dimension formula, we have that $\dim(\ker + W_0) \leq 3$ and

$$\dim(\ker \cap W_0) \geq \dim \ker + \dim W_0 - \dim(\ker + W_0) \geq 1,$$

contradicting (5.56). Therefore $\ker(D\tilde{\Psi}(z)) = \text{span}\{\tilde{\mathbf{G}}(z)\}$ and, by rank-nullity, $\text{rank}(D\tilde{\Psi}(z)) = 2$, matching the upper bound of Step 1. ■

For the rank-2 matrix $D\tilde{\Psi}(z) \in \mathbb{R}^{2 \times 3}$,

$$J_2\Psi(z) = \{\det(D\tilde{\Psi}(z) D\tilde{\Psi}(z)^\top)\}^{1/2} = \sigma_1(z) \sigma_2(z),$$

the product of the two nonzero singular values. The smallest nonzero singular value is $\sigma_2(z) = \min\{|D\tilde{\Psi}(z)v| : v \in W_z, |v| = 1\}$, the minimum being over the orthogonal complement $W_z = (\ker D\tilde{\Psi}(z))^\perp = (\text{span}\{\tilde{\mathbf{G}}(z)\})^\perp$. To transfer (5.56) from W_0 to W_z , let $v \in W_z$ with $|v| = 1$ and write $v = (v_s, \mathbf{v})$. Since $\tilde{\mathbf{G}}(z)$ has first component 1, the direct sum $\mathbb{R}^3 = \text{span}\{\tilde{\mathbf{G}}(z)\} \oplus W_0$ gives the unique decomposition

$$v = v_s \tilde{\mathbf{G}}(z) + w, \quad w \stackrel{\text{def}}{=} (0, \mathbf{v} - v_s \mathbf{G}(z)) \in W_0,$$

the coefficient being v_s by comparison of first components. As $\tilde{\mathbf{G}}(z) \in \ker(D\tilde{\Psi}(z))$ we have $D\tilde{\Psi}(z)v = D\tilde{\Psi}(z)w$. Moreover $v \perp \tilde{\mathbf{G}}(z)$ yields

$$\langle w, v \rangle = \langle v, v \rangle - v_s \langle \tilde{\mathbf{G}}(z), v \rangle = |v|^2 = 1,$$

so $1 = \langle w, v \rangle \leq |w| |v| = |w|$ by Cauchy–Schwarz. Combining with (5.56),

$$|D\tilde{\Psi}(z)v| = |D\tilde{\Psi}(z)w| \geq \lambda_D |w| \geq \lambda_D.$$

Hence $\sigma_1(z) \geq \sigma_2(z) \geq \lambda_D$ and

$$(5.57) \quad J_2\Psi(z) = \sigma_1(z)\sigma_2(z) \geq \lambda_D^2 \stackrel{\text{def}}{=} c_D > 0 \quad \text{for a.e. } z \in D.$$

which is (5.24). As $D \rightarrow \partial\mathbb{D}$ the transversality margin $\gamma_D \rightarrow 0$ (Lem. 4.9), hence $L_\tau(D) \rightarrow \infty$, $\lambda_D \rightarrow 0$ and $c_D \rightarrow 0$; this is the analytic origin of the strict requirement $D \Subset \mathbb{D}$. $\blacksquare$

Remark 5.16 (Independence of the representative of the forcing). *Every quantity used in Lems. 4.9 to 5.15 is invariant under modifying the forcing $b = \mathbf{a}_\ell - \alpha\zeta_\ell$ on an $\mathcal{L}^1$ -null set of times; the flow Φ and gap Ξ are Carathéodory integrals of $\mathbf{G}$; Cor. 4.10 integrates Λ over a window; the linearization coefficient $\partial_{\mathbf{y}}\mathbf{G}$ does not contain b (5.46); and Lems. 4.12 and 5.15 use only finite differences of Ξ and Φ . Consequently (5.57) is well posed for $b \in L^\infty$, with no continuity assumption on the leader acceleration $\mathbf{a}_\ell$.*

Bound on I_1 via the co-area formula.

We apply the Federer co-area formula [19, Thm. 3.2.12] to $\Psi : D \subset \mathbb{R}^3 \rightarrow Q_D \subset \mathbb{R}^2$ with $m = 3$, $n = 2$, $m - n = 1$. Since Ψ is Lipschitz (Thm. 4.13 and Thm. 4.4), $J_2\Psi$ exists $\mathcal{H}^3$ -a.e. by Rademacher’s theorem. Writing:

$$(5.58) \quad I_1 = \int_D |\mathbf{V}_\theta(\Psi(x))|^2 d\mu(x) = \int_D \frac{|\mathbf{V}_\theta(\Psi(x))|^2}{J_2\Psi(x)} \cdot J_2\Psi(x) d\mu(x),$$

and applying the co-area formula with $g(x) = |\mathbf{V}_\theta(\Psi(x))|^2 / J_2\Psi(x)$:

$$I_1 = \int_{Q_D} |\mathbf{V}_\theta(z)|^2 \underbrace{\int_{\Psi^{-1}(z) \cap D} \frac{1}{J_2\Psi(x)} d\ell(x)}_{(\star)} d\sigma(z),$$

where $d\ell$ denotes arc-length on the 1-dimensional fiber $\Psi^{-1}(z) \cap D$, and for $x \in \Psi^{-1}(z)$ we used $\Psi(x) = z$ to pull $|\mathbf{V}_\theta(z)|^2$ outside the inner integral.

By Thm. 5.9 (Equation (5.24)), $J_2\Psi(x) \geq c_D > 0$ for $\mathcal{H}^3$ -a.e. $x \in D$. For the length of the fibre, note first that Ψ is constant along OVM characteristics: by the dynamic programming principle (Lem. 5.1), $\Psi(\mathcal{F}_\kappa(s, \mathbf{y})) = \Psi(s, \mathbf{y})$ for every admissible κ , since flowing forward does not change the eventual collision point. Equivalently, $\tilde{\mathbf{G}} \in \ker D\tilde{\Psi}$, which is (5.44). Consequently Ψ is *not* injective, and the preimage $\Psi^{-1}(z)$ of a collision point z is a curve rather than a single state: it contains the whole characteristic through each of its points. Intersecting with D clips this curve to the portion lying in D . By Remark 5.17 below the fibre lies on a *single* characteristic, so it is the image of an elapsed-time interval $I_z \subseteq [0, V_{\max}(D)]$ under the injective Lipschitz parametrization $\kappa \mapsto (s + \kappa, \Phi(\kappa; s, \mathbf{y}))$, whose speed is $|\tilde{\mathbf{G}}| = \sqrt{1 + |\mathbf{G}|^2} \leq \sqrt{1 + M^2}$. Hence

$$(5.59) \quad \mathcal{H}^1(\Psi^{-1}(z) \cap D) = \int_{I_z} |\tilde{\mathbf{G}}(\gamma_z(r))| dr \leq \sqrt{1 + M^2} |I_z| \leq \sqrt{1 + M^2} V_{\max}(D),$$

where $V_{\max}(D) \stackrel{\text{def}}{=} \max_{(s, \mathbf{y}) \in D} V(s, \mathbf{y}) < \infty$ by continuity of V on the compact D (Thm. 4.13). $\blacksquare$

Remark 5.17 (The fibre is a single characteristic arc). The fiber $\Psi^{-1}(z) \cap \mathbf{D}$ lies on one OVM characteristic, and is the image of a single interval of elapsed times under an injective Lipschitz parametrization. Consequently (5.59) bounds its total $\mathcal{H}^1$ measure, with no multiplicity factor.

Indeed, let $x_1 = (s_1, \mathbf{y}_1)$ and $x_2 = (s_2, \mathbf{y}_2)$ belong to $\Psi^{-1}(z) \cap \mathbf{D}$, and write $z = (\sigma_z, \mathbf{w}_z)$ for the common collision point. By the definition (5.8) of Ψ ,

$$s_i + \tau(x_i) = \sigma_z, \quad \Phi(\tau(x_i); s_i, \mathbf{y}_i) = \mathbf{w}_z, \quad i = 1, 2,$$

so both trajectories pass through the same state $\mathbf{w}_z$ at the same absolute time σ_z . By uniqueness of solutions of the Carathéodory system (Thm. 4.4), backward in time from the common datum $(\sigma_z, \mathbf{w}_z)$ there is exactly one trajectory; hence x_1 and x_2 lie on it, i.e. on a single characteristic. Explicitly, writing $\mathbf{w}(r) \stackrel{\text{def}}{=} \Phi(-r; \sigma_z, \mathbf{w}_z)$ for the backward solution, we have $x_i = (\sigma_z - \tau(x_i), \mathbf{w}(\tau(x_i)))$.

The fiber is therefore contained in the curve

$$(5.60) \quad \gamma_z(r) \stackrel{\text{def}}{=} (\sigma_z - r, \mathbf{w}(r)), \quad r \geq 0,$$

and $\Psi^{-1}(z) \cap \mathbf{D} = \gamma_z(I_z)$ with $I_z \stackrel{\text{def}}{=} \{r \geq 0 : \gamma_z(r) \in \mathbf{D}\}$. The parametrisation (5.60) is injective, since its first coordinate $\sigma_z - r$ is strictly decreasing in r ; and it is Lipschitz with speed $|\tilde{\mathbf{G}}| = \sqrt{1 + |\mathbf{G}|^2} \leq \sqrt{1 + M^2}$. For an injective Lipschitz curve the one-dimensional Hausdorff measure of the image coincides with the classical arc length, whence (5.59) follows. Finally $I_z \subseteq [0, V_{\max}(\mathbf{D})]$: for $\gamma_z(r) \in \mathbf{D}$ the remaining time to collision is exactly r , so $r = \tau(\gamma_z(r)) \leq V_{\max}(\mathbf{D})$.

Remark 5.18 (I_z need not be an interval, and this does not matter). The set I_z is the set of times at which the backward trajectory (5.60) lies in $\mathbf{D}$; if $\mathbf{D}$ is not flow-invariant, the trajectory may leave and re-enter, so I_z is in general a finite or countable union of intervals rather than a single one. This affects neither (5.59) nor the resulting bound: the estimate uses only $|I_z| \leq V_{\max}(\mathbf{D})$, which holds for any measurable $I_z \subseteq [0, V_{\max}(\mathbf{D})]$, and the parametrization remains injective on all of I_z . What the argument above rules out — and what would genuinely break the estimate — is the fiber meeting several distinct characteristics, since the arc-length computation would then have to be repeated on each and $\mathcal{H}^1$ would pick up the corresponding multiplicity.

Therefore

$$(\star) \leq \frac{\sqrt{1 + M^2} V_{\max}(\mathbf{D})}{c_D} \stackrel{\text{def}}{=} C_D,$$

and

$$(5.61) \quad I_1 \leq C_D \int_{\mathcal{Q}_D} |\mathbf{V}_\theta(z)|^2 d\sigma(z) = C_D J_{\partial \mathcal{Q}}(\theta).$$

The constant $C_D = \sqrt{1 + M^2} V_{\max}(\mathbf{D})/c_D$ depends on $\mathbf{D}$ but not on θ , and diverges as $\mathbf{D} \rightarrow \partial \mathbb{D}$ (since $c_D \rightarrow 0$ while the numerator stays bounded), confirming the requirement $\mathbf{D} \Subset \mathbb{D}$.

Bound on I_2 . Let us start with some

Assumption 5.19 (Residual penalized on the forward tube). In [Def. 5.5](#) the interior loss is taken over the forward tube [\(5.7\)](#) rather than over $\mathbf{D}$ alone:

$$(5.62) \quad J_{\text{HJB}}(\theta) \stackrel{\text{def}}{=} \int_{\mathbf{D}^+} |\partial_s \mathbf{V}_\theta + \mathbf{G} \cdot \nabla_{\mathbf{y}} \mathbf{V}_\theta + 1|^2 d\mu(s, \mathbf{y}), \quad \mu = \mathcal{H}^3 \llcorner_{\mathbf{D}^+}.$$

It should be noted that $\mathcal{H}^3(\mathbf{D}^+) < \infty$, since $\mathbf{D}^+$ is contained in the bounded set $\{(s + \kappa, \Phi(\kappa; s, \mathbf{y})) : (s, \mathbf{y}) \in \mathbf{D}, 0 \leq \kappa \leq \tau_{\max}(\mathbf{D})\}$, the image of a bounded set under a Lipschitz map.

We set $\nu = \nu(s, \mathbf{y}) \stackrel{\text{def}}{=} \tau(s, \mathbf{y})$ and recall $\tau_{\max}(\mathbf{D}) \stackrel{\text{def}}{=} \max_{\mathbf{D}} \tau < \infty$, the maximum being attained by continuity of τ ([Lem. 4.5](#)) and compactness of $\mathbf{D}$. Recall also the flow map $\mathcal{F}_\kappa(s, \mathbf{y}) = (s + \kappa, \Phi(\kappa; s, \mathbf{y}))$ of [\(5.33\)](#).

Step 1: exchange the order of integration. The inner integral in [\(5.19\)](#) runs over $\kappa \in [0, \nu(s, \mathbf{y})]$, so by Tonelli's theorem (the integrand being nonnegative and measurable)

$$(5.63) \quad I_2 = \int_{\mathbf{D}} \int_0^{\nu(s, \mathbf{y})} |r_{\text{HJB}}(\mathcal{F}_\kappa(s, \mathbf{y}))|^2 d\kappa d\mu(s, \mathbf{y}) = \int_0^{\tau_{\max}(\mathbf{D})} \int_{\mathbf{D}_\kappa} |r_{\text{HJB}}(\mathcal{F}_\kappa(s, \mathbf{y}))|^2 d\mu(s, \mathbf{y}) d\kappa,$$

where $\mathbf{D}_\kappa \stackrel{\text{def}}{=} \{(s, \mathbf{y}) \in \mathbf{D} : \nu(s, \mathbf{y}) > \kappa\}$ is the (measurable, by continuity of τ) set of initial data surviving past elapsed time κ .

Step 2: change of variables at fixed κ . Fix $\kappa \in [0, \tau_{\max}(\mathbf{D})]$. On $\mathbf{D}_\kappa$ the map $\mathcal{F}_\kappa$ is a bi-Lipschitz bijection onto its image $\mathcal{F}_\kappa(\mathbf{D}_\kappa) \subset \mathbf{D}^+$; it is Lipschitz with constant $C_{\mathcal{F}}(\mathbf{D})$ by [\(5.35\)](#), injective by uniqueness of solutions of the Carathéodory system, and its inverse $\mathcal{F}_{-\kappa}$ obeys the same bound. Crucially, this is a map between sets of *equal* dimension 3, so the ordinary change of variables applies: by the area formula for injective Lipschitz maps, for every nonnegative measurable g ,

$$(5.64) \quad \int_{\mathbf{D}_\kappa} g(\mathcal{F}_\kappa(x)) d\mathcal{H}^3(x) = \int_{\mathcal{F}_\kappa(\mathbf{D}_\kappa)} \frac{g(z)}{J_3 \mathcal{F}_\kappa(\mathcal{F}_{-\kappa}(z))} d\mathcal{H}^3(z) \leq C_{\mathcal{F}}(\mathbf{D})^3 \int_{\mathcal{F}_\kappa(\mathbf{D}_\kappa)} g(z) d\mathcal{H}^3(z),$$

where the Jacobian of the bi-Lipschitz $\mathcal{F}_\kappa$ satisfies $J_3 \mathcal{F}_\kappa \geq C_{\mathcal{F}}(\mathbf{D})^{-3}$ a.e., by the two-sided Lipschitz bounds on $\mathcal{F}_{\pm\kappa}$.

Step 3: assemble. Applying [\(5.64\)](#) with $g = |r_{\text{HJB}}|^2$ and using $\mathcal{F}_\kappa(\mathbf{D}_\kappa) \subset \mathbf{D}^+$ together with [\(5.62\)](#),

$$\int_{\mathbf{D}_\kappa} |r_{\text{HJB}}(\mathcal{F}_\kappa(s, \mathbf{y}))|^2 d\mu(s, \mathbf{y}) \leq C_{\mathcal{F}}(\mathbf{D})^3 \int_{\mathbf{D}^+} |r_{\text{HJB}}(z)|^2 d\mu(z) = C_{\mathcal{F}}(\mathbf{D})^3 J_{\text{HJB}}(\theta),$$

uniformly in κ . Substituting into [\(5.63\)](#) and integrating the constant over an interval of length $\tau_{\max}(\mathbf{D})$,

$$(5.65) \quad I_2 \leq \tau_{\max}(\mathbf{D}) C_{\mathcal{F}}(\mathbf{D})^3 J_{\text{HJB}}(\theta) \stackrel{\text{def}}{=} C_{\mathbf{D}}^{(2)} J_{\text{HJB}}(\theta).$$

Substituting [\(5.61\)](#)–[\(5.65\)](#) into [\(5.17\)](#):

$$\|e\|_{L^2(\mathbf{D})}^2 \leq C_{\mathbf{D}} (J_{\partial \mathcal{Q}}(\theta) + \mathbf{T}_h J_{\text{HJB}}(\theta)) \leq C_{\mathbf{D}} (1 + \mathbf{T}_h) \delta.$$

Taking square roots, we have $\|\mathbf{V}_\theta - \mathbf{V}\|_{L^2(\mathbf{D})} \leq C_{\mathbf{D}} \sqrt{\delta}$, which completes the proof. ■

5.3. Regularity consistency at zero residual. The final results of this section is to show the consistency of regularity of $\mathbf{V}$ and $\mathbf{V}_\theta$ when the $J_{\text{HJB}} + J_{\partial Q} = 0$ over any compact set. In particular, zero optimization residual implies that $\mathbf{V}_\theta - \mathbf{V} = 0$ a.e. by [Thm. 5.6](#). However, by [Remark 5.3](#), in this case $\mathbf{V}_\theta$ is a viscosity solution and by uniqueness one expects that $\mathbf{V}$ is continuously differentiable over any compact set (as an interior set of a larger compact set). This is not consistent with the Lipschitz continuity we have shown [Thm. 4.13](#). Therefore, in what follows we show that in fact such regularity can be obtained.

Lemma 5.20 (State linearization on compact sub-tube; forcing-free and invertible). *Fix a compact $\mathbf{D} \in \mathbb{D}$ and $s \in [0, T_h)$, and set*

$$(5.66) \quad \mathcal{V}_s(\mathbf{D}) \stackrel{\text{def}}{=} \{(\kappa, \mathbf{y}) : (s, \mathbf{y}) \in \mathbf{D}, 0 \leq \kappa < \mathbf{V}_{\min}(\mathbf{D})\},$$

the compact-base sub-tube on which every trajectory survives with the margin $\mathbf{V}_{\min}(\mathbf{D}) - \kappa > 0$ uniformly.

Under [Asm. 5.10](#), for each $(\kappa_0, \mathbf{y}_0) \in \mathcal{V}_s(\mathbf{D})$ there is a radius $\varrho > 0$ such that the map $\mathbf{y}' \mapsto \Phi(\kappa; s, \mathbf{y}')$ is C^1 on $B(\mathbf{y}_0, \varrho)$ for every $\kappa \in [0, \kappa_0]$. In addition, the derivative $Y(\kappa; s, \mathbf{y}) \stackrel{\text{def}}{=} \partial_{\mathbf{y}} \Phi(\kappa; s, \mathbf{y}) \in \mathbb{R}^{2 \times 2}$ solves the linear Carathéodory system

$$(5.67) \quad \frac{d}{d\kappa} Y(\kappa; s, \mathbf{y}) = \partial_{\mathbf{y}} \mathbf{G}(\Phi(\kappa; s, \mathbf{y})) Y(\kappa; s, \mathbf{y}) \quad \text{a.e. in } \kappa, \quad Y(0; s, \mathbf{y}) = I_d,$$

on the interval $[0, \mathbf{V}(s, \mathbf{y}))$. Moreover $Y(\kappa; s, \mathbf{y})$ is invertible with the two-sided bound

$$(5.68) \quad C_{\text{Gr}}(\mathbf{D})^{-1}|v| \leq |Y(\kappa; s, \mathbf{y})v| \leq C_{\text{Gr}}(\mathbf{D})|v|, \quad \forall v \in \mathbb{R}^2,$$

uniformly for $(s, \mathbf{y}) \in \mathbf{D}$ and $\kappa \in [0, \mathbf{V}_{\min}(\mathbf{D}))$, where $\mathbf{V}_{\min}(\mathbf{D})$ is as in [Equation \(5.25\)](#) and $C_{\text{Gr}}(\mathbf{D})$ is the Gronwall constant as in [Lem. 5.12](#).

Proof. For any $(s, \mathbf{y}) \in \mathbf{D} \in \mathbb{D}$, compactness and $\mathbf{V} > 0$ on $\mathbb{D}$ implies that $\mathbf{V}(s, \mathbf{y}) \geq \mathbf{V}_{\min}(\mathbf{D}) > 0$. Therefore, fixing $(\kappa_0, \mathbf{y}_0) \in \mathcal{V}_s(\mathbf{D})$ and by continuity of $\mathbf{V}(s, \cdot)$, there are $r > 0$ and $\eta > 0$ with $\mathbf{V}(s, \mathbf{y}) > \kappa_0 + \eta$ for $\mathbf{y} \in B(\mathbf{y}_0, \varrho)$. Hence on $B(\mathbf{y}_0, \varrho)$ every trajectory survives past κ_0 , and the difference quotients below are well defined for all elapsed times up to κ_0 .

Differentiability in $\mathbf{y}$. Lipschitz dependence on the initial state ([Lem. 5.7\(iii\)](#)) gives, by Rademacher, existence of $\partial_{\mathbf{y}} \Phi$ a.e.; we upgrade this to genuine C^1 dependence on $B(\mathbf{y}_0, \varrho)$. By [\(5.46\)](#) the coefficient $\partial_{\mathbf{y}} \mathbf{G}$ is continuous in the state, bounded on compacts, and independent of the forcing b . Fix a direction v and small h with $\mathbf{y}_0 + hv \in B(\mathbf{y}_0, \varrho)$. Writing

$$(5.69) \quad \Phi \stackrel{\text{def}}{=} \Phi(\cdot; s, \mathbf{y}_0), \quad \Phi^h \stackrel{\text{def}}{=} \Phi(\cdot; s, \mathbf{y}_0 + hv)$$

and using that $\mathbf{G}(s, \cdot)$ is C^1 in the state with $\partial_{\mathbf{y}} \mathbf{G}$ independent of b ,

$$(5.70) \quad \mathbf{G}(r, \Phi^h(r)) - \mathbf{G}(r, \Phi(r)) = \partial_{\mathbf{y}} \mathbf{G}(\Phi(r))(\Phi^h(r) - \Phi(r)) + R(r, h),$$

where, integrating $\partial_{\mathbf{y}} \mathbf{G}$ along the segment joining $\Phi(r)$ and $\Phi^h(r)$.

In particular, set $x_\lambda \stackrel{\text{def}}{=} \Phi(r) + \lambda(\Phi^h(r) - \Phi(r))$, $\lambda \in [0, 1]$, the segment joining $\Phi(r)$ to $\Phi^h(r)$; it lies in the compact tube for h small. Since $\mathbf{G}(r, \cdot) \in C^1$, the fundamental theorem of

calculus applied to $\lambda \mapsto \mathbf{G}(r, x_\lambda)$ gives $\mathbf{G}(r, \Phi^h(r)) - \mathbf{G}(r, \Phi(r)) = (\int_0^1 \partial_{\mathbf{y}} \mathbf{G}(x_\lambda(r)) d\lambda)(\Phi^h(r) - \Phi(r))$, hence, subtracting the linearization,

$$(5.71) \quad R(r, h) = \left(\int_0^1 [\partial_{\mathbf{y}} \mathbf{G}(x_\lambda) - \partial_{\mathbf{y}} \mathbf{G}(x_0)] d\lambda \right) (\Phi^h(r) - \Phi(r)), \quad x_0 = \Phi(r).$$

Bounding the integrand (using Equation (5.46)) we have

$$|R(r, h)| \leq \omega(|\Phi^h(r) - \Phi(r)|) |\Phi^h(r) - \Phi(r)|,$$

with ω the modulus of continuity of $\partial_{\mathbf{y}} \mathbf{G}$ on the compact tube ($\omega(\rho) \rightarrow 0$ as $\rho \rightarrow 0$; for the OVM model $\omega(\rho) \leq C\rho$ since $\mathcal{V}''$ is bounded). Dividing the integrated flow identity by h and setting $\xi_h(r) \stackrel{\text{def}}{=} (\Phi^h(r) - \Phi(r))/h$, then using Equation (5.21) and Equation (5.70), we have

$$\xi_h(r) = v + \int_s^r \partial_{\mathbf{y}} \mathbf{G}(\Phi(r')) \xi_h(r') dr' + \underbrace{\frac{1}{h} \int_s^r R(r', h) dr'}_{:= E_h(r)}.$$

the residual $E_h(r)$ contributes at most $\kappa_0 C_{\text{Gr}}(\mathbf{D})|v| \omega(C_{\text{Gr}}(\mathbf{D})h|v|) \rightarrow 0$ as $h \rightarrow 0$ (using $|\Phi^h - \Phi| \leq C_{\text{Gr}}(\mathbf{D})h|v|$ from Lem. 5.7(iii) and Lem. 5.12). Using the Gronwall hypothesis, one can conclude ξ_h converges uniformly on $[s, s + \kappa_0]$ to the solution ξ of

$$(5.72) \quad \xi(r) = v + \int_s^r \partial_{\mathbf{y}} \mathbf{G}(\Phi(r')) \xi(r') dr',$$

a linear Carathéodory equation with L^∞ -in-time coefficient (admissible, since the roughness of forcing b enters only through Φ and not through $\partial_{\mathbf{y}} \mathbf{G}$; see also Equation (3.12)-(3.14)). Thus $\partial_{\mathbf{y}} \Phi(\kappa; s, \mathbf{y}_0)$ exists for every $\kappa \leq \kappa_0$, and by Equation (5.72) equals $\xi(s + \kappa) = Y(\kappa)v$ (along the direction v), and Y solves (5.67); continuity of Y in $\mathbf{y}$ follows from continuous dependence of the linear system on its data, so the map is C^1 on $B(\mathbf{y}_0, \varrho)$. Since $(\kappa_0, \mathbf{y}_0) \in \mathcal{V}_s(\mathbf{D})$ was arbitrary, the conclusion holds for every $(s, \mathbf{y}) \in \mathbf{D}$ and $\kappa \in [0, \mathbf{V}_{\min}(\mathbf{D})]$.

Two-sided bound. Fix $v \in \mathbb{R}^2$, write $w(\kappa) \stackrel{\text{def}}{=} Y(\kappa; s, \mathbf{y})v$, so that by (5.67)

$$(5.73) \quad \dot{w}(\kappa) = A(\kappa) w(\kappa) \text{ a.e.}, \quad w(0) = v, \quad A(\kappa) \stackrel{\text{def}}{=} \partial_{\mathbf{y}} \mathbf{G}(\Phi(\kappa; s, \mathbf{y})),$$

and $\|A\|_{L^\infty} \leq G_1(\mathbf{D})$ on the reachable tube (Lem. 5.12).

For the *upper* bound, Gronwall applied to $\rho(\kappa) \stackrel{\text{def}}{=} |w(\kappa)|$ gives $\rho(\kappa) \leq |v| e^{G_1(\mathbf{D})\kappa} \leq C_{\text{Gr}}(\mathbf{D})|v|$ for $\kappa \in [0, \mathbf{V}_{\min}(\mathbf{D})]$.

The *lower* bound is not immediate from a forward Gronwall inequality, which only ever produces an upper bound on $|w(\kappa)|$; we argue instead through an integrating factor on the squared norm, which sidesteps division by $|w|$ and does not presuppose $w \neq 0$. Set $\sigma(\kappa) \stackrel{\text{def}}{=} |w(\kappa)|^2$, a Lipschitz function with

$$\dot{\sigma}(\kappa) = 2\langle w(\kappa), \dot{w}(\kappa) \rangle = 2\langle w(\kappa), A(\kappa)w(\kappa) \rangle \quad \text{a.e.},$$

so by Cauchy–Schwarz and $\|A(\kappa)\| \leq G_1(\mathbf{D})$,

$$(5.74) \quad |\dot{\sigma}(\kappa)| \leq 2 |w(\kappa)| |A(\kappa)w(\kappa)| \leq 2 G_1(\mathbf{D}) \sigma(\kappa) \quad \text{a.e.}$$

The lower half of (5.74), $\dot{\sigma}(\kappa) \geq -2G_1(\mathbf{D})\sigma(\kappa)$, which provides

$$\frac{d}{d\kappa} \left(\sigma(\kappa) e^{2G_1(\mathbf{D})\kappa} \right) = (\dot{\sigma}(\kappa) + 2G_1(\mathbf{D})\sigma(\kappa)) e^{2G_1(\mathbf{D})\kappa} \geq 0 \quad \text{a.e.}$$

Since $w(\cdot) = Y(\cdot; s, \mathbf{y})v$ is a Carathéodory solution of the linear system (5.73) with $A \in L^\infty$, it is absolutely continuous (indeed Lipschitz); hence so are $\sigma = |w|^2$ and $g(\kappa) \stackrel{\text{def}}{=} \sigma(\kappa) e^{2G_1(\mathbf{D})\kappa}$, being smooth functions and products of such. For an absolutely continuous function the fundamental theorem of calculus applies,

$$g(\kappa) - g(0) = \int_0^\kappa g'(t) dt \geq 0,$$

the integrand being nonnegative a.e. by the display above; the inequality therefore holds at the genuine endpoints κ and 0, not merely almost everywhere. Consequently

$$\sigma(\kappa) e^{2G_1(\mathbf{D})\kappa} \geq \sigma(0) = |v|^2, \quad \text{i.e.} \quad |w(\kappa)| \geq |v| e^{-G_1(\mathbf{D})\kappa} \geq C_{\text{Gr}}(\mathbf{D})^{-1} |v|,$$

where we are using $e^{-G_1(\mathbf{D})\kappa} \geq e^{-G_1(\mathbf{D})\mathbf{V}_{\min}(\mathbf{D})} = C_{\text{Gr}}(\mathbf{D})^{-1}$ for $\kappa < \mathbf{V}_{\min}(\mathbf{D})$. As v was arbitrary, this is the lower bound in (5.68); in particular $Y(\kappa; s, \mathbf{y})$ has trivial kernel, hence is invertible with $\|Y(\kappa; s, \mathbf{y})^{-1}\| \leq C_{\text{Gr}}(\mathbf{D})$. $\blacksquare$

Globally on $\mathbb{D}$ the value function is only Lipschitz, $\mathbf{V} \in \mathcal{W}^{1,\infty}(\mathbf{D})$ (Thm. 4.13), since transversality degrades as $\mathbf{D} \rightarrow \partial\mathbb{D}$. On any compact $\mathbf{D} \Subset \mathbb{D}$, however, the uniform transversality margin of Lem. 4.9 together with the C^1 -in-initial-data regularity of the OVM flow (Lem. 5.20) upgrades this to genuine C^1 smoothness, via the implicit function theorem.

Remark 5.21 (Convention: C^1 on compact sets). For a compact $\mathbf{K} \Subset \mathbb{D}$ we write $f \in C^1(\mathbf{K})$ to mean that f is continuously differentiable on some open neighborhood of $\mathbf{K}$; since $\mathbb{D}$ is open and $\mathbf{K} \Subset \mathbb{D}$, this is equivalent to $f \in C^1(\mathbf{U})$ for an open $\mathbf{U}$ with $\mathbf{K} \subset \mathbf{U} \subseteq \mathbb{D}$. The bounds on the derivative uniform, through the constants $\gamma_{\mathbf{K}}$ and $C_{\text{Gr}}(\mathbf{K})$.

Lemma 5.22 (C^1 regularity of $\mathbf{V}$ on compacts). Let $\mathbf{D} \Subset \mathbb{D}$ be compact. Then the predictive collision time $\tilde{T}$ and the value function $\mathbf{V}(t_o, \mathbf{x}_o) = \tilde{T}(t_o, \mathbf{x}_o) - t_o$ are of class C^1 on $\mathbf{D}$, with

$$(5.75) \quad \nabla_{\mathbf{x}_o} \tilde{T}(t_o, \mathbf{x}_o) = - \frac{\nabla_{\mathbf{x}_o} \Xi(\tilde{T}; t_o, \mathbf{x}_o)}{\Lambda(\tilde{T}; t_o, \mathbf{x}_o)},$$

the denominator being bounded away from 0 uniformly on $\mathbf{D}$.

Proof. Fix $\mathbf{z}^0 = (t_o^0, \mathbf{x}_o^0) \in \mathbf{D}$ and write $\nu^0 \stackrel{\text{def}}{=} \tau(\mathbf{z}^0)$, $\nu \stackrel{\text{def}}{=} \tau(\mathbf{z})$ for $\mathbf{z} \in \mathbf{D}$ near $\mathbf{z}^0$. Set

$$\gamma \stackrel{\text{def}}{=} -\Lambda(\nu^0; \mathbf{z}^0) \geq \gamma_{\mathbf{D}} > 0$$

by the uniform transversality margin (4.29). By definition (4.17) of the collision time,

$$(5.76) \quad \Xi(\nu; \mathbf{z}) = 0 = \Xi(\nu^0; \mathbf{z}^0).$$

All evaluations below occur at times not exceeding $\min\{\nu, \nu^0\}$, hence strictly before the collision of either trajectory; no trajectory is continued past its collision time.

Step 1: a finite-difference identity at a common time. Assume first $\nu \geq \nu^0$. Since $\Xi(\cdot; \mathbf{z})$ is absolutely continuous on $[0, \nu]$ with $\frac{d}{d\theta} \Xi = \Lambda$ a.e. (integrate (4.6)), and since $[\nu^0, \nu] \subseteq [0, \nu]$,

$$0 = \Xi(\nu; \mathbf{z}) - \Xi(\nu^0; \mathbf{z}) + \int_{\nu^0}^{\nu} \Lambda(\theta; \mathbf{z}) d\theta.$$

Subtracting $\Xi(\nu^0; \mathbf{z}^0) = 0$ from $\Xi(\nu^0; \mathbf{z})$ and using the last display,

$$(5.77) \quad \Xi(\nu^0; \mathbf{z}) - \Xi(\nu^0; \mathbf{z}^0) = - \int_{\nu^0}^{\nu} \Lambda(\theta; \mathbf{z}) d\theta.$$

If instead $\nu < \nu^0$, the same computation with the roles of $\mathbf{z}$ and $\mathbf{z}^0$ interchanged—evaluating both gaps at the common time ν —gives (5.77) with the integral over $[\nu, \nu^0]$ and the opposite sign; the estimates below are unaffected, so we treat only the case $\nu \geq \nu^0$.

Step 2: expansion of the left-hand side. By Lem. 5.20 the map $\mathbf{z} \mapsto \Phi(\nu^0; \mathbf{z})$ is C^1 on a neighbourhood of $\mathbf{z}^0$ in the survival set — note $\nu^0 < \tau(\mathbf{z})$ is guaranteed for $\mathbf{z}$ near $\mathbf{z}^0$ by Lem. 4.5 in the case $\nu \geq \nu^0$ — with derivative $Y(\nu^0; \mathbf{z}^0)$. Taking the first (gap) component,

$$(5.78) \quad \Xi(\nu^0; \mathbf{z}) - \Xi(\nu^0; \mathbf{z}^0) = \nabla_{\mathbf{z}} \Xi(\nu^0; \mathbf{z}^0) \cdot (\mathbf{z} - \mathbf{z}^0) + o(\|\mathbf{z} - \mathbf{z}^0\|), \quad \nabla_{\mathbf{z}} \Xi(\nu^0; \mathbf{z}^0) = \mathbf{e}_1^\top Y(\nu^0; \mathbf{z}^0).$$

Step 3: expansion of the right-hand side. The map $(\theta, \mathbf{z}) \mapsto \Lambda(\theta; \mathbf{z})$ is jointly continuous, being M -Lipschitz in θ by (4.7) and C_F -Lipschitz in $\mathbf{z}$ by Thm. 4.4. Hence, for $\theta \in [\nu^0, \nu]$,

$$|\Lambda(\theta; \mathbf{z}) - \Lambda(\nu^0; \mathbf{z}^0)| \leq M|\theta - \nu^0| + C_F\|\mathbf{z} - \mathbf{z}^0\| \leq M|\nu - \nu^0| + C_F\|\mathbf{z} - \mathbf{z}^0\| \stackrel{\text{def}}{=} \varepsilon(\mathbf{z}),$$

with $\varepsilon(\mathbf{z}) \rightarrow 0$ as $\mathbf{z} \rightarrow \mathbf{z}^0$, using $\nu \rightarrow \nu^0$ from Lem. 4.5. Therefore

$$(5.79) \quad - \int_{\nu^0}^{\nu} \Lambda(\theta; \mathbf{z}) d\theta = -\Lambda(\nu^0; \mathbf{z}^0) (\nu - \nu^0) + R, \quad |R| \leq \varepsilon(\mathbf{z}) |\nu - \nu^0|.$$

Step 4: the remainder is $o(\|\mathbf{z} - \mathbf{z}^0\|)$. By Lem. 4.12, $|\nu - \nu^0| \leq L_\tau(\mathbf{D})\|\mathbf{z} - \mathbf{z}^0\|$ for $\|\mathbf{z} - \mathbf{z}^0\|$ small. Consequently $\varepsilon(\mathbf{z}) \leq (ML_\tau(\mathbf{D}) + C_F)\|\mathbf{z} - \mathbf{z}^0\| \rightarrow 0$ and

$$|R| \leq (ML_\tau(\mathbf{D}) + C_F)L_\tau(\mathbf{D})\|\mathbf{z} - \mathbf{z}^0\|^2 = o(\|\mathbf{z} - \mathbf{z}^0\|).$$

Step 5: conclusion. Substituting (5.78) and (5.79) into (5.77) and collecting the $o(\|\mathbf{z} - \mathbf{z}^0\|)$ terms,

$$\nabla_{\mathbf{z}} \Xi(\nu^0; \mathbf{z}^0) \cdot (\mathbf{z} - \mathbf{z}^0) = -\Lambda(\nu^0; \mathbf{z}^0) (\tau(\mathbf{z}) - \tau(\mathbf{z}^0)) + o(\|\mathbf{z} - \mathbf{z}^0\|).$$

Dividing by $-\Lambda(\nu^0; \mathbf{z}^0) = \gamma \geq \gamma_D > 0$, which is legitimate and stable by (4.29),

$$(5.80) \quad \tau(\mathbf{z}) - \tau(\mathbf{z}^0) = - \frac{\nabla_{\mathbf{z}} \Xi(\nu^0; \mathbf{z}^0)}{\Lambda(\nu^0; \mathbf{z}^0)} \cdot (\mathbf{z} - \mathbf{z}^0) + o(\|\mathbf{z} - \mathbf{z}^0\|).$$

Thus τ is differentiable at $\mathbf{z}^0$ with the gradient (5.75). Finally $\nabla \tau$ is continuous on $\mathbf{D}$: the numerator $\mathbf{z} \mapsto \nabla_{\mathbf{z}} \Xi(\tau(\mathbf{z}); \mathbf{z})$ is continuous, being the composition of the continuous τ (Lem. 4.5) with the continuous Y (Lem. 5.20), and is bounded by $C_{\text{Gr}}(\mathbf{D})$; the denominator $\mathbf{z} \mapsto \Lambda(\tau(\mathbf{z}); \mathbf{z})$ is continuous by Lem. 4.6 and bounded away from 0 by γ_D . Hence $\tau \in C^1(\mathbf{D})$, and so are $\tilde{\tau} = \tau + t_o$ and $\mathbf{V} = \tau$. ■

5.4. Zero residual implies the strong fixed point.

Proposition 5.23 (Exact fixed point in the zero-residual limit). *Let $\mathbf{V}_\theta \in C^1(\mathbf{D}^+) \cap C(\overline{\mathbf{D}^+})$ (see Remark 5.21) satisfy $J_{\text{HJB}}(\theta) = 0$ and $J_{\partial\mathcal{Q}}(\theta) = 0$. Then $\mathbf{V}_\theta = \mathbf{V}$ pointwise on $\mathbf{D}$. Consequently $\mathbf{V}_\theta$ is a strong fixed point in the sense of (5.6)—it satisfies the dynamic programming identity along every OVM characteristic through $\mathbf{D}^+$ —and coincides with the viscosity solution of the HJB equation (Lem. 5.1).*

Proof. *Step 1: the residual vanishes everywhere in state, a.e. in time.* Since $J_{\text{HJB}}(\theta) = \|\mathcal{R}[\mathbf{V}_\theta]\|_{L^2(\mathbf{D}^+, \mu)}^2 = 0$ and μ is equivalent to Lebesgue measure, the residual $\mathcal{R}[\mathbf{V}_\theta] = \partial_s \mathbf{V}_\theta + \mathbf{G} \cdot \nabla_{\mathbf{y}} \mathbf{V}_\theta + 1$ vanishes $\mathcal{H}^3$ -a.e. on $\mathbf{D}^+$. Because $\mathbf{V}_\theta \in C^1$, the maps $\partial_s \mathbf{V}_\theta$ and $\nabla_{\mathbf{y}} \mathbf{V}_\theta$ are continuous, so $\mathcal{R}[\mathbf{V}_\theta]$ is continuous in the state $\mathbf{y}$; only the coefficient $\mathbf{G}$ carries the L^∞ -in-time forcing $\mathbf{a}_\ell, \zeta_\ell$. By Fubini, for a.e. fixed s the slice $\mathbf{y} \mapsto \mathcal{R}[\mathbf{V}_\theta](s, \mathbf{y})$ is continuous and vanishes for a.e. $\mathbf{y}$, hence vanishes for every $\mathbf{y}$ in that slice. Therefore

$$(5.81) \quad \mathcal{R}[\mathbf{V}_\theta](s, \mathbf{y}) = 0 \quad \text{for every } \mathbf{y}, \text{ a.e. } s.$$

(One cannot upgrade the “a.e. s ” to “every s ”: the forcing is genuinely L^∞ in time, so $\mathbf{V}$ itself solves the PDE only a.e. in time. This is the correct Carathéodory sense, matching the regularity of $\mathbf{V}$.)

Step 2: integration along OVM characteristics. Fix $(s, \mathbf{y}) \in \mathbf{D}$ and let $r \mapsto \Phi(r; s, \mathbf{y})$ be the OVM characteristic through it, defined up to the collision time $\tilde{\mathcal{T}}$. Since $\mathbf{V}_\theta \in C^1(\mathbf{D}^+)$ and the trajectory is Lipschitz in r (Lem. 5.7(i)), the composition

$$g(r) \stackrel{\text{def}}{=} \mathbf{V}_\theta(s + r, \Phi(r; s, \mathbf{y}))$$

is absolutely continuous, and by the chain rule, for a.e. r ,

$$(5.82) \quad g'(r) = \partial_s \mathbf{V}_\theta + \mathbf{G} \cdot \nabla_{\mathbf{y}} \mathbf{V}_\theta = \mathcal{R}[\mathbf{V}_\theta](s + r, \Phi(r; s, \mathbf{y})) - 1 = -1,$$

the last equality by (5.81) applied at the state $\Phi(r; s, \mathbf{y})$ (valid for a.e. r). Integrating the a.e. identity (5.82) over $[0, \kappa]$ —legitimate since g is absolutely continuous, so the fundamental theorem of calculus applies and the $\mathcal{L}^1$ -null set of exceptional times does not contribute—gives, for every κ with $(s + \kappa, \Phi(\kappa; s, \mathbf{y})) \in \text{cl}[\mathbb{D}]$,

$$(5.83) \quad \mathbf{V}_\theta(s + \kappa, \Phi(\kappa; s, \mathbf{y})) - \mathbf{V}_\theta(s, \mathbf{y}) = -\kappa.$$

This is precisely the strong fixed-point identity (5.6).

Step 3: matching the boundary value gives $\mathbf{V}_\theta = \mathbf{V}$. Take $\kappa = \tilde{\mathcal{T}}(s, \mathbf{y}) - s$, so that $\Phi(\kappa; s, \mathbf{y})$ reaches the collision boundary $\mathcal{Q}_\mathbf{D}$, where $\Xi = 0$. Since $J_{\partial\mathcal{Q}}(\theta) = \|\mathbf{V}_\theta\|_{L^2(\mathcal{Q}_\mathbf{D}, \sigma)}^2 = 0$, the nonnegative integrand vanishes σ -a.e. on $\mathcal{Q}_\mathbf{D}$. This upgrades to every point; the set $\{\mathbf{V}_\theta \neq 0\}$ is relatively open in $\mathcal{Q}_\mathbf{D}$ by continuity of $\mathbf{V}_\theta$, and every nonempty relatively open subset of the two-dimensional set $\mathcal{Q}_\mathbf{D} \subset \{\Xi = 0\}$ contains a relative ball of positive $\mathcal{H}^2$ -measure; a σ -null open set is therefore empty. Hence $\mathbf{V}_\theta \equiv 0$ on $\mathcal{Q}_\mathbf{D}$, and in particular $\mathbf{V}_\theta(s + \kappa, \Phi(\kappa; s, \mathbf{y})) = 0$. Substituting into (5.83),

$$\mathbf{V}_\theta(s, \mathbf{y}) = \kappa + 0 = \mathbf{V}(s, \mathbf{y}).$$

As $(s, \mathbf{y}) \in \mathbf{D}$ was arbitrary, $\mathbf{V}_\theta = \mathbf{V}$ pointwise on $\mathbf{D}$. Finally, (5.83) is the strong fixed-point property, and by Lem. 5.1 the value function—now equal to $\mathbf{V}_\theta$ —is the viscosity solution of the HJB equation, so $\mathbf{V}_\theta$ is that viscosity solution. $\blacksquare$

6. Numerical Validation. The analysis above is constructive in two respects that admit direct numerical assessment. [Thm. 3.9](#) excludes collision by contradiction, and therefore does not exhibit the avoidance behavior it guarantees; and [Thm. 5.6](#) bounds the error of a learned value function by the training loss, without indicating how small that loss becomes in practice. This section reports simulations addressing both. The definition of different types of simulation is detailed in [Appendix B](#). Implementation details — the architectures of V_θ , $\mathcal{N}_V$ and $\mathcal{N}_X$, the biased sampling strategy, the training procedure and its performance metrics — are deferred to [Appendices C, F and H](#), and the full set of trajectory comparisons to [Appendix J](#).

Learning the TTC map and the full trajectory. The value network V_θ is a feedforward multi-layer perceptron trained on the total loss (5.12), with sampling concentrated near the collision boundary where the transversality margin γ_D — and hence every constant in [Sections 3 and 4](#) — is small. Compared against direct numerical integration of the relative OVM dynamics (4.1), the learned map reproduces the level-set structure of τ with small absolute error away from the boundary and growing error as it is approached, consistent with the degeneration of γ_D ([Figure 4, Table 1 in Appendix C](#)). A purely black-box network trained without the analytical structure fails to capture the dynamics in the near-collision regime at all ([Figure 5](#)), which is the practical counterpart of the observation that the strong-form residual is only meaningful where the solution is regular ([Lem. 5.22](#)). Composing V_θ with the trajectory network $\mathcal{N}_X$ yields the two-stage map $\mathcal{N}_{\text{sol}}$, whose predictions of τ , spacing and relative velocity track the delay-free reference simulation over the full horizon across the initial conditions tested ([Appendix J](#)).

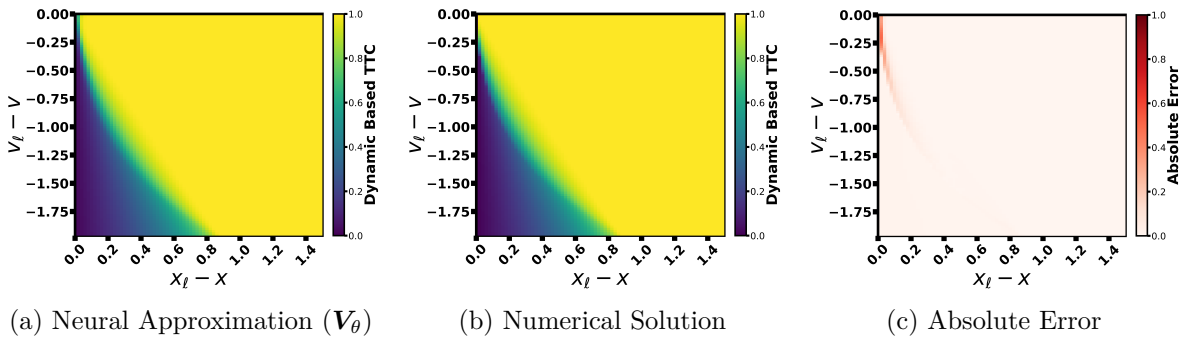


Figure 4: Comparison between the dynamic-based TTC values approximated by V_θ and dynamic-based TTC values computed through the direct numerical simulation of the relative OVM dynamics (4.1) with $\alpha = 2$ and $T_h = 1$.

Platoon behavior. To examine the closed loop rather than the approximation in isolation, we simulate a six-vehicle platoon in which the leader follows a non-constant acceleration profile and the pair (0, 1) is driven by $\mathcal{N}_{\text{sol}}$, the remaining pairs being integrated numerically. The initial spacing of the first pair is set to 0.1, deliberately close to the collision boundary, with the remaining vehicles started at their OVM equilibrium spacing. [Figure 14](#) and [Figure 15](#) show the resulting spacing and relative velocity along the platoon. Two features are worth noting.

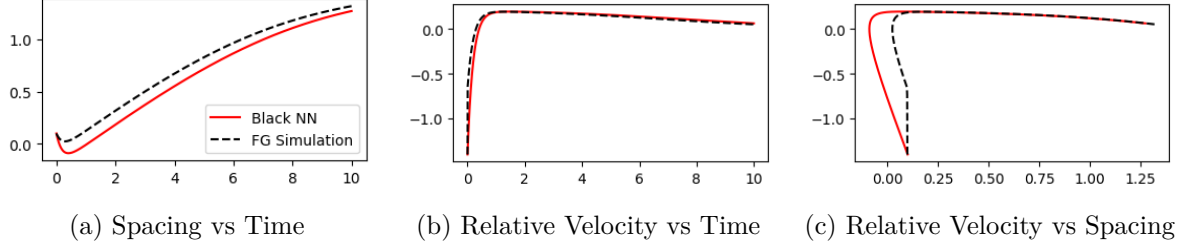


Figure 5: Black Box Neural Network Performance compared with the delay-free trajectory of the proposed model (FG-simulation; see [Appendix B](#) for the definition), illustrating the limitation of non-physically informed training to capture the proposed dynamics in the near-collision regime. Same parameters, simulation time, and x profile as in [Figure 2](#), with $a_\ell = 0$. Initial Conditions: $x_\ell = 1.21$, $v = 1.52$, $v_\ell = 0.13$.

Spacing remains bounded away from zero throughout, despite the tight initial configuration — the behavior that [Thm. 3.9](#) predicts, here visible rather than inferred. And the disturbance introduced by the leader attenuates as it propagates downstream, with later pairs exhibiting smaller excursions than earlier ones, so the learned controller inherits the string-stability behavior of the underlying model (cf. [Appendix C.1](#)).

Scope of these results. These simulations illustrate the theory rather than test it: a finite set of scenarios can neither confirm nor falsify the theorems above, and the reported behavior is consistent with, but not evidence for, the analytical guarantees. Two limitations should be stated. The evaluation uses the same leader acceleration profile as training, so the results speak to in-distribution accuracy only; assessing out-of-distribution leader behavior is left to future work. And the analysis of [Subsection 5.2](#) is at the continuum level, bounding the error by the *population* loss, whereas training minimizes an empirical loss over finitely many collocation points. Closing that gap requires a generalization bound, which would inherit the same dependence on γ_D as the estimates established here; we regard this as the principal open problem on the theoretical side.

7. Conclusions. A dynamically informed TTC-based feedback control framework for non-linear microscopic traffic dynamics is proposed in this paper. Unlike conventional TTC formulations based on kinematic prediction, the proposed controller computes dynamic-based TTC using the OVM dynamics in this work, producing a predictive feedback law that anticipates vehicle interactions.

We established several analytical properties of the resulting closed-loop system. First, we rigorously proved well-posedness of the proposed model using a Picard-type fixed-point argument. We then showed the collision-freeness using a barrier-domain argument, near-collision boundary analysis, and a maximum principle. We additionally established the Lipschitz regularity of the dynamic-based TTC map, providing the basis for neural network approximation.

Building on these analytical results, we developed a Hamilton-Jacobi reachability-based learning framework for approximating dynamic-based TTC map for delay-free control law,

preserving the safety and stability of the model. In addition, we introduced a two-stage semi-supervised physics-informed neural network architecture for approximating the full system trajectories, considering the near-collision singularities and oscillatory dynamics.

Future work will focus on extending the present framework to more complex dynamical systems involving interactions among multiple control agents. In particular, it is important to generalize the proposed reachability-based analysis to systems in which agents interactions are not limited to one-dimensional car-following behavior, but may include lane changing, different vehicle types, network-level interactions, and more general multi-agent control structures. Such extensions would broaden the applicability of the framework to safety-critical transportation systems and other interacting-agent systems where control decisions, dynamic predictions, and collision-avoidance constraints are strongly coupled.

Appendix A. Schematics: From OVM to HJB equation. The relation between OVM properties, HJB presentation, and proof of convergence of PINN approximation can be illustrated in Figure 6

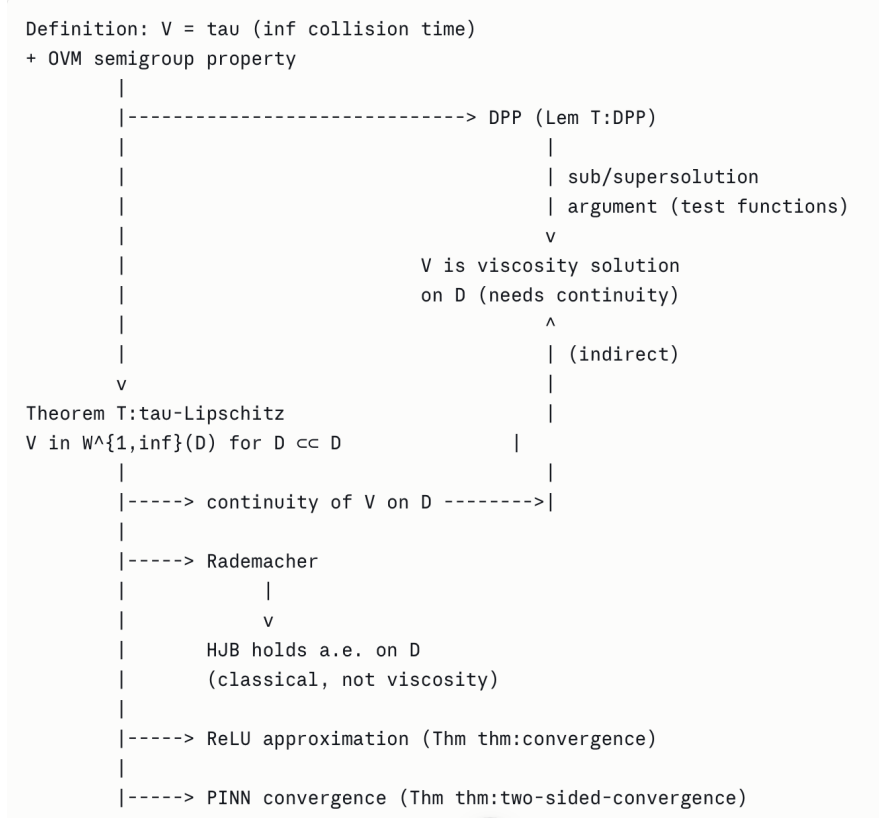


Figure 6: From regularity of OVM to approximation of viscosity solution of HJB equation.

Appendix B. Simulation: definitions.

Definition B.1 (Simulation Term).

Throughout this work, we use several notions of simulation to support our results and discussions. For clarity, we define and distinguish the following terms:

- (i) **Freeze-and-Go Simulation (FG-Simulation)** refers to calculating the solution step by step without taking the delay of TTC-approximation into account. In particular, at each time step, we freeze the dynamical model, approximate TTC, and update the numerical solution for the next time step. See Figure 2 for trajectories.
- (ii) **Neural-Numerical Simulation (NeuNum-Simulation)** refers to a hybrid computational framework that integrates the conventional numerical simulation of vehicle dynamics with the trained NN for dynamic-based TTC approximation, thereby avoiding delay in TTC computation. See Figure 14.
- (iii) **NN-based Approximation** is designed to provide the solution learned by a DNN. More

precisely, in this approach, the TTC, spacing, and relative solution are approximated through DNNs. See [Figure 13](#) for the corresponding behavior of trajectories.

Appendix C. Architecture and performance of V_θ .

Validated by the analytical results, the total loss (5.12), the biased sampling strategy, and the well-established stability properties of the OVM (see e.g., [4, 3]), we employ the neural network $V_\theta(s, \mathbf{y})$ as a standard feedforward MLP and trained according to [Algorithm G.1](#) to approximate the TTC mapping τ . The architecture and training procedure are described in this section and detailed in [Appendix H.1](#) in [Appendix H](#).

Remark C.1 (Local Stability and Justification for the Neural Network Framework). *The local stability of the non-delayed system is fundamental to justify the accuracy of NN-learning, specifically in regularity of the dynamic systems. Stability ensures that trajectories starting in some neighborhood of the initial conditions evolve in a consistent and predictable manner, allowing the neural network to accurately learn the system dynamics. As a result, the network can generalize effectively to unseen initial conditions within the stability region. Conversely, in an unstable system, even small differences in initial states can lead to significantly divergent trajectories, making neural approximation unreliable for out-of-sample predictions. A local stability analysis of the class of car-following dynamics encompassing the proposed model is presented in [61, 59], while the numerical stability of our proposed model is examined in [Subsection F.4](#).*

The architecture of V_θ is informed by the analytical properties established earlier. Training performance metrics summarized in [Table 1](#) and the comparison between V_θ and numerical simulation in [Figure 4](#) show high predictive reliability and robustness of the trained model.

Table 1: Training performance summary for V_θ , $\mathcal{N}_V$, and $\mathcal{N}_X$. All models trained on Apple M2 Max with 96GB RAM. Full training configurations are detailed in [Appendix H](#).

Model	Epochs	Final Loss	R^2 (Train)	R^2 (Val)	R^2 (Test)	Primary Variable
V_θ	80,000	2.391E-4	–	0.9790	0.9845	TTC τ
$\mathcal{N}_V$	3,000	0.2389	–	0.9994	–	Relative velocity V
$\mathcal{N}_V$	3,000	0.2389	–	0.9997	–	Spacing X
$\mathcal{N}_X$	3,000	2321.265	–	0.9999	–	Spacing X

Appendix D. Leading Vehicle Acceleration. To evaluate the general performance of the neural network framework, it is important to consider the cases of non-zero acceleration of the leading vehicle, causing the leading vehicle to move at non-constant velocity. Although we can train the proposed neural network framework with various $\mathbf{a}_\ell$ profiles, it requires a lot of training efforts and time to do so. Therefore, in this paper, we present the performance of the proposed neural network frameworks with a particular leading vehicle’s acceleration profile.

For the general evaluation, we choose the acceleration profile such that the profile consists

of a series of accelerations and decelerations. The chosen acceleration profile is as follows:

$$(D.1) \quad a_\ell(t) = \begin{cases} 3.5 \exp(-\ln(100)t) \sin(8\pi t), & \forall t \in [0, 1], \\ 0, & \text{otherwise} \end{cases}$$

where the visualization of the acceleration profile and the resulting profile of the $v_\ell(t) - v_\ell(t_o)$ is illustrated in Figure 7. This acceleration profile a_ℓ is the attenuating oscillation function, making the leading vehicle accelerate and decelerate while its magnitude decreases within the time horizon. With the setup, the neural network needs to learn the effect of the leading vehicle's dynamics on the following vehicle's positions and velocity, especially the oscillating dynamics.

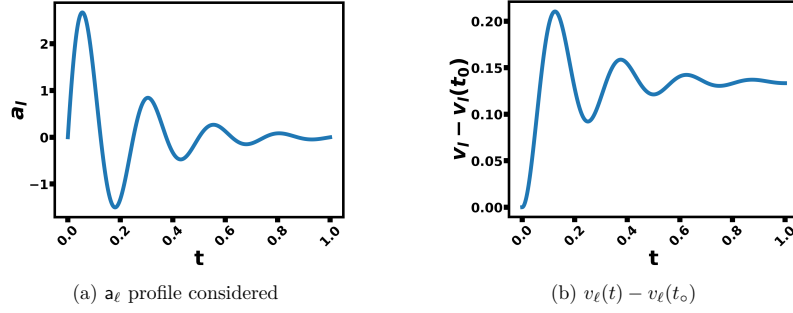


Figure 7: Visualization of a_ℓ profile and $v_\ell(t) - v_\ell(t_o)$ profile.

Appendix E. Flow Charts.

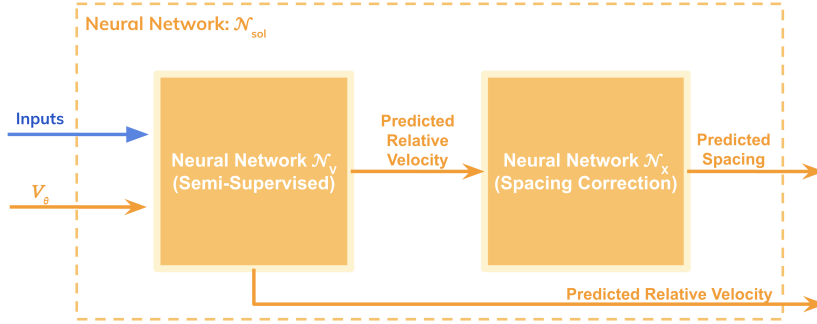


Figure 8: Overall Framework for Non-Delayed Solution with Neural Network

Appendix F. Deep Neural Network Approximation of the solution.

Building on V_θ , we now extend the framework to learn the entire system dynamics. While V_θ provides instantaneous dynamic-based TTC estimates τ , a well-coordinated vehicle platoon requires non-delayed, long-horizon prediction of spacing and relative velocity, as each follower

depends on its leader's acceleration, velocity, and position. Learning the full trajectory thus enables real-time, delay-free prediction of leader dynamics, ensuring stable platoon operation.

However, a non-physically informed learning approach, without incorporating analytical structure, proves ineffective in this context. Due to strong irregularities and nonlinearities (see Section 3), a black-box neural network fails to capture the proposed dynamics or to maintain stability across varying initial conditions, as shown in Figure 5. This limitation highlights the need for an analytically guided, biased-sampling framework that aligns the learning process with the system's underlying physical and stability properties.

To address the limitation of non-physically informed black-box learning in capturing the proposed irregular and nonlinear dynamics, we proposed a semi-supervised Physics-Informed Neural Network (semi-supervised PINN) framework that applies data-driven supervision with biased sampling in the safety-critical region and a PINN formulation in the remaining region. The overall architecture of the proposed framework is shown in Figure 9.

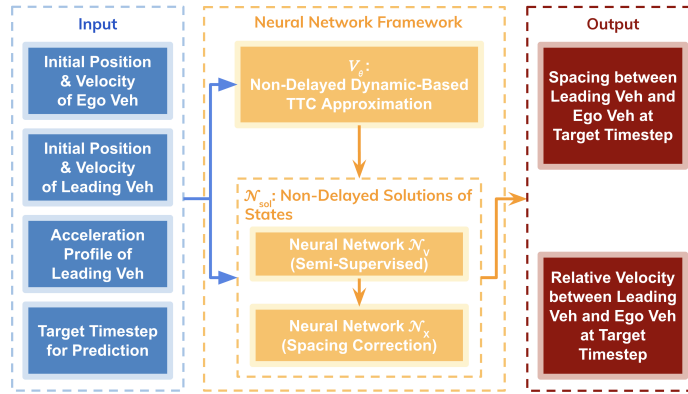


Figure 9: Overall Framework using Neural Network

As illustrated in Figure 9, the proposed neural network framework $\mathcal{N}_{\text{sol}}$ takes the following inputs: (i) the initial positions and velocities, (ii) the acceleration profile of the leading vehicle, and (iii) the timestep of interest for prediction. By leveraging these inputs along with the dynamic-based TTC approximation from the pre-trained $\mathbf{V}_\theta$, $\mathcal{N}_{\text{sol}}$ predicts the full state evolution of spacing X and relative velocity V , enabling delay-free, physically consistent trajectory prediction across any forecast horizon. The detailed framework of $\mathcal{N}_{\text{sol}}$ is shown in Figure 8 in Appendix E.

One motivation for the semi-supervised PINN framework is to address loss-imbalance in purely physics-informed training, as illustrated in Figure 10 for the single-stage unsupervised PINN with joint spacing–relative velocity prediction. In the near-collision regime, the proposed model generates collision-free yet aggressive maneuvers with abrupt changes in relative-velocity, producing larger residuals in relative velocity than in spacing and causing training to prioritize the relative-velocity profile over spacing [57, 58].

To capture the coupled spacing and relative velocity dynamics, $\mathcal{N}_{\text{sol}}$ adopts a two-stage architecture with $\mathcal{N}_V$ and $\mathcal{N}_X$. The first-stage model $\mathcal{N}_V$ predicts the relative velocity profiles through a semi-supervised strategy that integrates ODE-based physical constraint with

data-driven supervision, conditioned on τ . Then, the second-stage model $\mathcal{N}_X$ refines the corresponding spacing profile using the output of $\mathcal{N}_V$. This architecture ensures consistency between dynamic states and achieves stable, delay-free prediction with high accuracy and robustness in the safety-critical region.

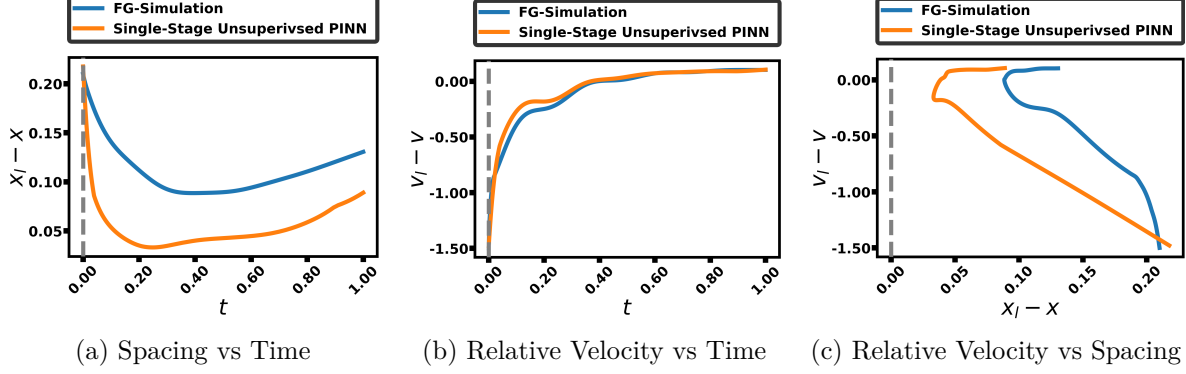


Figure 10: Predictions from trained single-stage unsupervised PINN, showing degraded spacing prediction in the near-collision regime due to loss-imbalance issue in purely physics-informed training. Same parameters, simulation time, a_ℓ , and x profile as in Figure 2. Initial Conditions: $x_\ell = 0.21$, $v = 1.65$, $v_\ell = 0.15$.

While the proposed two-stage learning framework mitigates imbalance-related errors, uniform random sampling remains insufficient as it ignores the underlying system behavior, which is particularly characterized by sharp decelerations, abrupt transitions, and irregular non-uniform behavior near the collision boundary. Consequently, the safety-critical region with small τ is not emphasized enough during training, leading to degraded prediction accuracy (see Figure 11 for an example).

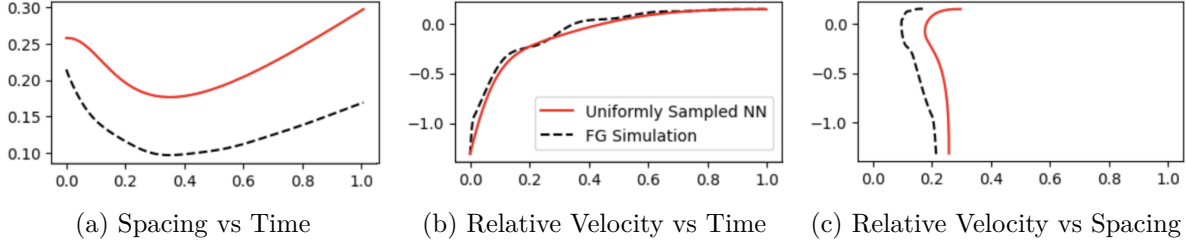


Figure 11: Uniform Random Sampling and Performance of NN, illustrating degraded prediction in the near-collision regime due to insufficient emphasis on safety-critical cases. Same parameters, simulation time, a_ℓ , and x profile as in Figure 2. Initial Conditions: $x_\ell = 0.26$, $v = 1.53$, $v_\ell = 0.21$.

To improve the prediction reliability and robustness in the safety-critical region, we utilize

a biased sampling strategy analogous to that for $\mathbf{V}_\theta$ in Section 5, increasing the representation of near-collision cases according to their safety importance.

Having established the analytical foundations, safety-critical region, local stability analysis, the concept of semi-supervised PINN, biased sampling strategy, and two-stage learning, we now detail each stage of the proposed framework, $\mathcal{N}_V$ and $\mathcal{N}_X$.

F.1. Neural Network $\mathcal{N}_V$. We begin with the first-stage network, $\mathcal{N}_V$, implemented using a semi-supervised PINN formulation to capture the nonlinear coupling between X and V . Within this framework, $\mathcal{N}_V$ acts as a neural operator parameterized by θ , mapping inputs including $X_\circ$, $V_\circ$, $v_{\ell,\circ}$, $\mathbf{a}_\ell$, τ from $\mathbf{V}_\theta$, and a query time t to the corresponding system states. This setup enables $\mathcal{N}_V$ to model the full solution map and predict the state evolution for any $t \in [0, T]$.

To train this operator, θ of $\mathcal{N}_V$ is learned by minimizing the following composite loss:

$$(F.1) \quad J_{\mathcal{N}_V} = \lambda_r J_{r,\mathcal{N}_V} + \lambda_{ic} J_{ic,\mathcal{N}_V} + \lambda_{data} J_{data,\mathcal{N}_V},$$

where $J_{r,\mathcal{N}_V}$, $J_{ic,\mathcal{N}_V}$, and $J_{data,\mathcal{N}_V}$ denote the residual, initial condition, and data-driven, respectively. Their weights are set to $\lambda_r = 6$, $\lambda_{ic} = 75$, and $\lambda_{data} = 10$, as determined by grid search. Each loss component is described below.

ODE Loss $J_{r,\mathcal{N}_V}$. The ODE loss is formulated as the residual between predicted and true derivatives, promoting physically consistent trajectories that adhere to the governing dynamical equations. For each sample i and randomly selected time step t_j , the residuals are computed as follows:

$$(F.2) \quad \begin{cases} r_X^{(i,j)} = \frac{d\hat{X}^{(i)}(t_j)}{dt} - \hat{V}^{(i)}(t_j) \\ r_V^{(i,j)} = \frac{d\hat{V}^{(i)}(t_j)}{dt} - \left\{ \mathbf{a}_\ell - \left(\alpha \left\{ \mathcal{V}(\hat{X}^{(i)}(t_j)) + \hat{V}^{(i)}(t_j) - v_\ell^{(i)}(t_j) \right\} + \beta \frac{\hat{V}^{(i)}(t_j)}{(\hat{\tau}^{(i)}(t_j)/T)^2} \right) \right\}. \end{cases}$$

Here, $\hat{V}^{(i)}(t_j)$ and $\hat{X}^{(i)}(t_j)$ are the predictions from $\mathcal{N}_V$, while $\hat{\tau}^{(i)}(t_j)$ is the prediction from $\mathbf{V}_\theta$.

In the semi-supervised PINN framework, these residuals are enforced selectively, with the ODE loss computed only when the dynamic-based TTC exceeds a specified threshold $\underline{\tau}$:

$$(F.3) \quad J_{r,\mathcal{N}_V} = \frac{1}{|\mathcal{I}_{ode}|} \sum_{(i,j) \in \mathcal{I}_{ode}} \left(r_X^{(i,j)} \right)^2 + \left(r_V^{(i,j)} \right)^2,$$

where $\mathcal{I}_{ode} = \{(i, j) : \tau^{(i)}(t_j) \geq \underline{\tau}\}$ and $\underline{\tau}$ is set to be 0.5 in this article.

Data-Driven Loss $J_{data,\mathcal{N}_V}$. The data-driven loss acts as a stabilizing regularizer, aligning predictions with data and enhancing robustness by providing supervision in a safety-critical region with small dynamic-based TTC, where sharp decelerations, stiffness, or singularities occur. This loss is computed for timesteps where the dynamic-based TTC falls below a threshold $\bar{\tau}$:

$$(F.4) \quad J_{data,\mathcal{N}_V} = \frac{1}{|\mathcal{I}_{data}|} \sum_{(i,j) \in \mathcal{I}_{data}} \left(\hat{X}^{(i)}(t_j) - X^{(i)}(t_j) \right)^2 + \left(\hat{V}^{(i)}(t_j) - V^{(i)}(t_j) \right)^2,$$

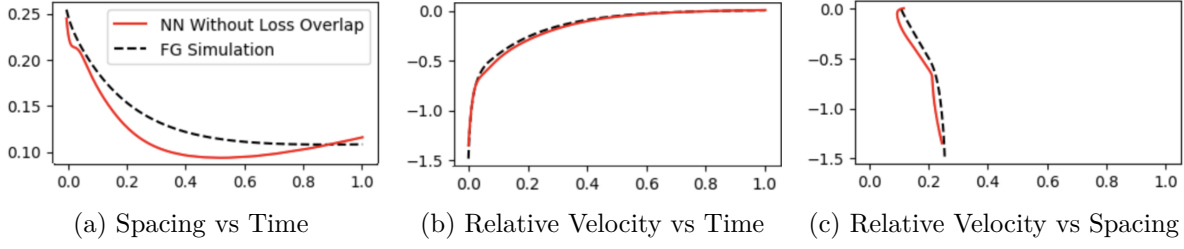


Figure 12: Example trajectory trained without overlap between the ODE and data-driven losses, resulting in a jagged transition in the middle of the trajectories. Same parameters, simulation time, and x profile as in Figure 2, with $a_\ell = 0$. Initial Conditions: $x_\ell = 0.25$, $v = 1.63$, $v_\ell = 0.14$.

where $\mathcal{I}_{\text{data}} = \{(i, j) : \tau^{(i)}(t_j) \leq \bar{\tau}\}$ and $\bar{\tau} = 0.7$ in this article.

From (F.3) and (F.4), some timesteps can be included in both the ODE and data-driven loss domains. This overlap improves sampling stability while preserving physical consistency. Without it, the transition between the two loss regimes can produce inaccurate and jagged behavior in the middle of the trajectory, as illustrated in Figure 12.

Initial Condition Loss, $J_{\text{ic}, \mathcal{N}_V}$. Given the existence and uniqueness of the proposed model solution, $\mathcal{N}_V$ must satisfy the initial condition at time $t = 0$. We enforce this using a band-penalty initial-condition loss, which penalizes deviation only when they exceed a tolerance $\delta_{\text{tol}} > 0$. This accounts for finite neural-network precision and avoids hindering convergence due to negligible deviations.

We set $\delta_{\text{tol}} = 10^{-4}$, and define the loss as:

$$(F.5) \quad J_{\text{ic}, \mathcal{N}_V} = \frac{1}{N} \sum_{i \in [1, N]} \left(\rho_{\delta_{\text{tol}}} \left(\hat{X}^{(i)}(0) - X^{(i)}(0) \right) \right)^2 + \left(\rho_{\delta_{\text{tol}}} \left(\hat{V}^{(i)}(0) - V^{(i)}(0) \right) \right)^2,$$

where N denotes the number of samples, and $\rho_{\delta_{\text{tol}}}(\cdot)$ denotes the band penalty function defined by:

$$(F.6) \quad \rho_{\delta_{\text{tol}}}(e) = \max(|e - \delta_{\text{tol}}| + |e + \delta_{\text{tol}}| - 2\delta_{\text{tol}}, 0).$$

Here, the band-penalty function is zero for $|e| \leq \delta_{\text{tol}}$ and grows linearly beyond this tolerance band. Analogous to the Lagrange multiplier method, it balances initial-condition enforcement with differentiability. Tuning λ_{ic} avoids over-constraining optimization while guiding $\mathcal{N}_V$ toward a unique, physically meaningful solution branch consistent with the Picard–Lindelöf theorem.

With all loss components defined, the neural network $\mathcal{N}_V$ is trained as a standard sequential feedforward MLP; the architecture is summarized in Algorithm G.2 in Appendix G and the implementation details are provided in Appendix H.2. Its performance is reported in Table 1 in Appendix C. The results show that the neural operator formulation effectively learns the nonlinear characteristics of the proposed system dynamics, even in the near-collision cases.

F.2. Neural Network $\mathcal{N}_X$. Having fully trained $\mathcal{N}_V$, we now introduce the second-stage network $\mathcal{N}_X$ to address the spacing prediction degradation of $\mathcal{N}_V$ caused by residual loss imbalance. Specifically, $\mathcal{N}_X$ refines spacing predictions from $\mathcal{N}_V$ by enforcing the dynamical consistency between spacing and relative velocity.

The loss function for $\mathcal{N}_X$ consists of two components: 1) a physics-based loss using V from $\mathcal{N}_V$, and 2) an initial condition loss ensuring well-posedness:

$$(F.7) \quad J_{\mathcal{N}_X} = \lambda_r J_{r, \mathcal{N}_X} + \lambda_{ic} J_{ic, \mathcal{N}_X}$$

where $\lambda_r = 2$ and $\lambda_{ic} = 50,000$ are selected through grid search to balance physical fidelity and numerical stability. Each loss component is described below.

ODE Loss Based on $\hat{V}$ from $\mathcal{N}_V$, $J_{r, \mathcal{N}_X}$. Similar to $\mathcal{N}_V$, the ODE loss enforces a physical consistency of $\mathcal{N}_X$ with the dynamical relationship $\dot{X} = V$ by minimizing the residual between the derivative of the predicted spacing and the predicted relative velocity. For each sample i and randomly selected time step t_j , the residual is computed as :

$$(F.8) \quad r_X^{(i,j)} = \frac{d\hat{X}^{(i)}(t_j)}{dt} - \hat{V}_{\mathcal{N}_V}^{(i)}(t_j).$$

The total residual loss is then given by

$$(F.9) \quad J_{r, \mathcal{N}_X} = \sum_{i,j} \left(r_X^{(i,j)} \right)^2.$$

Initial Condition Loss, $J_{ic, \mathcal{N}_X}$. To ensure uniqueness and stability, the initial condition loss for $\mathcal{N}_X$ is enforced through the band-penalty formulation analogous to that imposed for $\mathcal{N}_V$, as defined in (F.5) and (F.6), with $\delta_{tol} = 10^{-4}$:

$$(F.10) \quad J_{ic, \mathcal{N}_X} = \sum_{i=1}^N \left(\rho_{\delta_{tol}} \left(\hat{X}^{(i)}(0) - X^{(i)}(0) \right) \right)^2.$$

The loss function of $\mathcal{N}_X$ therefore shares the quadratic and convex structure as $\mathcal{N}_V$, ensuring differentiability, convexity, and stable optimization. With the loss function (F.7), $\mathcal{N}_X$ employs a sequential feedforward MLP architecture detailed in Algorithm G.3. Then, $\mathcal{N}_X$ is trained using the strategy described in Appendix H.3. Table 1 in Appendix C summarizes the performance of $\mathcal{N}_X$. The results show that $\mathcal{N}_X$ exhibits a clear improvement in spacing accuracy, validating the effectiveness of the two-stage framework in refining the spacing prediction of $\mathcal{N}_V$ by enforcing the physical relationship $\dot{X} = V$.

F.3. Performance of $\mathcal{N}_{sol}$. With the two-stage framework validated by the improved spacing prediction from $\mathcal{N}_X$, we now evaluate the overall performance of the complete neural network framework $\mathcal{N}_{sol}$, composed of the trained $\mathcal{N}_V$ and $\mathcal{N}_X$. The evaluation compares trajectories predicted by the *NN-based approximation* ($\mathcal{N}_{sol}$) with the reference trajectories from the *FG-Simulation*. Three key quantities are examined: the dynamic-based TTC τ , spacing X , and relative velocity V , under the same non-zero leading vehicle acceleration profile $\mathbf{a}_\ell$ defined in (D.1) in Appendix D.

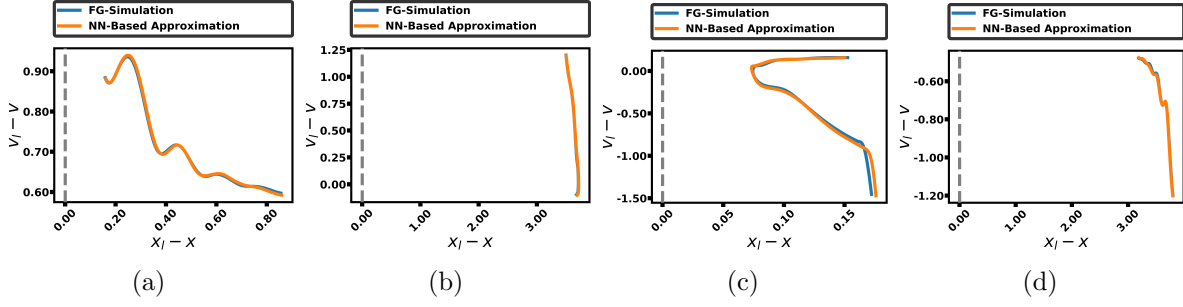


Figure 13: *NN-based Approximation* vs *FG-Simulation* Framework, Parameters: $\alpha = 2$, $\beta = 3$, $x = 0$, Simulation Time = 1, $\mathbf{a}_\ell$ as (D.1) in Appendix D. Initial Conditions: (a) $x_\ell = 0.16$, $v = 0.59$, $v_\ell = 1.47$; (b) $x_\ell = 3.5$, $v = 0.3$, $v_\ell = 1.5$; (c) $x_\ell = 0.17$, $v = 1.74$, $v_\ell = 0.29$; (d) $x_\ell = 3.8$, $v = 1.7$, $v_\ell = 0.5$.

As shown in Figure 13, the trajectories predicted by $\mathcal{N}_{\text{sol}}$ closely match those from the *FG-Simulation* across all scenarios, exhibiting strong long-horizon consistency and convergence to the reference solution, confirming its ability to capture the delay-free nonlinear dynamics. Moreover, Figure 19 in Appendix J highlights that $\mathbf{V}_\theta$ maintains accurate TTC prediction throughout each scenario, reinforcing the reliability of the proposed neural network framework for safety-critical applications.

F.4. Numerical Stability Analysis of the Model with Dynamic-Based Time-to-Collision: Platoon Simulation. To further assess the stability of the proposed model (2.1) under $\mathcal{N}_{\text{sol}}$, we conduct a numerical platoon simulation involving six vehicles as a representative example. Note that the proposed model is applicable to arbitrary platoon sizes. The leading vehicle follows the same non-constant acceleration $\mathbf{a}_\ell$ defined in (D.1) in Appendix D, consistent with the training setup of $\mathbf{V}_\theta$, $\mathcal{N}_V$, and $\mathcal{N}_X$, while the following vehicles $n \in [1, 5]$ evolve according to the proposed dynamics. The initial conditions are specified as:

$$(F.11) \quad \begin{cases} v_\ell(t_o) = 1.5, \\ v_n(t_o) = 0.4, \quad n \in [1, 5], \\ x_\ell(t_o) - \mathbf{x}_1(t_o) = 0.1, \\ x_{n-1} - x_n = \mathcal{V}^{-1}(v_n(t_o)), \quad n \in [2, 5]. \end{cases}$$

To examine disturbance propagation, we track spacing and relative velocity along the platoon over $t \in [0, 30]$. In the simulation, the leader-follower pair ($n = 0, 1$) employs the *NN-based approximation* ($\mathcal{N}_{\text{sol}}$) while remaining pairs use the *NeuNum-Simulation*.

As shown in Figure 14, fluctuations in spacing and relative velocity decay along the platoon, and the states converge toward equilibrium, demonstrating stable dynamics. This numerical evidence is consistent with local stability results for car-following dynamics as presented in [61, 59] and confirms that the neural network implementation preserves system stability properties.

To further evaluate the neural network framework, we perform the same simulation using the reference *FG-Simulation* for comparison. As shown in Figure 14 and Figure 15, trajectories

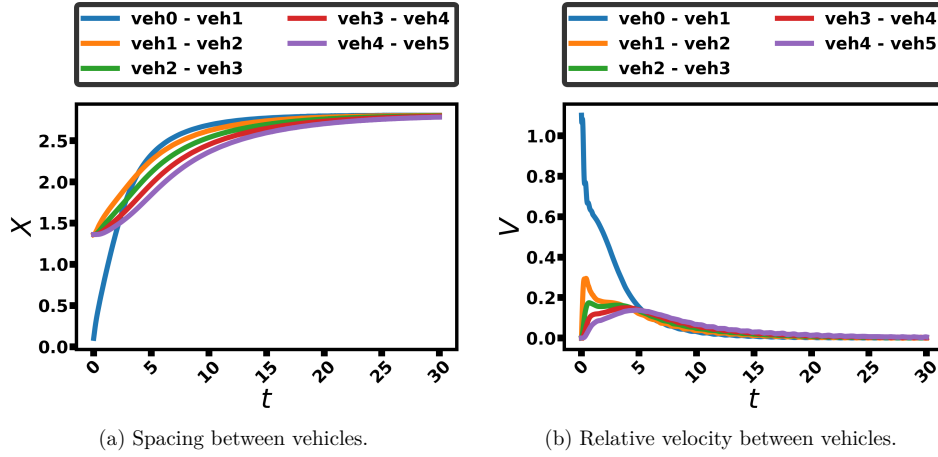


Figure 14: Simulated nonlinear stability of the proposed model through *NN-based Approximation* and *NeuNum-Simulation* Approach: (a) spacing $X(t)$ and (b) relative velocity $V(t)$ between vehicles. Same parameters, simulation time, and a_ℓ as in Figure 2. Initial Conditions as stated in (F.11) with $x_5 = 0$.

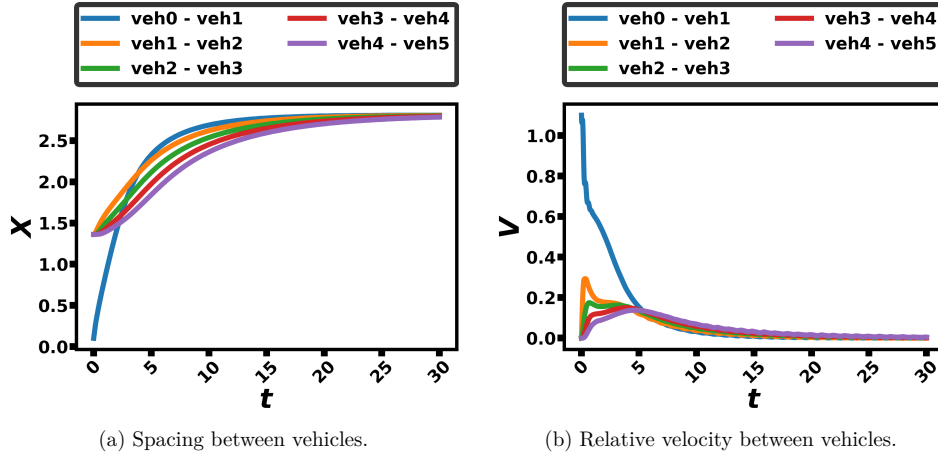


Figure 15: Simulated nonlinear stability of the proposed model through *FG-Simulation Offline* Approach: (a) spacing $X(t)$ and (b) relative velocity $V(t)$ between vehicles. Same parameters, simulation time, and a_ℓ as in Figure 2. Initial Conditions as stated in (F.11) with $x_5 = 0$.

from the *NN-based Approximation* and *NeuNum-Simulation* are nearly indistinguishable, validating that the proposed framework reproduces system dynamics accurately while eliminating delays.

Appendix G. Pseudo-codes for architectures of V_θ , $\mathcal{N}_{\text{sol}}$, $\mathcal{N}_V$, and $\mathcal{N}_X$.

The details of NN-architecture for the HJB solution V_θ is detailed in the following algo-

rithm:

Algorithm G.1 Neural Network Framework $\mathbf{V}_\theta$ for Value Function Approximation

Input: States $s \in [t_o, T_h]$, $\Delta\xi$, $\Delta\zeta$, and v_ℓ

Output: Approximated value function $\mathbf{V}_\theta(\Delta\xi, \Delta\zeta, v_\ell, s)$

Initialize a sequential neural network model

Normalize s , $\Delta\xi$, $\Delta\zeta$, and v_ℓ

Add Dense layer with 256 neurons, activation `tanh`

Add Dense layer with 128 neurons, activation `tanh`

Add Dense layer with 256 neurons, activation `swish`

Add Dense layer with 512 neurons, activation `tanh`

Add Dense layer with 256 neurons, activation `tanh`

Add Dense layer with 128 neurons, activation `swish`

Add Dense layer with 64 neurons, activation `tanh`

Apply `tanh` after each residual addition

Add Dense layer with 1 neuron

Apply the sigmoid function for output transformation

return Value function approximation $\mathbf{V}_\theta(s, \Delta\xi, \Delta\zeta, v_\ell) \in [0, T_h - s]$

The NN-architecture for approximating the relative velocity V is elaborated in the following algorithm:

Algorithm G.2 Neural Network Framework $\mathcal{N}_V$ for Proposed Model Solution Approximation

Input: Initial conditions $x_\ell(0) - x(0)$, $v_\ell(0) - v(0)$, $v_\ell(0)$, $\tau(0)$, t .

Output: Spacing X and Relative Velocity V at t

Initialize a sequential neural network model

Add Dense layer with 256 neurons, activation **LeakyReLU** with slope $\alpha = 0.001$

Add Dense layer with 128 neurons, activation **tanh**

Add Dense layer with 256 neurons, activation **sinusoidal** with frequency $\omega_0 = 10$

Add Dense layer with 128 neurons, activation **swish**

Add Dense layer with 128 neurons, activation **tanh**

Add Dense layer with 128 neurons, activation **swish**

Add Dense layer with 2 neurons

Apply **softplus** to the first output (spacing X) to ensure strict positivity

return Spacing X and Relative Velocity V Approximation.

The NN-architecture for approximating the Spacing X is elaborated in the following algorithm:

Algorithm G.3 Neural Network Framework $\mathcal{N}_X$ for Proposed Model Solution Approximation

Input: Initial conditions $x_\ell(0) - x(0)$, $v_\ell(0) - v(0)$, $v_\ell(0)$, $\tau(0)$, t .

Output: Corrected Spacing X at t

Initialize a sequential neural network model

Add Dense layer with 256 neurons, activation **tanh**

Add Dense layer with 256 neurons, activation **swish**

Add Dense layer with 256 neurons, activation **tanh**

Add Dense layer with 256 neurons, activation **swish**

Add Dense layer with 256 neurons, activation **tanh**

Add Dense layer with 256 neurons, activation **swish**

Add Dense layer with 1 neuron

return Corrected Spacing X

Appendix H. Neural Network Training Strategies.

Training of the Neural Network is done on an Apple M2 Max ARM Processor with a maximum clock frequency of 3696 MHz, 38 GPU cores, 12 CPU cores (8 Performance and 4 Efficiency), and 96 GB of RAM.

H.1. V_θ Training Implementation.

H.1.1. V_θ Training Setup. Training is carried out using the Adam optimizer with gradient clipping. Instead of employing a validation-loss-based learning rate reduction rule, the learning rate follows a warmup–cosine decay schedule: it increases linearly over the first 300 training

steps to a peak value 3×10^{-4} , and then decreases according to a cosine decay down to 10^{-6} over the course of training.

In the interior loss, the Bellman residual J_{HJB} is scaled by a constant weight $\lambda_{\text{HJB}} = 10$. In addition, the boundary loss $J_{\partial\mathcal{Q}}$ is multiplied by a separate coefficient $\lambda_{\partial\mathcal{Q}}$, set to 1 in the implementation. Consequently, the training objective places stronger emphasis on satisfying the HJB equation in the compact interior region while still enforcing the boundary condition.

H.1.2. V_θ Training Dataset. With $T_h = 1$, each training step uses two distinct batches: one interior batch and one boundary batch. The interior batch contains 65,536 samples and the boundary batch contains 8,192 samples. This procedure is repeated for 80,000 training steps. Before training, we first identify the dynamic collision boundary using the line method. For each sampled combination of $\Delta\zeta$, v_ℓ , and s , the line method determines the dynamic boundary: the critical spacing value separating states for which a dynamic-based TTC exists from states outside the collision-relevant region.

The interior batch is then sampled only from the compact collision-relevant region. Specifically, values of $\Delta\xi$ are extracted between the collision boundary $\Delta\xi = 0$ and the dynamic boundary found by the line method, $\Delta\zeta \in [-v_{\max}, 0]$, $v_\ell \in [0, v_{\max} + \Delta\zeta]$, and $s \in [0, T_h)$. The boundary batch is sampled on the collision boundary itself $\Delta\xi = 0$, together with the same feasible sampling logic for $\Delta\zeta$, v_ℓ , and s . The boundary batch is introduced to enforce the near-collision zero-value condition, while the interior batch is used to learn the value function over the admissible safe domain.

For both batches, the samples are generated on-the-fly at every training step rather than being drawn from a fixed offline dataset. This design allows the training procedure to continuously expose V_θ to fresh states while focusing sampling effort on the safety-critical region. For the boundary-batch, candidate points are first sampled from the feasible domain and are then retained only if they lie inside the compact set determined by the line method.

To ensure that the empirical loss approximates the continuum objective, the interior samples are drawn according to Monte-Carlo quadrature over the compact interior region. In particular, $\Delta\zeta$ and v_ℓ are sampled according to the Lebesgue measure on the feasible triangular set $\{(\Delta\zeta, v_\ell) : -v_{\max} \leq \Delta\zeta \leq 0, 0 \leq v_\ell \leq v_{\max} + \Delta\zeta\}$ and s is sampled uniformly over $[0, T_h)$. The spacing variable $\Delta\xi$ is then sampled within the admissible interval determined by the dynamic boundary, with the margin of 10^{-5} . Consequently, the HJB residual is evaluated only in the compact collision-relevant region, while the boundary condition is imposed only on the collision boundary $\Delta\xi = 0$.

In summary, each training step combines one interior batch and one boundary batch, both generated on-the-fly. The interior batch is restricted to the compact region identified by the line method, and the boundary batch enforces the zero-value condition on $\Delta\xi = 0$. This sampling design aligns the empirical training objective with the continuum formulation by applying the HJB loss in the compact interior region and the boundary loss on the collision boundary.

H.2. $\mathcal{N}_V$ Training Implementation.

H.2.1. $\mathcal{N}_V$ Training Setup. Training is carried out using the Adam optimizer with a learning rate of 10^{-3} . A learning rate scheduler reduces the rate by a factor of 0.75 whenever

the loss does not improve for 200 consecutive iterations. Exponential moving averages of gradients and squared gradients are used with decay rates of 0.9 and 0.999. The batch size is set to $N = 256$.

H.2.2. $\mathcal{N}_V$ Training Dataset. With the architecture and loss established, we now describe the training procedure for $\mathcal{N}_V$. The training dataset is composed of full-trajectory samples generated under various initial conditions. During each epoch, the data are divided into mini-batches of size N . For every trajectory $i \in \{1, \dots, N\}$ within a batch, the neural network is evaluated at M randomly sampled time points $t_j \in [0, T]$ where $j \in \{1, \dots, M\}$. To ensure temporal boundary accuracy, we always include both $t_1 = 0$ and $t_M = T$. This randomized temporal sampling allows $\mathcal{N}_V$ to learn the full operator mapping from initial conditions to continuous trajectories, rather than overfitting to a fixed time grid. In this study, we set $T = 1$ with a fine timestep size Δt being $1e-5$, resulting in a temporal grid of 100,000 discrete steps. At each training iteration, $M = 17,500$ timesteps are randomly selected from this grid to promote temporal robustness and reduce overfitting.

To improve predictive performance in the safety-critical region, the initial condition sampling is intentionally biased toward near-collision scenarios. This strategy is consistent with the sampling approach used for training $\mathbf{V}_\theta$ and is guided by the safety-critical region, following local stability analysis for car-following dynamics presented in [61, 59]. Accordingly, initial spacings are sampled uniformly over intervals from $[0.00, 0.01]$ up to $[0.30, 0.31]$, with 4500 samples per interval. Beyond this range, for intervals from $[0.30, 0.4]$ up to $[\delta_f - 0.1, \delta_f]$ with $\delta_f = 4.1$, the sampling density is reduced progressively by a factor of 0.8 per interval, while ensuring no fewer than 1,000 samples in any interval.

Full trajectories are then simulated for each selected initial condition. All generated trajectories are simulated under the same non-zero leading vehicle acceleration profile $\mathbf{a}_\ell$ specified in (D.1) in Appendix D, identical to the setup used for training $\mathbf{V}_\theta$ in Section 5. This consistency ensures that the learned operator remains aligned with the dynamic environment for which the dynamic-based TTC is defined and approximated.

H.3. $\mathcal{N}_X$ Training Implementation.

H.3.1. $\mathcal{N}_X$ Training Setup. The training setup for $\mathcal{N}_X$ closely follows that of $\mathcal{N}_V$, employing the same Adam optimizer, learning-rate schedule, number of epochs, and mini-batch size ($N = 256$). See Appendix H.2.1 for the corresponding $\mathcal{N}_V$ setup.

H.3.2. $\mathcal{N}_X$ Training Dataset. Similar to the training setup, the training dataset of $\mathcal{N}_X$ is generated based on the same approach as for $\mathcal{N}_V$ — a randomized temporal sampling strategy, where 17,500 random steps are selected from a temporal grid of 100,000 timesteps over the interval $[0, 1]$ with resolution $\Delta t = 10^{-5}$. Likewise, the same biased sampling approach is applied to initial conditions to ensure reliable performance in the safety-critical region. Throughout training, $\mathcal{N}_V$ provides the predicted relative velocity profiles used in the residual computation. All models are trained under the same non-zero leading vehicle acceleration $\mathbf{a}_\ell$ defined in (D.1) in Appendix D, consistent with the setup used for $\mathbf{V}_\theta$ in Section 5 and Appendix H.1, and for $\mathcal{N}_V$ in Appendix H.2.

Appendix I. Neural Network Training Performance.

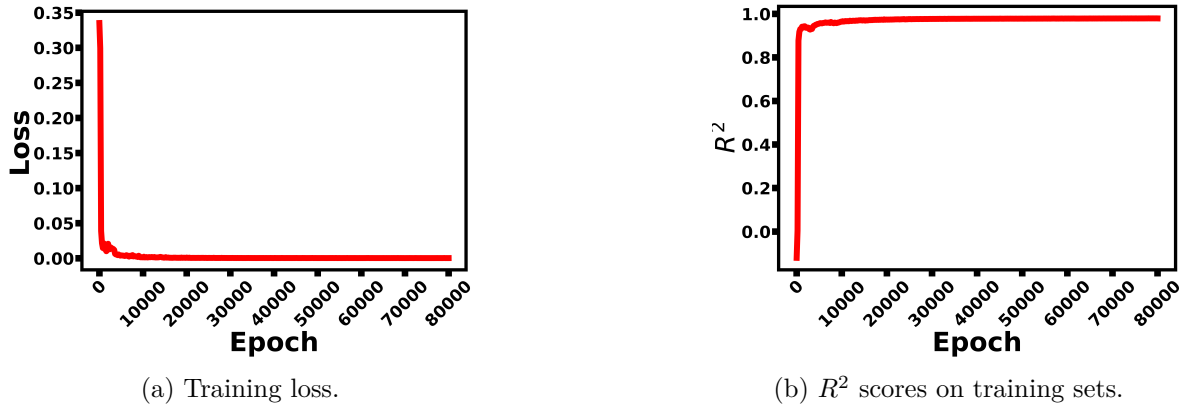


Figure 16: Training performance of V_θ : (a) loss curves and (b) R^2 scores.

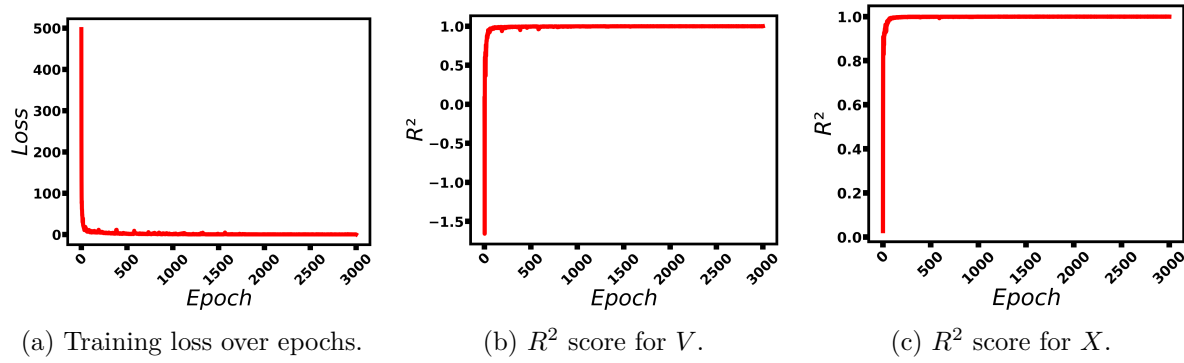


Figure 17: Training performance of $\mathcal{N}_V$ with $\mathbf{a}_\ell$ defined by (D.1) in Appendix D: (a) training loss, (b) R^2 score for relative velocity V , and (c) R^2 score for spacing X .

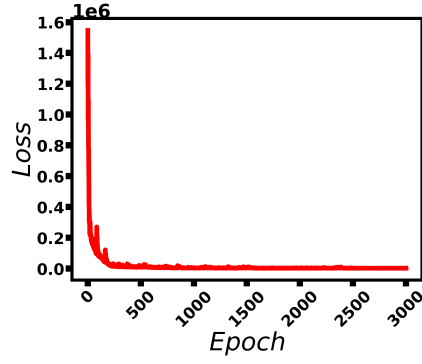
Appendix J. Dynamic-Based TTC for Each Region.

Code Availability. The code used to perform the simulations, train and validate the neural network models, and generate the numerical results and figures in this paper is available from <https://github.com/yuneily97/hjb-ttc-neural-workflow>.

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(a) Training loss over epochs.

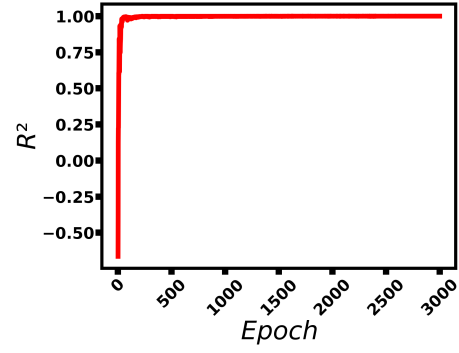
(b) R^2 score for X during training.

Figure 18: Training performance of $\mathcal{N}_X$ with $\mathbf{a}_\ell$ defined by (D.1) in Appendix D: (a) training loss and (b) R^2 score for spacing X .

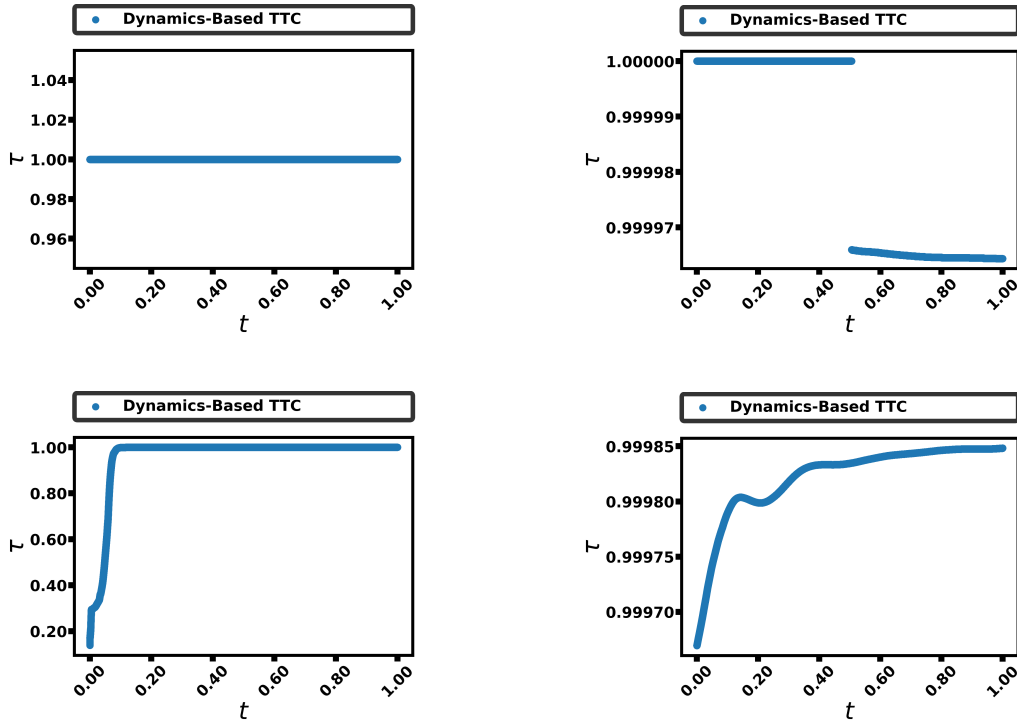


Figure 19: Dynamic-Based TTC from $\mathbf{V}_\theta$ for different initial conditions, Parameters: $\alpha = 2$, $\beta = 3$, $x = 0$, Simulation Time = 1, $\mathbf{a}_\ell$ as (D.1) in Appendix D. Initial Conditions: (a) $x_\ell = 0.16$, $v = 0.59$, $v_\ell = 1.47$; (b) $x_\ell = 3.5$, $v = 0.3$, $v_\ell = 1.5$; (c) $x_\ell = 0.17$, $v = 1.74$, $v_\ell = 0.29$; (d) $x_\ell = 3.8$, $v = 1.7$, $v_\ell = 0.5$

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